Sorbonne Université
Institut de Minéralogie, de Physique des Matériaux et de Cosmochimie
Ecole Doctorale 564 : Physique Ile de France

Green-Function Approaches to Lattice Vibrations in Disordered Materials

Presented by ALESSANDRO CIAVATTA

PhD thesis in Physics

October 2026

Président du Jury :	ELEONORA LUPPI	LCT - Sorbonne Université
Rapporteurs :	MATTEO COCOCCIONI	University of Pavia
	IVANA SAVIC	King's College London
Examinateurs :	KONSTANTINOS TERMENTZIDIS	INSA - Lyon
	SEVERINE GOMES	INSA - Lyon
	FRANÇOIS-XAVIER COUDERT	IRCP - Université PSL
Directeur :	LORENZO PAULATTO	IMPMC - Sorbonne Université

ABSTRACT

This thesis develops a Green-function framework for describing the interplay between lattice vibrations and disorder. The perturbing potential incorporates mass and force-constant disorder parameters calculated using density-functional perturbation theory (DFPT). A formalism combining Dyson's equation with configurational averaging in reciprocal space is developed. The Fourier-interpolation framework for the perturbing potential is extended to incorporate a generalized centering procedure, long-range electrostatic corrections to the force constants, and associated sum rules. The key approximations used to treat disorder are examined, including the virtual crystal approximation (VCA), the average T-matrix approximation (ATA), the coherent potential approximation (CPA), and the dynamical cluster approximation (DCA). The ATA and DCA are implemented in an optimized MPI-parallelized code, which uses tetrahedron integration in the complex plane. The methods are applied to Te-substituted monolayer WSe_2, where defect and anharmonic self-energies are used together with Raman tensors to calculate shifted and broadened spectra and identify impurity states. B- and P-doped silicon are then studied, with particular attention to shallow dopants, charged configurations, and force-constant perturbations. $Si_{1-x}Ge_x$ alloys are finally investigated, with a focus on the interplay between mass and force-constant disorder in the lattice thermal conductivity. Overall, this work bridges first-principles lattice dynamics and effective-medium and cluster descriptions of disorder, and provides tools for interpreting phonon linewidths, spectral functions, and defect-induced Raman features in real materials.

Contents

Introduction **5**

1 Theory of lattice vibrations and disorder **9**
1.1 Density Functional Theory 9
1.1.1 Adiabatic approximation 9
1.1.2 Hohenberg-Kohn Theorems 10
1.1.3 Kohn-Sham system 11
1.1.4 Density Functional Perturbation Theory (DFPT) 13
1.1.5 2n+1 theorem 14
1.1.6 Periodic systems 15
1.1.7 Electric field response 16
1.1.8 Raman spectroscopy 19
1.1.9 Anharmonicity and phonon-phonon interactions 20
1.2 Modeling of the defective potential 21
1.2.1 Defect force constants 21
1.2.2 Centering of force constants 24
1.2.3 Configuration potential 27
1.2.4 Sum rule and detailed balance 28
1.2.5 New ASR 29
1.2.6 Long range interactions 30
1.2.7 Dipole-dipole constants corrections 32
1.3 Green's function formalism 34
1.3.1 Dyson expansion 34
1.3.2 T-matrix and self-energy 37
1.3.3 Localized states 37
1.3.4 Resonant states 39
1.3.5 Configurational Average 40
1.3.6 Higher order diagrammatic expansions 41
1.3.7 Real and reciprocal space 44
1.3.8 Normalization of the potential 46
1.4 Disorder approximations 47
1.4.1 Virtual Crystal Approximation (VCA) 47
1.4.2 Average T-matrix Approximation (ATA) 50
1.4.3 Coherent Potential Approximation (CPA) 52
1.4.4 Dynamical cluster approximation (DCA) 57

2 Numerical implementations **63**
2.1 Tetrahedron Methods . . . 63
2.1.1 Non-diagonal operators . . . 66
2.1.2 Three-phonon surfaces and self-energy interpolation . . . 68
2.2 Numerical implementation of ATA . . . 68
2.2.1 Optical Theorem and scattering rate . . . 69
2.2.2 Mass disorder and Tamura model . . . 70
2.2.3 Aliasing in Fourier transforms . . . 71
2.2.4 Self-energy normalization . . . 73
2.3 Numerical implementation of DCA . . . 75
2.3.1 Broyden's mixing . . . 75
2.3.2 Configurational average . . . 77
2.3.3 Coarsening and interpolation . . . 78

3 Results and Discussion **83**
3.1 Vacancies and inclusions . . . 83
3.1.1 Vacancies in silicon . . . 84
3.2 Substitutional doping . . . 85
3.2.1 Raman spectrum of WSe_2 with Te substitutions . . . 85
3.2.2 P and B doping in Silicon . . . 90
3.3 Lattice thermal conductivity and thermoelectricity . . . 93
3.4 Alloying and thermal conductivity . . . 97
3.4.1 Silicon-germanium alloys . . . 97
3.4.2 Scattering from virtual crystal . . . 98

Introduction

The transport of heat and charge in crystalline materials is governed by a delicate balance between order and disorder. In a perfect harmonic crystal, phonons are normal modes with infinite lifetime: they propagate without decay, and the only information needed to describe them is their frequency and polarization. Real materials are never in this ideal limit. Phonons are scattered by other phonons, by electrons, by boundaries, by isotopes and by structural defects. These scattering mechanisms determine phonon linewidths in spectroscopy, phonon mean free paths in thermal transport, and the lattice contribution to the thermal conductivity.

This problem is especially important for thermoelectric materials. A good thermoelectric material should conduct charge efficiently while conducting heat poorly, as summarized by the figure of merit $ZT = S^2\sigma T/(\kappa_e + \kappa_l)$. Defects and dopants enter this problem in two opposite ways. On one hand, they are often necessary to tune the carrier concentration and therefore the electrical conductivity σ and the Seebeck coefficient S. On the other hand, they perturb the crystal lattice and scatter phonons, reducing the lattice κ_l and electronic κ_e thermal conductivities. This reduction can improve thermoelectric performance, but the same defects may also degrade electronic mobility, introduce localized vibrational modes, or produce resonant scattering features. A predictive theory must therefore describe the defect not merely as a mass change, but as a full perturbation of the lattice dynamics.

The simplest models of phonon-defect scattering treat disorder as a weak mass perturbation. They are useful for isotopic mixtures and for dilute limits, but they miss several effects that become essential in real doped materials. A substitutional atom changes the local bonding environment, relaxes the surrounding atoms and modifies the interatomic force constants. Vacancies and interstitials change the number and character of the local degrees of freedom. Charged and shallow defects may also generate long-range electrostatic responses that converge slowly in supercell calculations. In these cases the perturbation theory can fail, and the defect may create resonant or localized phonon states. The central motivation of this work is to build a framework that retains these first-principles ingredients while still allowing one to average over disorder and compute quantities relevant to spectroscopy and transport.

The direct supercell approach provides an intuitive route: one places defects in a large periodic cell, computes the force constants and extracts the vibrational properties. However, such a calculation describes a periodic array of defects rather than a random alloy or a randomly doped crystal. Increasing the supercell size reduces the artificial interaction between periodic images, but it quickly becomes computationally prohibitive, and it still samples only one configuration. The approach followed in this thesis separates the problem into two parts. First, density-functional perturbation the-

ory is used to construct the pristine and defective force constants and therefore the localized perturbing potential produced by a defect. Second, Green-function methods are used to embed this perturbation into the infinite crystal and to perform the configurational average over defect positions.

In this language, disorder modifies the phonon propagator through a self-energy. The real part of the self-energy shifts phonon frequencies, while the imaginary part gives the linewidths and lifetimes associated with phonon-defect scattering. Once these quantities are known, they can be combined with other scattering channels, such as phonon-phonon, electron-phonon and boundary scattering, to evaluate thermal and thermoelectric properties. The same formalism also gives access to spectral functions and density of states, making Raman spectra and impurity modes natural tests of the theory. This dissertation is organized as follows.

Chapter 1 presents the theoretical framework. Its first part reviews the first-principles methods used in lattice dynamics: density-functional theory (DFT), density-functional perturbation theory (DFPT), harmonic phonons, electric field responses, Raman tensors and anharmonic phonon-phonon interactions. The second part is devoted to the construction of the defect perturbing potential. Starting from pristine and defective interatomic force constants, the mass and force-constant contributions are combined into a dynamical-matrix perturbation. Particular attention is paid to Fourier interpolation, centering procedures, acoustic sum rules, detailed balance and the treatment of long-range electrostatic interactions in polar or charged systems. The last part of the chapter develops the Green-function description of disorder. Dyson expansions, localized and resonant states, the T-matrix, self-energies, diagrammatic expansions and configurational averaging are introduced in the notation used throughout the thesis. The chapter ends by comparing the main approximations for treating disorder: the virtual crystal approximation (VCA), the average T-matrix approximation (ATA), the coherent potential approximation (CPA) and the dynamical cluster approximation (DCA).

Chapter 2 describes the numerical implementation of these ideas. The chapter begins with the tetrahedron methods used to evaluate Brillouin-zone integrals of Green functions in the complex plane. It then presents the ATA implementation, with emphasis on the optical theorem, the connection with the Tamura model for mass disorder, the treatment of Fourier aliasing, the real-space support of the perturbing potential and the normalization of the self-energy for equivalent defect sites. The DCA implementation is then discussed as the finite-concentration extension of the single-defect T-matrix picture. The self-consistent solution requires a stable mixing procedure, implemented through a complex Broyden algorithm, and a careful sampling of defect configurations. The final part analyzes the coarsening and interpolation steps that connect the cluster self-energy to the lattice Green function.

Chapter 3 applies the formalism to representative classes of defects. The first section discusses how vacancies and interstitial inclusions can be written as perturbations compatible with the same Green-function machinery. The following sections focus on substitutional doping. Te substitutions in monolayer WSe_2 are used to study defect-induced Raman broadening and frequency shifts, and to compare ATA and DCA descriptions of impurity states in a two-dimensional material. B and P substitutional defects in silicon

are then analyzed as prototypical shallow dopants. Their different masses, local relaxations and charge states lead to different phonon self-energies, densities of states and localized or resonant vibrational features. The final section connects these results to lattice thermal conductivity and thermoelectricity, showing how the phonon-defect self-energy developed in this work can be combined with the other microscopic scattering mechanisms needed for first-principles transport calculations. $Si_{1-x}Ge_x$ alloys are then chosen as the prototypical example of a system in which disorder strongly affects the thermal transport.

The aim of the thesis is therefore not only to compute the effect of a single defect in a specific material, but to provide a transferable route from first-principles force constants to disorder-averaged phonon lifetimes, spectral functions and transport properties. By treating phonon-defect scattering beyond the weak-scattering limit, the methods developed here make it possible to describe the same defects that are used to engineer thermal and thermoelectric performance in real materials.

1 Theory of lattice vibrations and disorder

1.1 Density Functional Theory

1.1.1 Adiabatic approximation

All the fundamental properties of a material, such as its electronic structure, optical response, and vibrational properties, can be derived from the Hamiltonian of the system, a function of the positions of the nuclei and the electronic degrees of freedom. At $T = 0\,\mathrm{K}$, the system is found in its ground state, defined as the lowest energy eigenstate of the Hamiltonian. The full, non-relativistic Hamiltonian of a system composed of nuclei and electrons $H = T_e + V_e + V_{ext} + T_N + V_N$ has the following contributions:

- electron kinetic energy $T_e = -\frac{1}{2}\sum_i \nabla_i^2$;
- electron-electron interaction $V_e = \frac{1}{2}\sum_{ij} |\boldsymbol{r}_i - \boldsymbol{r}_j|^{-1}$;
- electron-nucleus interaction $V_{ext} = -\sum_{iI} Z_I |\boldsymbol{r}_i - \boldsymbol{R}_I|^{-1}$;
- nuclear kinetic energy $T_N = -\frac{1}{2}\sum_I \frac{\nabla_I^2}{M_I}$;
- nucleus-nucleus interaction $V_N = \frac{1}{2}\sum_{IJ} Z_I Z_J |\boldsymbol{R}_I - \boldsymbol{R}_J|^{-1}$.

Atomic Hartree units are used throughout the entire dissertation, fixing the constants $\hbar$, m_e, e, $4\pi\epsilon_0$ to unity.

The most common approximation, known as the *Born–Oppenheimer* or *adiabatic* approximation, involves first solving the Hamiltonian that includes the degrees of freedom of the electrons, while treating the positions of the ions, denoted as $\boldsymbol{R}_I$, as fixed parameters. Since the electron mass is a fraction of the nuclear mass, the electrons relax into the equilibrium configuration much faster than the ions. The kinetic term of the nuclei can be treated as a perturbation ($m_e/M_I \ll 1$). The potential V_N does not depend on the electronic degrees of freedom and can be determined simply by considering the positions of the nuclei. Its effect is to shift the energies of the system, with the positions of the ions kept frozen. The following discussion focuses on this system, which can be described by the Hamiltonian $H = T_e + V_e + V_{ext}$. Electrons deep in the core levels of the atomic species are known to not participate significantly in chemical bonding. In the isolated atom, they possess a very high binding energy and hence are not expected to be greatly perturbed by the presence of other atoms. They can be considered as rigidly attached to the nucleus, as a fixed charge distribution that follows the ion. The electronic degrees of freedom are then reduced to the valence electrons: each ionic potential V_{ext} is modified to account for the effect of the core electrons, resulting in a

smoother *pseudopotential*, constructed to reproduce the scattering properties of the full potential [1].

1.1.2 Hohenberg-Kohn Theorems

The full many-body Schrödinger equation for a system of N electrons is solved by determining the many-body eigenfunctions $\Psi(\boldsymbol{r}_i)$, which involve all $3N$ electronic degrees of freedom. Due to the Coulomb repulsion between electrons, the Hamiltonian cannot be separated into single-particle terms, and the wavefunction Ψ cannot be factorized. Hohenberg and Kohn [2] demonstrated that all the relevant properties of a system, including its excited state electronic properties, can be derived from the ground state electron density $n_0(\boldsymbol{r})$. Hohenberg and Kohn formulated two theorems that establish the effectiveness of this approach for addressing the many-body electron problem:

- **Theorem I:** The external potential $V_{ext}(\boldsymbol{r})$ of a system of interacting electrons is uniquely determined, up to a constant, by the ground state electron density $n_0(\boldsymbol{r})$. The electronic Hamiltonian $H = T_e + V_e + V_{ext}$ is fully determined by the external potential since the forms of the kinetic and Coulomb repulsion terms are independent from the system's structure. By constructing the corresponding Hamiltonian from $n_0(\boldsymbol{r})$, one can determine the ground and excited states and energies by solving the many-body Schrödinger equation.
  Proof: Suppose that there are two systems with external potentials V_{ext} and V'_{ext}, associated with the same ground state density $n_0(\boldsymbol{r})$. The two systems are then characterized by different Hamiltonians H and H', with different ground states Ψ and Ψ'. Thus,

  $$E = \langle\Psi|H|\Psi\rangle < E' + \langle\Psi'|H - H'|\Psi'\rangle$$
  $$E' = \langle\Psi'|H'|\Psi'\rangle < E - \langle\Psi|H - H'|\Psi\rangle\,.$$

  The inequality is strict only in the non-degenerate case, but the demonstration can be extended to encompass these situations as well. The difference between the Hamiltonians boils down to the discrepancy in external potentials, which yields the same outcome when evaluated for both Ψ and Ψ':

  $$\langle\Psi|H - H'|\Psi\rangle = \langle\Psi'|H - H'|\Psi'\rangle = \int d^3r\,\left[V_{ext}(\boldsymbol{r}) - V'_{ext}(\boldsymbol{r})\right] n_0(\boldsymbol{r}).$$

  Adding the two strict inequalities gives the contradiction $E + E' < E + E'$. Therefore, two distinct external potentials cannot yield the same non-degenerate ground-state density.

- **Theorem II:** $T_e + V_e$ can be explicitly constructed as functionals of the electron density $n(\boldsymbol{r})$. The ground state electron density $n_0(\boldsymbol{r})$ minimizes the functional.

  $$E[n] = T_e[n] + V_e[n] + \int d^3r\, V_{ext}(\boldsymbol{r})n(\boldsymbol{r}) \tag{1.1}$$

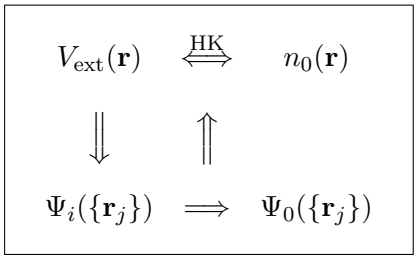


Figure 1.1: *Schematic representation of the Hohenberg–Kohn mapping. The left vertical arrow is given by solving the Hamiltonian for a given external potential, the ground state density is built from the ground state wavefunction. The theorem connects the density to a unique external potential.*

and the minimum value corresponds to the ground state energy.

Proof: The construction of the energy functional is derived from Levy and Lieb [3, 4]. The ground state energy is obtained by minimizing the energy with respect to every possible wavefunction Ψ. Alternatively, one can first minimize within the restricted space $A(n)$, which consists of all wavefunctions with the same density $n(\boldsymbol{r})$, and then minimize with respect to the densities. As a result, the universal functional arises from the initial minimization process: $T_e[n] + V_e[n] = \min_{A(n)} \langle\Psi|T_e + V_e|\Psi\rangle$. Each density can then be associated with the wavefunction Ψ that minimizes the universal functional. In particular,

$$E_0 = E[n_0] = \langle\Psi_0|H|\Psi_0\rangle < \langle\Psi|H|\Psi\rangle = E[n]$$

1.1.3 Kohn-Sham system

We first consider a non-interacting system. Its many-electron wavefunction is a Slater determinant built from single-particle orbitals ψ_i. The density and the kinetic term are expressed as:

$$n(\boldsymbol{r}) = \sum_i |\psi_i(\boldsymbol{r})|^2 \tag{1.2}$$

$$T_s = -\frac{1}{2}\sum_i \langle\psi_i|\nabla^2|\psi_i\rangle\,. \tag{1.3}$$

The most natural way to treat charged particles as if they were independent is the *Hartree* approximation: each particle is subject to a Coulomb potential generated by all the other particles. The contribution to the total energy is given by the integral

$$E_H[n] = \frac{1}{2}\int d^3r d^3r'\,\frac{n(\boldsymbol{r})n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}, \tag{1.4}$$

which includes a spurious interaction between an electron and itself. While this term is an explicit functional of the density, the non-interacting kinetic energy $T_s[n]$ is defined

through the constrained-search construction. This approach is missing two important ingredients:

- *Exchange:* The ground state many-body fermionic wavefunction is expected to be antisymmetric under the exchange of particles. One can remedy this inadequacy by explicitly antisymmetrizing the trial wavefunction, which yields the correct exchange contribution and a qualitatively better approximation to the true ground-state wavefunction.
- *Correlation:* Within this picture, the only electronic configuration retained is the one in which the lowest-energy orbitals are occupied. Electron-electron interactions admix excited configurations into the many-body wavefunction. Accounting for this admixture lowers the energy beyond the exchange-only description.

The Hartree energy contains the electrostatic interaction of the total electronic density with itself, and therefore includes unphysical self-interaction terms. In Hartree-Fock theory these terms are cancelled exactly by the exchange energy for each occupied orbital. In exact DFT the exchange-correlation functional should provide the corresponding cancellation, but approximate functionals generally do not. In periodic systems this issue is also related to the character of the one-electron states: for perfectly delocalized Bloch orbitals the orbital self-interaction vanishes in the thermodynamic limit, while localized orbitals have a finite self-Coulomb energy. As a consequence, approximate functionals may spuriously favor or penalize localization and can lead to symmetry-broken solutions. The density functional approach groups these many-body effects into a single exchange-correlation energy:

$$E_{xc}[n] = T_e[n] - T_s[n] + V_e[n] - E_H[n]. \tag{1.5}$$

It is important to notice that each term is a functional of the density. The true energy functional in Eq. (1.1) can be rewritten as

$$E[n] = T_s[n] + E_H[n] + \int d^3r\, V_{ext}(\boldsymbol{r}) n(\boldsymbol{r}) + E_{xc}[n]. \tag{1.6}$$

The central idea of Kohn and Sham [5] is based on the following assumption: a system of interacting electrons described by an external potential can be mapped to an equivalent system of non-interacting particles with a different effective potential that shares the same ground state density. The effective potential results from the minimization of the energy with respect to the orbitals:

$$V_{SCF}(\boldsymbol{r}) = V_{ext}(\boldsymbol{r}) + \int d^3r' \, \frac{n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} + \frac{\delta E_{xc}}{\delta n(\boldsymbol{r})} \tag{1.7}$$

$$H_{SCF}(\boldsymbol{r}) = -\frac{1}{2}\nabla^2 + V_{SCF}(\boldsymbol{r}); \quad H_{SCF}\psi_i(\boldsymbol{r}) = \epsilon_i \psi_i(\boldsymbol{r}), \tag{1.8}$$

with the last term of the potential also called $v_{xc}(\boldsymbol{r})$. These equations are solved self-consistently, starting with a guess for the density, calculating the effective potential,

solving for the orbitals and evaluating the density through Eq. (1.2). The energy functional should give lower and lower results at each cycle, eventually changing only negligibly once convergence is achieved. Although the functional dependence of the exchange-correlation functional is still unknown, some approximations have proven to be successful. The simplest one is the local density approximation (LDA): at each point $\boldsymbol{r}$ the exchange-correlation energy per particle is approximated by that of a homogeneous electron gas at density $n(\boldsymbol{r})$. While the exchange functional is known analytically, the correlation functional is parameterized using Quantum Monte Carlo data for the homogeneous electron gas. A generally more predictive approximation is the generalized-gradient approximation (GGA), in which the energy functional depends on both the density and its gradient.

1.1.4 Density Functional Perturbation Theory (DFPT)

The electronic ground state density [6], [7] is the key ingredient for lattice dynamics. The lattice Hamiltonian, within the adiabatic approximation, is written in terms of $n_0(\boldsymbol{r})$ in the actual lattice configuration $\{\boldsymbol{R}\}$:

$$H = T_N + V_N(\{\boldsymbol{R}\}) + E[n_0, \{\boldsymbol{R}\}]. \tag{1.9}$$

The nuclei move under the action of forces, which are simply given by the negative gradient of the energy expectation value in the classical picture. A major simplification in lattice dynamics follows from the *Hellmann–Feynman theorem*, which states that knowledge of the variational ground state at position $\{\boldsymbol{R}\}$ is sufficient to calculate the gradients:

$$\boldsymbol{F}_I = -\frac{\partial E}{\partial \boldsymbol{R}_I} = -\frac{\partial \langle \Psi_{\{\boldsymbol{R}\}} | H | \Psi_{\{\boldsymbol{R}\}} \rangle}{\partial \boldsymbol{R}_I} = -\langle \Psi_{\{\boldsymbol{R}\}} | \frac{\partial H}{\partial \boldsymbol{R}_I} | \Psi_{\{\boldsymbol{R}\}} \rangle . \tag{1.10}$$

The same theorem can be written in functional form and applied to the DFT ground state: the contribution coming from the differentiation of the electronic energy with respect to the density is zero at the ground state density, hence only the explicit derivative remains:

$$\boldsymbol{F}_I = -\frac{\partial V_N}{\partial \boldsymbol{R}_I} - \frac{\partial E}{\partial \boldsymbol{R}_I} = -\frac{\partial V_N}{\partial \boldsymbol{R}_I} - \int d^3r \, \frac{\partial V_{ext}(\{\boldsymbol{R}\}, \boldsymbol{r})}{\partial \boldsymbol{R}_I} n_0(\boldsymbol{r}). \tag{1.11}$$

Another central quantity in lattice dynamics is the second derivative of the energy with respect to two atomic displacements:

$$\frac{\partial^2 E}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} = \frac{\partial^2 V_N}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} + \int d^3r \left[\delta_{IJ} \frac{\partial^2 V_{ext}}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} n_0(\boldsymbol{r}) + \frac{\partial V_{ext}}{\partial \boldsymbol{R}_I} \frac{\partial n(\boldsymbol{r})}{\partial \boldsymbol{R}_J} \right]. \tag{1.12}$$

These derivatives define the interatomic force constants (IFCs), which determine the harmonic restoring forces near equilibrium. As a consequence of Eq. (1.11), the IFCs depend on the response of the ground state electronic density to the atomic displacement

only up to *linear* order, which dramatically simplifies the problem. Within perturbation theory, first-order variations are obtained by linearizing the Kohn–Sham equations in Eqs. (1.2), (1.7)–(1.8):

$$\Delta n(\boldsymbol{r}) = 4\mathrm{Re}\sum_{n=1}^{N/2}\psi_n^*(\boldsymbol{r})\Delta\psi_n(\boldsymbol{r}) \tag{1.13}$$

$$\Delta V_{SCF}(\boldsymbol{r}) = \Delta V_{ext}(\boldsymbol{r}) + \int d^3r' \frac{\Delta n(\boldsymbol{r'})}{|\boldsymbol{r}-\boldsymbol{r'}|} + \frac{dv_{xc}}{dn}\Delta n(\boldsymbol{r}) \tag{1.14}$$

$$(H_{SCF}-\epsilon_n)\,|\Delta\psi_n\rangle = -(\Delta V_{SCF}-\Delta\epsilon_n)\,|\psi_n\rangle\,, \tag{1.15}$$

where Δ stands for a variation with respect to any set of parameters. One can use the standard perturbation theory formula for $\Delta\psi_n(\boldsymbol{r})$ to write the density as:

$$\Delta n(\boldsymbol{r}) = 4\mathrm{Re}\sum_{n=1}^{N/2}\sum_{m\neq n}\psi_n^*(\boldsymbol{r})\psi_m(\boldsymbol{r})\frac{\langle\psi_m|\Delta V_{SCF}|\psi_n\rangle}{\epsilon_n-\epsilon_m}. \tag{1.16}$$

It is easy to see that when the index m runs over the valence band, we have two contributions that always cancel each other. Since the projector over the conduction band P_c commutes with the Hamiltonian, and $P_c\,|\Delta\psi_n\rangle = |\Delta\psi_n\rangle$, projecting both sides of Eq. (1.15) we obtain:

$$(H_{SCF}-\epsilon_n)\,|\Delta\psi_n\rangle = -P_c\Delta V_{SCF}\,|\psi_n\rangle \tag{1.17}$$

Expressions of this kind are called *Sternheimer equations*. The operator $(H_{SCF}-\epsilon_n)$ cannot be directly inverted, as it becomes singular when evaluated at $|\psi_n\rangle$. To avoid this singularity, the projection is usually introduced in equations as $P_c(H_{SCF}-\epsilon_n)P_c$. The main disadvantage in Eq. (1.16) is that we need to solve the unperturbed Hamiltonian for an arbitrary amount of unperturbed empty orbitals to evaluate the perturbed density. On the other hand, the Sternheimer equation can be directly solved on occupied states: $P_c = 1 - P_v = 1 - \sum_v |\psi_v\rangle\langle\psi_v|$, where v stands for valence states.

1.1.5 2n+1 theorem

Hellmann–Feynman theorem and the expression of second-order derivatives of the energy in Eq. (1.12) are particular cases of the well-known $2n+1$ theorem, which states that the $(2n+1)$-th energy derivative can be evaluated from the eigenfunction derivatives of order n. The forces, being first energy derivatives, are calculated from the eigenfunctions of the electronic Hamiltonian. Moving the atoms in the direction of the forces, we can relax the system until their magnitudes fall below a chosen threshold. We then study small oscillations about this equilibrium position. Within the harmonic approximation, the potential is expanded to second order. There are two main procedures to obtain this result: the first is to perform several plane wave calculations for configurations slightly shifted with respect to the equilibrium position. The class of methods implementing this procedure is generally called finite-difference methods. The second

method takes advantage of the $2n+1$ theorem: we calculate the first derivatives of the eigenfunctions through the Sternheimer equation to access the second (and possibly third) derivative of the energy. This method is called Density Functional Perturbation Theory (DFPT) and is implemented in the Quantum ESPRESSO package [8, 9], in the PHonon routine.

1.1.6 Periodic systems

A periodic system is defined by a unit cell spanned by $(\boldsymbol{a}_1,\ \boldsymbol{a}_2,\ \boldsymbol{a}_3)$ which forms a lattice $\boldsymbol{R} = l_1\boldsymbol{a}_1 + l_2\boldsymbol{a}_2 + l_3\boldsymbol{a}_3$, where l_1, l_2, and l_3 are integers. Inside the unit cell, atomic positions are described by the basis $\boldsymbol{\tau}$. Consequently, in a basis of simultaneous eigenstates of the Hamiltonian and translation operator, electronic states are classified by the wave-vector $\boldsymbol{k}$ and the band index n. $\boldsymbol{k}$ belongs to the reciprocal lattice $(\boldsymbol{b}_1,\ \boldsymbol{b}_2,\ \boldsymbol{b}_3)$, defined by the orthogonality condition $\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}$. It labels a solution of the translation operator eigenvalue equation; n is an integer that labels different solutions of the stationary Schrödinger equation. These states can be written in the so-called Bloch form as $\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\phi_{n\boldsymbol{k}}(\boldsymbol{r})$, where ϕ is a lattice-periodic function. The vector space of atomic displacements is indexed by the cell index $\boldsymbol{R}$, the atom coordinate τ and the cartesian direction of the displacement α. The translation invariance imposes that the displacement u, written in the basis of the translation operator, indexed by the wavevector $\boldsymbol{q}$, follows the Bloch condition

$$u_{\boldsymbol{\tau}\alpha\boldsymbol{R}}(\boldsymbol{q}) = e^{i\boldsymbol{q}\cdot\boldsymbol{R}} u_{\boldsymbol{\tau}\alpha\boldsymbol{0}}(\boldsymbol{q}), \tag{1.18}$$

where the displacement in the right part of the equation refers to the unit cell at the origin. The wrapped notation $\eta = (\boldsymbol{\tau}\alpha)$ is used throughout the entire dissertation. These vectors are solutions of the ionic Hamiltonian in the harmonic approximation, where Eq. (1.9) is expanded up to second-order as:

$$H = T_N + E_0 + \frac{1}{2}\sum_{\boldsymbol{R}\eta\boldsymbol{R'}\eta'} \left.\frac{\partial^2 E}{\partial u_\eta(\boldsymbol{R})\partial u_{\eta'}(\boldsymbol{R'})}\right|_{u=0} u_\eta(\boldsymbol{R}) u_{\eta'}(\boldsymbol{R'}). \tag{1.19}$$

The translation invariance is reflected in a strong constraint on the energy derivatives, which must respect

$$\Phi_{\eta\eta'}(\boldsymbol{R},\boldsymbol{R'}) = \left.\frac{\partial^2 E}{\partial u_\eta(\boldsymbol{R})\partial u_{\eta'}(\boldsymbol{R'})}\right|_{u=0} = \Phi_{\eta\eta'}(\boldsymbol{R}+\boldsymbol{T},\boldsymbol{R'}+\boldsymbol{T}) \tag{1.20}$$

If we take $\boldsymbol{T} = -\boldsymbol{R'}$, the IFC become a single-valued matrix in the variable $\boldsymbol{R}-\boldsymbol{R'}$, by defining $\Phi(\boldsymbol{R}-\boldsymbol{R'}) := \Phi(\boldsymbol{R}-\boldsymbol{R'},\boldsymbol{0})$. The Hamiltonian can be decoupled in reciprocal space, by defining a monochromatic displacement with wavevector $\boldsymbol{q}$ as in Eq. (1.18) and diagonalizing the Fourier transformed IFCs:

$$\Phi_{\eta\eta'}(\boldsymbol{q}) = \sum_{\boldsymbol{R}} e^{-i\boldsymbol{q}\cdot\boldsymbol{R}} \Phi_{\eta\eta'}(\boldsymbol{R}). \tag{1.21}$$

Each normal mode is labeled by a specific wavevector $\boldsymbol{q}$ and the branch index j, which counts the number of modes for a specific $\boldsymbol{q}$. The advantage of working with DFPT is that the perturbation and the response share the respective wavelength, thanks to the linearity of the Eqs. (1.13)–(1.15). To solve the phonon equation for a specific wavevector $\boldsymbol{q}$, one can calculate the derivative of the potential as follows:

$$\frac{\partial V_{ext}(\boldsymbol{r})}{\partial u_\eta(\boldsymbol{q})} = -\sum_{\boldsymbol{R}} \frac{\partial V_{ext}(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\eta)}{\partial \boldsymbol{r}} e^{i\boldsymbol{q}\cdot\boldsymbol{R}}. \tag{1.22}$$

The resulting variation is in the Bloch form: $\Delta V^{\boldsymbol{q}}_{ext}(\boldsymbol{r}+\boldsymbol{R}) = e^{i\boldsymbol{q}\cdot\boldsymbol{R}}\Delta V^{\boldsymbol{q}}_{ext}(\boldsymbol{r})$. The perturbation carries a momentum $\boldsymbol{q}$, since it shares the orthogonality property of the electronic Bloch functions:

$$\langle \psi_{n'\boldsymbol{k'}} | \Delta V^{\boldsymbol{q}}_{ext} | \psi_{n\boldsymbol{k}} \rangle = \sum_{\boldsymbol{R}} e^{i(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k'})\cdot\boldsymbol{R}} f(\boldsymbol{k},\boldsymbol{k'},n,n',\boldsymbol{q}) \tag{1.23}$$

$$= N\delta_{\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k'}} f(\boldsymbol{k},n,n',\boldsymbol{q}), \tag{1.24}$$

where N represents the number of cells and f denotes the integral of the periodic parts of the three Bloch functions. As a result, the perturbed eigenfunctions have defined momentum $\boldsymbol{k}+\boldsymbol{q}$:

$$|\Delta\psi_{n\boldsymbol{k}}\rangle = \sum_{n'} |\psi_{n'\boldsymbol{k}+\boldsymbol{q}}\rangle \frac{\langle \psi_{n'\boldsymbol{k}+\boldsymbol{q}} | \Delta V^{\boldsymbol{q}}_{ext} | \psi_{n\boldsymbol{k}}\rangle}{\epsilon_{n\boldsymbol{k}} - \epsilon_{n'\boldsymbol{k}+\boldsymbol{q}}} \tag{1.25}$$

and the density in Eq. (1.13) is also expressed in Bloch form as

$$\Delta n_{\boldsymbol{q}}(\boldsymbol{r}) = e^{i\boldsymbol{q}\cdot\boldsymbol{r}} \Delta\tilde{n}_{\boldsymbol{q}}(\boldsymbol{r}), \tag{1.26}$$

where $\tilde{n}$ represents a lattice periodic function. Similarly, the entire SCF potential in Eq. (1.14) takes on the same Bloch form. The Sternheimer equation in Eq. (1.17) is solved by expanding in plane waves of reciprocal lattice vectors, just as in the unperturbed case.

1.1.7 Electric field response

In metals, valence electrons are free to move. Under the action of an external electric field $\boldsymbol{\mathcal{E}}$, the electrons experience the potential $V = -e\boldsymbol{\mathcal{E}}\cdot\mathbf{r}$ and flow until the induced charge distribution screens the applied field. After a short relaxation time, the static electric field inside the bulk vanishes. Therefore, in a metal, macroscopic polarization is essentially a surface effect: charge accumulates at the boundaries of the sample rather than producing a well-defined static bulk polarization. This contrasts with insulating crystals, for which a homogeneous electric field is able to produce a bulk polarization $\boldsymbol{\mathcal{P}}$, which only partially screens the electric field. It can be defined through the Berry phase of the occupied Bloch wavefunctions, as in the modern theory of polarization [10]. It has two distinct contributions: the first originates from the redistribution of

electron density and is governed by the electronic susceptibility $\chi^{\infty}_{\alpha\beta} = \partial\mathcal{P}_\alpha/\partial\mathcal{E}_\beta$, which is calculated with the ions held fixed. The second contribution arises from the ionic displacements that induce a macroscopic polarization. Some insulators exhibit a polar behavior, where the displacement of an ion generates an electric dipole within the unit cell, developing a first order macroscopic polarization. Furthermore, dipoles generated by longitudinal displacements (LO phonons) are unable to cancel out in the long-wavelength limit and give rise to an additional contribution to the polarization, governed by the response $Z^{*\alpha\beta}_{\boldsymbol{\tau}} = \partial\mathcal{P}_\alpha/\partial u_{\boldsymbol{\tau}\beta}(\mathbf{0})$, calculated at vanishing external electric fields. These responses are also known as *Born effective charges*. We first analyse the electronic contribution to the polarization:

$$\boldsymbol{\mathcal{P}} = -\frac{1}{V}\int d^3r\, \Delta n(\boldsymbol{r})\boldsymbol{r}, \tag{1.27}$$

where the variation of the density, from now on, is intended to come from the variation of the electric field $\boldsymbol{\mathcal{E}}$. This expression becomes ill-defined in periodic systems, as the choice of the origin can affect the polarization. However, in the linear regime, we can accurately define the position operator $\boldsymbol{r}$ in terms of lattice-periodic operators. This can be done using the commutation relation:

$$\langle\psi_m|\boldsymbol{r}|\psi_n\rangle = \frac{\langle\psi_m|[H_{SCF},\boldsymbol{r}]|\psi_n\rangle}{\epsilon_m-\epsilon_n} = -\frac{\hbar^2}{m}\frac{\langle\psi_m|\partial/\partial\boldsymbol{r}|\psi_n\rangle}{\epsilon_m-\epsilon_n}, \tag{1.28}$$

where ϵ_m and ϵ_n are the energies of the respective states. Since we are only interested in off-diagonal elements to calculate linear responses, the denominator is always nonzero. Eq. (1.27), written in KS orbitals through Eq. (1.13), becomes:

$$\mathcal{P}_\alpha = -\frac{4}{V}\sum_n \langle\psi_n|r_\alpha|\Delta\psi_n\rangle \tag{1.29}$$

$$= -\frac{4}{V}\sum_n \left[\sum_{m\neq n}\frac{\langle\psi_m|[H_{SCF},r_\alpha]|\psi_n\rangle}{\epsilon_m-\epsilon_n}\langle\psi_m|\right]|\Delta\psi_n\rangle \tag{1.30}$$

$$= -\frac{4}{V}\sum_n \langle\tilde{\psi}^\alpha_n|\Delta\psi_n\rangle \tag{1.31}$$

The quantity inside the square brackets, denoted as $\langle\tilde{\psi}^\alpha_n|$, takes the form of a linear response of $\langle\psi_n|$ under the perturbing potential $[H_{SCF},r_\alpha]$. Therefore, it can be expressed in the Sternheimer form as shown in Eq. (1.17).

$$(H_{SCF}-\epsilon_n)\,|\tilde{\psi}^\alpha_n\rangle = -P_c[H_{SCF},r_\alpha]\,|\psi_n\rangle\,, \tag{1.32}$$

The perturbing potential for the $|\Delta\psi_n\rangle$ state consists of the electric contribution $V = -e\boldsymbol{\mathcal{E}}\cdot\boldsymbol{r}$ and the Hartree and exchange-correlation contribution ΔV_{Hxc}, which depends on the perturbed density. The corresponding linear equation is then given by:

$$(H_{SCF}-\epsilon_n)\,|\Delta\psi_n\rangle = -\sum_\alpha \mathcal{E}_\alpha\,|\tilde{\psi}^\alpha_n\rangle - \Delta V_{Hxc}\,|\psi_n\rangle\,. \tag{1.33}$$

These equations must be solved self-consistently by updating the screened field $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 - 4\pi\boldsymbol{\mathcal{P}}$. At each iteration, the polarization in Eq. (1.31) is calculated, and the screened field is used to find the perturbed density in Eq. (1.13) and wavefunctions in Eq. (1.33), until self-consistency is achieved. The electronic susceptibility $\chi^{\infty}_{\alpha\beta}$ can then be obtained directly as:

$$\chi^{\infty}_{\alpha\beta} = \frac{\partial \mathcal{P}_\alpha}{\partial \mathcal{E}_\beta} = -\frac{4}{V}\sum_n \langle \tilde{\psi}^{\alpha}_n | \frac{\partial \psi_n}{\partial \mathcal{E}_\beta} \rangle \,. \tag{1.34}$$

Born effective charges have both electronic and ionic contributions. The ionic contribution is straightforward, as the polarization created after an ion displacement is $\boldsymbol{\mathcal{P}} = Z_{\boldsymbol{\tau}}\boldsymbol{u}_{\boldsymbol{\tau}}$ and the derivative is simply the charge of the ion $Z_{\boldsymbol{\tau}}$. The electronic contribution is trivial once the linear responses of the wavefunctions to the density $\partial\psi_n/\partial\boldsymbol{u}_{\boldsymbol{\tau}}$ are known. In this case, the electronic polarization in Eq. (1.27) is calculated with respect to the displacement $u_{\boldsymbol{\tau}\beta}(\mathbf{0})$. By summing up the two terms, we have:

$$Z^{*\alpha\beta}_{\boldsymbol{\tau}} = \frac{\partial \mathcal{P}^{tot}_\alpha}{u_{\boldsymbol{\tau}\beta}} = Z_{\boldsymbol{\tau}} + \frac{4}{V}\sum_n \langle \tilde{\psi}^{\alpha}_n | \frac{\partial \psi_n}{\partial u_{\boldsymbol{\tau}\beta}} \rangle \,. \tag{1.35}$$

These two responses are the only ingredients needed to determine the correct frequency of LO phonons in the limit $\boldsymbol{q} \to 0$ in polar insulators. The restoring force acting on ions, once the phonon frequency is known, is of the form $\boldsymbol{F} = -M\omega^2\boldsymbol{u}$. LO phonons are capable of generating equally oriented dipoles, which give rise to an electric field and an additional electrostatic term $Z^*\boldsymbol{\mathcal{E}}$ to the force. The third Maxwell's equation $\boldsymbol{q} \times \boldsymbol{\mathcal{E}} = 0$, ensures that this field is absent for transverse modes, where the only valid solution is $\boldsymbol{\mathcal{E}} = 0$. In the absence of free charges, the divergence of the polarization field $\boldsymbol{\mathcal{D}} = \boldsymbol{\mathcal{E}} + 4\pi\boldsymbol{\mathcal{P}}$ is zero. In highly symmetric systems, $\boldsymbol{q} \cdot \boldsymbol{\mathcal{D}} = 0$ gives $\boldsymbol{\mathcal{D}} = 0$, although this result can be generalized to all crystals [11]. Expanding the polarization

$$\boldsymbol{\mathcal{P}} = \sum_\tau \frac{\partial \boldsymbol{\mathcal{P}}}{\partial \boldsymbol{u}_\tau}\boldsymbol{u}_{\boldsymbol{\tau}} + \frac{\partial \boldsymbol{\mathcal{P}}}{\partial \boldsymbol{\mathcal{E}}}\boldsymbol{\mathcal{E}} = \sum_\tau Z^*_\tau \boldsymbol{u}_\tau + \chi^{\infty}\boldsymbol{\mathcal{E}} \tag{1.36}$$

and defining the electronic contribution of the dielectric function as $\epsilon^{\infty} = 1 + 4\pi\chi^{\infty}$ we can determine the electric field as:

$$\boldsymbol{\mathcal{E}} = -\epsilon^{-1}_{\infty}\sum_\tau Z^*_\tau \boldsymbol{u}_\tau \tag{1.37}$$

which is responsible for the increased stiffness in longitudinal oscillations. This is why LO phonons have higher frequency in the long-wavelength limit. IFCs also are available from the reciprocal version of Eq. (1.12) once $\partial\psi_n/\partial\boldsymbol{u}_\tau(\boldsymbol{q})$ are known. After IFCs are computed for a coarse grid of $\boldsymbol{q}$ points, one can verify the convergence by examining the IFCs in real space using Fast Fourier Transform (FFT). In the real space, the IFCs on a grid of N $\boldsymbol{q}$ points correspond to the force constants in a supercell composed of N unit cells. Typically, the force constants decay exponentially. Assuming that IFCs in real space are exactly zero outside the supercell, one can obtain by inverse FFT a

smooth phonon dispersion by performing an inverse FFT, a procedure known as *Fourier interpolation*. In polar insulators near the Γ point, as mentioned earlier, the dipoles give rise to a long-range polynomial interaction, which introduces an additional non-analytic term in the reciprocal IFCs. The divergence can be resolved by calculating the non-analytical part of the force constants from ϵ^{∞} and Z^{*} [12, 13]. This term is subtracted from the calculated reciprocal IFC before performing FFT, and then re-added later [14].

1.1.8 Raman spectroscopy

Raman spectroscopy probes lattice vibrations through the inelastic scattering of light. In a crystal, the incident electromagnetic field induces a macroscopic polarization whose magnitude depends on the electronic susceptibility χ, or equivalently on the dielectric tensor ϵ. Atomic vibrations modulate this susceptibility, so that phonons can exchange energy with the optical field. The interaction between the electric field and existing atomic vibrations can be resonant with a frequency of a new phonon mode. These frequency shifts happen in the infrared range, where the phonon spectrum lies. At the microscopic level, the incoming photon is scattered inelastically by the phonon, the energy difference is then adsorbed by creating another phonon [15]. The following energy and momentum conservation laws must be satisfied:

$$\omega_{out} = \omega_{in} \pm \omega_{\boldsymbol{q}j} \tag{1.38}$$

$$\boldsymbol{k}_{out} = \boldsymbol{k}_{in} \pm \boldsymbol{q}. \tag{1.39}$$

The second equation fixes a strict limit on the magnitude of the phonon wavevector $\boldsymbol{q}$: the maximum momentum transfer is at the angle π between incoming and outgoing photons, where $\Delta k = 2k = 2\omega/c$. For visible or near-infrared light, the photon wavevector is small compared to the size of the Brillouin zone: the scattering is concentrated around the Γ point.

The interaction between the electric field and the electronic cloud is complex. In principle, the photon is capable of exciting the crystal into a virtual state, which is described in terms of the crystal's excited states. However, a simpler approach, known as the *Placzek approximation* [16], assumes that the electrons are in their respective ground state in both the initial and final states. The ions do not directly respond to the external electric field due to its rapid oscillation. The coupling between phonons and light occurs in two steps: first, the electrons couple with light, and their response is characterized by the electronic susceptibility χ^{∞}; then, the change induced in the electronic density couples with nuclear motion. In this approach, the susceptibility is calculated for each ionic configuration while keeping the ions fixed in their positions. The Raman intensity depends then on the Raman tensor

$$R^{j}_{\alpha\beta} = \frac{\partial^3 E}{\partial \mathcal{E}_{\alpha} \partial \mathcal{E}_{\beta} \partial u_{\mathbf{0}j}} = \frac{\partial \chi_{\alpha\beta}}{\partial u_{\mathbf{0}j}}, \tag{1.40}$$

by the polarizations of the light $\boldsymbol{e}$, as

$$I_j \propto |\boldsymbol{e_\alpha} \cdot R^j_{\alpha\beta} \cdot \boldsymbol{e_\beta}|^2. \tag{1.41}$$

From a phenomenological point of view, the crystal responds to an oscillating field $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 \cos(\omega t)$ by developing a polarization

$$\boldsymbol{\mathcal{P}} = \chi \boldsymbol{\mathcal{E}} = \left(\chi_0 + \frac{\partial \chi}{\partial \boldsymbol{u}} \cos(\omega_{\boldsymbol{q}} t)\right) \boldsymbol{\mathcal{E}}_0 \cos(\omega t) \tag{1.42}$$

$$= \chi_0 \boldsymbol{\mathcal{E}}_0 \cos(\omega t) + \frac{1}{2} \frac{\partial \chi}{\partial \boldsymbol{u}} \boldsymbol{\mathcal{E}}_0 [\cos(\omega + \omega_{\boldsymbol{q}})t + \cos(\omega - \omega_{\boldsymbol{q}})t]. \tag{1.43}$$

The second term in the equation is responsible for the frequency shift of the incident light. The Raman tensor is evaluated directly using $\psi^{\alpha\beta}$ [17] as follows:

$$\frac{\partial^3 E}{\partial \mathcal{E}_\alpha \partial \mathcal{E}_\beta \partial u_j} = \int d^3 r \, \frac{\partial V_{ext}(\boldsymbol{r})}{\partial u_j} \frac{\partial^2 n}{\partial \mathcal{E}_\alpha \mathcal{E}_\beta}(\boldsymbol{r}), \tag{1.44}$$

which is obtained by differentiating the force in Eq. (1.11) with respect to the two fields. The external potential is the ionic one, hence it does not depend explicitly on the electric fields. This approach needs six second order wavefunctions through a Sternheimer equation. The entire problem can be reformulated in terms of the density matrix [18], equal to the projection operator onto the valence band.

1.1.9 Anharmonicity and phonon-phonon interactions

In a crystal, under the adiabatic approximation, we can study the ion displacements by solving the ionic Hamiltonian, with a potential given by the electronic energy functional in the ground state of a certain ionic configuration. At the equilibrium, the forces acting on the ions are zero, the potential can be expanded in a Taylor series around the equilibrium position. Phonons are computed in the harmonic approximation, where the potential is truncated at the second order. The phonons, being normal modes of the harmonic Hamiltonian, are stationary states of the system, and do not interact with each other. The third order can be treated perturbatively from the harmonic solution, it is responsible for scattering processes involving three phonons, such as the decay of one phonon into two, or the combination of two phonons into one. In second quantization, the harmonic Hamiltonian is diagonal on the basis of phonon creation and annihilation operators $a^\dagger_{\boldsymbol{q}j}$ and $a_{\boldsymbol{q}j}$:

$$H = \sum_{\boldsymbol{q}j} \omega_{\boldsymbol{q}j} (a^\dagger_{\boldsymbol{q}j} a_{\boldsymbol{q}j} + \frac{1}{2}). \tag{1.45}$$

The phonon displacement operator can be written as

$$u_{\boldsymbol{q}j} = -i \frac{1}{\sqrt{2\omega_{\boldsymbol{q}j}}} (a^\dagger_{\boldsymbol{q}j} - a_{-\boldsymbol{q}j}). \tag{1.46}$$

In this basis, the anharmonic term is then

$$V_3 = \frac{1}{3!} \sum_{\boldsymbol{q}j,\boldsymbol{q'}j',\boldsymbol{q''}j''} \delta_{\boldsymbol{G},\boldsymbol{q}+\boldsymbol{q'}+\boldsymbol{q''}} \frac{\partial^3 E}{\partial u_{\boldsymbol{q}j} u_{\boldsymbol{q'}j'} u_{\boldsymbol{q''}j''}} \tag{1.47}$$

$$\times (a^\dagger_{\boldsymbol{q}j} - a_{-\boldsymbol{q}j})(a^\dagger_{\boldsymbol{q'}j'} - a_{-\boldsymbol{q'}j'})(a^\dagger_{\boldsymbol{q''}j''} - a_{-\boldsymbol{q''}j''}), \tag{1.48}$$

where $\boldsymbol{G}$ is the reciprocal lattice vector that ensures momentum conservation. For Raman scattering, one of the phonons must be at Γ point: the scattering which adds a $\boldsymbol{G}$, called Umklapp process, is allowed only for phonons with wavevector at the Brillouin-zone boundary, hence it is highly suppressed. The process which occupies the majority of the scattering space is then $\boldsymbol{0} \to \boldsymbol{q} + (-\boldsymbol{q})$ and $\boldsymbol{q} + (-\boldsymbol{q}) \to \boldsymbol{0}$. The scattering rate, linked with the phonon linewidth, can be calculated using Fermi's golden rule:

$$P_i^f = 2\pi \left|\langle i|V_3|f\rangle\right|^2 \delta(E_f - E_i). \tag{1.49}$$

The delta gives the energy conservation, while the matrix element is calculated by evaluating it between initial and final states. For a crystal at thermal equilibrium, the occupation of the phonon states is given by the Bose-Einstein distribution. The electric field creates/annihilates a phonon in a branch j at Γ, giving an initial occupation state written as $|n_{\boldsymbol{0}j} \pm 1, n_{\boldsymbol{q}j'}, n_{-\boldsymbol{q}j'}\rangle$, where n is the equilibrium occupation. The final state is instead $|n_{\boldsymbol{0}j}, n_{\boldsymbol{q}j'} \pm 1, n_{-\boldsymbol{q}j'} \pm 1\rangle$. The full scattering rate is then given by the sum over all the possible final states, as:

$$P_j = 2\pi \sum_{\boldsymbol{q}j'j''} |\tilde{V}_3(\boldsymbol{0}j, \boldsymbol{q}j', -\boldsymbol{q}j'')|^2 [2(n_{\boldsymbol{q}j''} - n_{\boldsymbol{q}j'})\delta(\omega_{\boldsymbol{0}j} + \omega_{\boldsymbol{q}j'} - \omega_{-\boldsymbol{q}j''}) \tag{1.50}$$

$$-(n_{\boldsymbol{q}j'} + n_{\boldsymbol{q}j''} + 1)\delta(\omega_{\boldsymbol{0}j} - \omega_{\boldsymbol{q}j'} - \omega_{-\boldsymbol{q}j''})]. \tag{1.51}$$

The third order derivative $\tilde{V}_3$ is obtained through the $2n+1$ theorem, as in [19].

1.2 Modeling of the defective potential

1.2.1 Defect force constants

A defect perturbs both the electronic and phononic Hamiltonians. For electrons, the perturbation is the local change in the Kohn–Sham potential around the defective site [20, 21]. For phonons, it comprises changes in bond stiffness and local structural relaxation, which are encoded in the IFCs [22]. The starting point of non-perturbative methods is a perturbing potential that acts on the phonon Hamiltonian. The unperturbed Hamiltonian of the pristine periodic system is described by the IFCs, denoted by Φ_0. They can be obtained from DFPT in reciprocal space on a desired $\boldsymbol{q}$-grid, giving fast-decaying real-space IFCs on the corresponding supercell. The presence of a defect breaks the periodicity of the system. In a supercell large enough to make interactions between replicas of the defect negligible, the defective force constants Φ are expected to change only near the defect, so that the perturbation $\Delta\Phi$ is localized in real space.

This locality is essential for a useful perturbing potential. Φ is obtained with the same DFPT procedure, but directly for the defect-containing supercell. Sampling only the Γ point is sufficient, since the bands at other $\boldsymbol{q}$-points are folded into the center of the supercell Brillouin zone.

We now discuss the properties of the IFCs and their relation with the perturbing potential. IFCs are defined as the second derivatives of the total energy, as in Eq. (1.20). They encompass the bond stiffness and the strain fields around the defect. The pure system possesses translational invariance, expressed as $\Phi_0(\boldsymbol{R}, \boldsymbol{R'}) = \Phi_0(\boldsymbol{R} - \boldsymbol{T}, \boldsymbol{R'} - \boldsymbol{T})$. After a coordinate change, the equation enforcing this symmetry would change accordingly. Specifically, in the space $\boldsymbol{R}_+ = \boldsymbol{R} + \boldsymbol{R'}$ and $\boldsymbol{R}_- = \boldsymbol{R} - \boldsymbol{R'}$, the translation symmetry imposes that the potential is constant in $\boldsymbol{R}_+$. Intuitively, the same condition in the reciprocal would imply that $\boldsymbol{q} + \boldsymbol{q'} = \boldsymbol{0}$: the potential is diagonal in the $(\boldsymbol{q}, -\boldsymbol{q'})$ and the force constants matrix can be diagonalized for blocks $\Phi_\eta^{\eta'}(\boldsymbol{q}, -\boldsymbol{q})$ of specific momentum $\boldsymbol{q}$. It is common to adopt the mass-weighted representation, in which the effect of mass inertia that comes from the kinetic energy term is adsorbed into the force constants. This new operator is called the dynamical matrix and is defined as:

$$D = M^{-1/2}\Phi M^{-1/2}, \tag{1.52}$$

where M is a diagonal matrix containing the masses of the species at each site. The displacements now are $\tilde{u} = M^{1/2}u$, and the Newton's equation reads

$$M\ddot{u} = -\Phi u \Rightarrow \ddot{\tilde{u}} = -D\tilde{u}. \tag{1.53}$$

The kinetic term in the Hamiltonian is now invariant and the information about the system is contained in the dynamical matrix:

$$\frac{1}{2}\dot{u}^\dagger M\dot{u} + \frac{1}{2}u^\dagger \Phi u \rightarrow \frac{1}{2}\dot{\tilde{u}}^\dagger\dot{\tilde{u}} + \frac{1}{2}\tilde{u}^\dagger D\tilde{u}. \tag{1.54}$$

The vector $e_{\boldsymbol{q}j}(\eta)$ is the j-th eigenvector of the D matrix, it gives the normal displacement on the cartesian basis of the unit cell. This new object includes the mass perturbation, previously separated in the kinetic energy term. The displacements in the other cells follow from the Bloch equations as in Eq. (1.18):

$$\langle \eta\boldsymbol{R}|\boldsymbol{q}j\rangle = e_{\boldsymbol{q}j}(\eta)e^{i\boldsymbol{q}\cdot\boldsymbol{R}}. \tag{1.55}$$

The presence of a single defect breaks translational invariance and adds an additional coordinate in the potential, the one in which the defect is placed. A more general translation invariance can be restated as

$$\Phi(\boldsymbol{R}, \boldsymbol{R'}, \boldsymbol{R''}) = \Phi(\boldsymbol{R} - \boldsymbol{T}, \boldsymbol{R'} - \boldsymbol{T}, \boldsymbol{R''} - \boldsymbol{T}). \tag{1.56}$$

The function stil does not acquire dependence on $\boldsymbol{R}_+ = \boldsymbol{R} + \boldsymbol{R'} + \boldsymbol{R''}$, since this axis is perpendicular to both $\boldsymbol{R_1} = \boldsymbol{R} - \boldsymbol{R'}$ and $\boldsymbol{R_2} = \boldsymbol{R'} - \boldsymbol{R''}$. These last two axes do not change under the translation operation, so the function does not. For this reason, the total momentum $\boldsymbol{q} + \boldsymbol{q'} + \boldsymbol{q_L} = \boldsymbol{0}$, where $\boldsymbol{q_L}$ is the lattice momentum. This scattering

potential allows a phonon with momentum $\boldsymbol{q}$ to scatter into a state with momentum $\boldsymbol{q'}$, giving the impulse $\boldsymbol{q'} - \boldsymbol{q}$ to the crystal itself. The potential can be seen as a function of the two phonon momenta in reciprocal space. The next section focuses on how to properly perform the Fourier transform for potentials with this shape. Here, we define the perturbing potential, which will be extensively used within the perturbation-theory framework. To do so, we seek a Hamiltonian H in a cleanly separable form $H = H_0 + V$, where H_0 is the Hamiltonian of the pure system and V is the perturbing potential. For our problem, the effective Hamiltonian we want to diagonalize would be a dynamical matrix, including all the force constants changes in the vicinity of the position of each defect placed randomly in a supercell. The perturbing potential must then be the difference between the dynamical matrices of the defective and the pure system, formally:

$$V = D - D_0. \tag{1.57}$$

For now, let us study a simpler case of one defect placed at the origin. The time Fourier transform of Newton's equation reads

$$\sum_{\eta' \boldsymbol{R'}} [M_\eta(\boldsymbol{R})\delta_{\eta\eta'}\delta_{\boldsymbol{R}\boldsymbol{R'}}\omega^2 - \Phi_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})]u_{\eta'}(\boldsymbol{R'}) = 0, \tag{1.58}$$

where M is the species mass and ω the reference frequency. This equation looks similar to an eigenvalue equation for the quantity Φ, apart from the presence of the mass term. It is instead a pure eigenvalue equation for the quantity

$$D_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) = M_\eta^{-1/2}(\boldsymbol{R})\Phi_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})M_{\eta'}^{-1/2}(\boldsymbol{R'}). \tag{1.59}$$

We can separate the mass perturbation into $M = M_0 - \epsilon M_0$ and $\Phi = \Phi_0 + \Delta\Phi$, so that

$$\sum_{\eta' \boldsymbol{R'}} [M_{0,\eta}(\boldsymbol{R'})\delta_{\eta\eta'}\delta_{\boldsymbol{R}\boldsymbol{R'}}\omega^2 - \Phi_{0,\eta\eta'}(\boldsymbol{R} - \boldsymbol{R'}) \tag{1.60}$$

$$- M_{0,\eta}^{1/2}(\boldsymbol{R})V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})M_{0,\eta'}^{1/2}(\boldsymbol{R'})]u_{\eta'}(\boldsymbol{R'}) = 0, \tag{1.61}$$

where

$$V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) = M_{0,\eta}^{-1/2}(\boldsymbol{R})\Delta\Phi_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})M_{0,\eta'}^{-1/2}(\boldsymbol{R'}) + \epsilon_\eta(\boldsymbol{R})\delta_{\eta\eta'}\delta_{\boldsymbol{R}\boldsymbol{R'}}\omega^2. \tag{1.62}$$

In general, the defect introduces three kinds of perturbations: the mass change is responsible for a local perturbation at the defective site. Changes in the IFCs are caused by a rearrangement of the electronic density around the defect, which modifies the electronic response to atomic displacements. Its effect is more extended in space, mainly because of the presence of electrostatic long-range interactions, discussed in Section (1.2.6). The last perturbation is the induced strain field, caused by a mismatch between host and defective covalent radii. It is well accounted by relaxing the host and defective systems to the equilibrium configuration. Strain fields can be able to break some of the symmetries of the host system, this issue is discussed in Section (3.1.1).

1.2.2 Centering of force constants

Force constants are intrinsically periodic: they are invariant under translation of the supercell's lattice vector $\boldsymbol{T}$. The same periodicity is expected to be present in reciprocal space; hence, they are exact only for $\boldsymbol{q}$-grids commensurate with the supercell's size. To increase the grid size indefinitely in real space, it is common to apply the centering procedure, also known as minimal image. It works under the assumption that force constants decay exponentially with the distance between the two atoms and consists in retaining the $(\boldsymbol{R}+\boldsymbol{T}, \boldsymbol{R'}+\boldsymbol{T'})$ couple which minimizes the norm of $\boldsymbol{\tau}+\boldsymbol{R}+\boldsymbol{T}-\boldsymbol{\tau'}-\boldsymbol{R'}-\boldsymbol{T'}$. Once these couples are known, it is assumed that all the other points in a real grid of arbitrary length are zero, if the decay is fast enough so that near the borders of this centered zone the force constants are almost zero. The choice of keeping couples near the origin ensures that the obtained function is as smooth as possible. This procedure is needed when we want to Fourier transform the force constants in a differently shaped grid and is called Fourier interpolation, because it is correct for the reciprocal counterpart of the supercell in which the force constants are calculated, while it interpolates with a minimal amount of sinusoidal functions the points in between. The (unit cell) translational invariance for a pristine cell reads $\Phi_0(\boldsymbol{R}, \boldsymbol{R'}) = \Phi_0(\boldsymbol{R}-\boldsymbol{R'}, \boldsymbol{0})$, so it is advisable to work with only the first coordinate. The distance to be minimized becomes $||\boldsymbol{\tau}+\boldsymbol{R}-\boldsymbol{R'}+\boldsymbol{T}-\boldsymbol{\tau}'||$. For a defective cell, the translation reads $\Phi(\boldsymbol{R}, \boldsymbol{R'}, \boldsymbol{T}) = \Phi(\boldsymbol{R}-\boldsymbol{T}, \boldsymbol{R'}-\boldsymbol{T}, \boldsymbol{0})$: even after putting the defect at the origin, two indices are still needed. The force constants difference is supposed to decay rapidly far from the defect, since the force experienced by a far atom due to the displacement of another far atom should reflect the pristine environment in which they're embedded. It should be then noted that only this difference possess this property: the force constants Φ alone only approach Φ_0 for a couple of atoms far from the defect. There are then multiple ways, depicted in Fig. (1.2) of centering these force constants. We define $\boldsymbol{r} := \boldsymbol{\tau}+\boldsymbol{R}$ and $\boldsymbol{r'} = \boldsymbol{\tau}'+\boldsymbol{R}'$ for clarity purposes.

a) $\boldsymbol{T}$ is chosen to minimize $||\boldsymbol{r}+\boldsymbol{T}-\boldsymbol{r'}||$, while $\boldsymbol{T'}$ minimizes $||\boldsymbol{r'}+\boldsymbol{T'}||$. This ensures that the maximum distance between the two atoms is the same as the Φ_0 centering, but makes the force constants no more symmetric under derivative exchange $(\boldsymbol{r}+\boldsymbol{T}) \leftrightarrow (\boldsymbol{r'}+\boldsymbol{T'})$.

b) All the coordinates are chosen to stay inside the (6-dimensional) Wigner-Seitz cell generated by $(\boldsymbol{r}, \boldsymbol{r'})$. This approach increases the maximum distance dramatically, leaving non-exponential long tails.

c) Minimizing the perimeter of the triangle $(\boldsymbol{r}, \boldsymbol{r'}, \boldsymbol{0})$. $\boldsymbol{T}$ and $\boldsymbol{T'}$ are chosen to minimize $||\boldsymbol{r}+\boldsymbol{T}||+||\boldsymbol{r'}+\boldsymbol{T'}||+||\boldsymbol{r}+\boldsymbol{T}-\boldsymbol{r'}-\boldsymbol{T'}||$. The maximum distance decreases with respect to b), but still is greater than the method used for the pure systems. This method is used to center third order force constants, since the three coordinates have the same priority in the minimization scheme.

d) The Wigner-Seitz cell is generated by $((\boldsymbol{r}-\boldsymbol{r'})/2, \boldsymbol{r}+\boldsymbol{r'})$: the maximum distance is the same as in case a), but the derivative exchange is retained. The norms

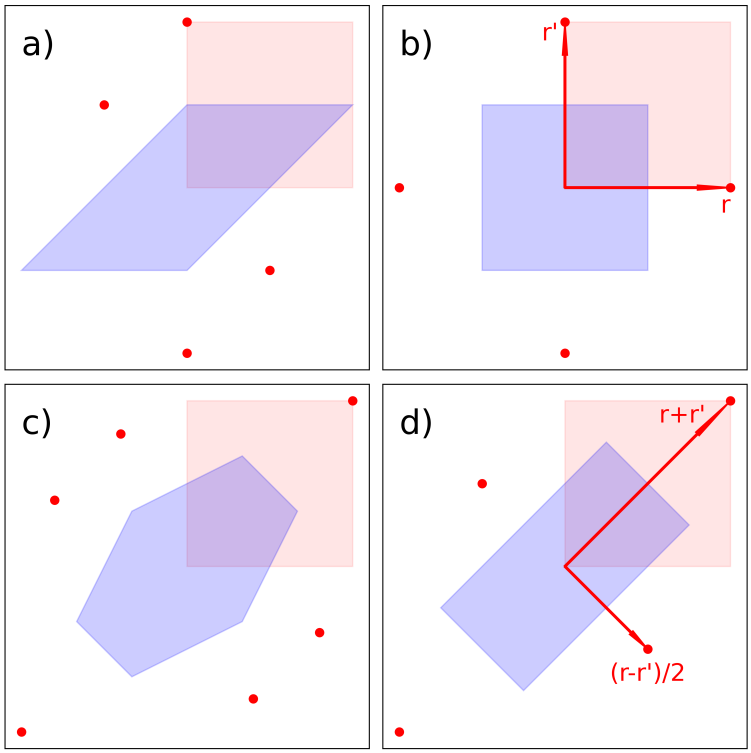


Figure 1.2: *Various centering methods for a one-dimensional system. The red area corresponds to all the couples $(\boldsymbol{r}, \boldsymbol{r'})$ inside the supercell, repeated periodically along the entire space. The blue area represent the centered cell. The perpendicular bisector between the red points and the origin generates the edge of the cell centered cell. In case b) and d) the red points are also lattice points of the (2-dimensional) Wigner-Seitz cell. Derivative exchange symmetry is equivalent to mirror symmetry along $\boldsymbol{r} = \boldsymbol{r'}$. The maximum distance $||\boldsymbol{r} - \boldsymbol{r'}||$ is respectively $\sqrt{2}/4$, $\sqrt{2}/2$, $\sqrt{3}/2$ and $\sqrt{2}/4$.*

$||\boldsymbol{r} - \boldsymbol{r'} + \boldsymbol{T} - \boldsymbol{T'}||$ and $||\boldsymbol{r} + \boldsymbol{r'} + \boldsymbol{T} + \boldsymbol{T'}||$ are minimized independently and the maximum distance in the $\boldsymbol{r} + \boldsymbol{r'}$ axis is the same as c).

We'll chose to implement the method d), because we want to make sure to have the same maximum distance $||\boldsymbol{r} - \boldsymbol{r'}||$ as in the pure system, to better compare force constants decay with distance. On the other hand, it's probably the hardest to implement correclty: the main difficulty to take into account is the asymmetry in the axes length. The method still gives a unique centered cell, excluding some unphysical solution of $\boldsymbol{T}_- = \boldsymbol{T} - \boldsymbol{T'}$ and $\boldsymbol{T}_+ = \boldsymbol{T} + \boldsymbol{T'}$. It's obvious that sums and differences of (supercell) lattice vectors are lattice vectors, but the contrary is not true. With this definition, $\boldsymbol{T} = (\boldsymbol{T}_+ + \boldsymbol{T}_-)/2$ and $\boldsymbol{T'} = (\boldsymbol{T}_+ - \boldsymbol{T}_-)/2$ can be a fraction of a lattice vector. We need to find more solutions and exlcude the unphysical ones. In other words, we need to relax the constraint of a unique $\boldsymbol{T}_+$ which minimizes $||\boldsymbol{r} + \boldsymbol{r'} + \boldsymbol{T}_+||$. If we just take all

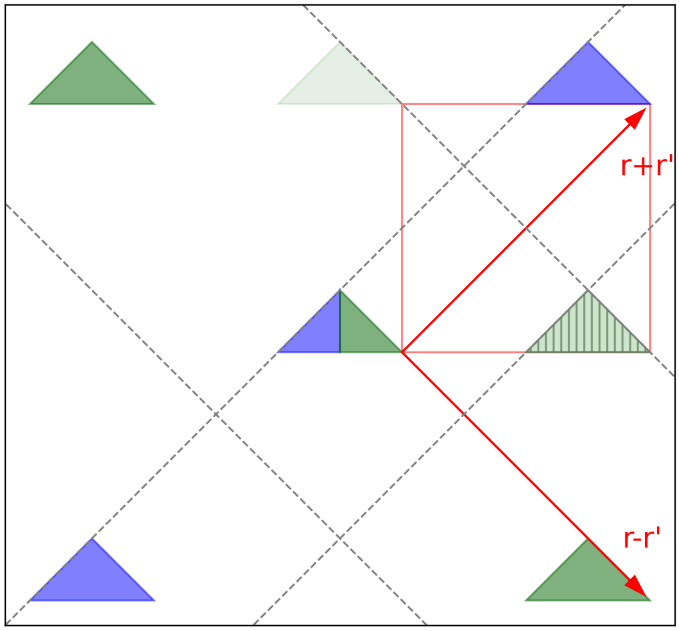


Figure 1.3: *Showcase of centering method d) for a one-dimensional system. The red square delimits the supercell's area, the hatched triangle is to be shifted according to the algorithm. This triangle is outside the centered Wigner-Seitz cell for the difference, but not for the sum. For the sum, the two classes of solutions are depicted in light green for* $\boldsymbol{T}_+ = 0$ *and dark green for* $\boldsymbol{T}_+ = -1$*. The only class of solutions for the difference are blue and correspond to* $\boldsymbol{T}_- = -1$*. They intersect for* $\boldsymbol{T} = -1$ *and* $\boldsymbol{T'} = 0$*, which is the only solution for which both* $\boldsymbol{T}$ *and* $\boldsymbol{T'}$ *are lattice vectors. In the figure, the centered triangle is dark green and blue.*

the vectors inside the Wigner-Seitz cell, it can be shown that these vectors are 2^d times more, where d is the number of dimensions. The reason comes from the fact that the Wigner-Seitz cell in the $\boldsymbol{r} + \boldsymbol{r'}$ axis is twice the size, so we get two solutions for each coordinate. This ambiguity in the definition can be exploited to always pick the only $\boldsymbol{T}_+$ which yields pure lattice translation vectors. In a crystalline coordinate system, $\boldsymbol{T}$ and $\boldsymbol{T'}$ have both integer or half-integer coordinates, because we're dividing by two in the definition. Since the coordinates of the 2^d solutions are consecutive numbers, we're sure to find the unique solution for which $\boldsymbol{T}$ and $\boldsymbol{T'}$ have both integer coordinates: in other words, the two vectors are really lattice vectors. A one-dimensional example of the algorithm for a small part of the supercell is depicted in Fig. (1.3). Due to the way we constructed the Wigner-Seitz, we would expect a longer tail for the $\boldsymbol{r}+\boldsymbol{r'}$ distances. The perfect tradeoff would be to have tails of equal lengths. This is reached with the coordinate change $((\boldsymbol{r}-\boldsymbol{r'})/\sqrt{2}, (\boldsymbol{r}+\boldsymbol{r'})/\sqrt{2})$, but in this cell the vector $\boldsymbol{T}_+ = (\boldsymbol{T}+\boldsymbol{T'})/\sqrt{2}$ is not integer or half-integer. The last option could be to use the difference as the longer

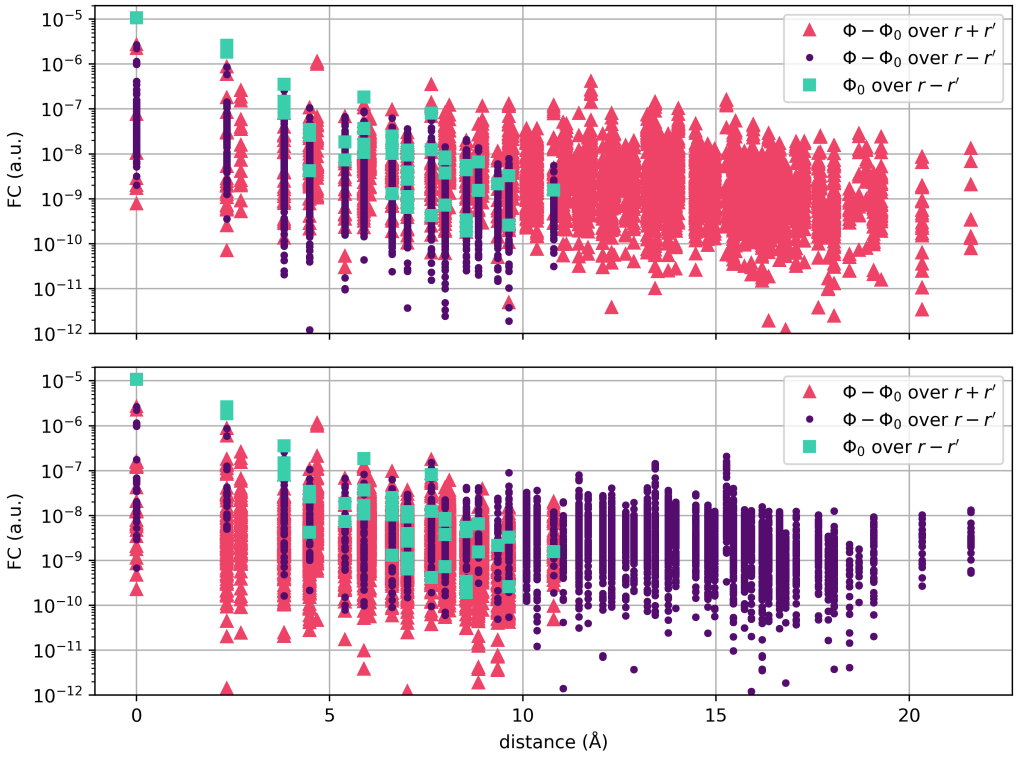


Figure 1.4: *Centered force constants for P-doped silicon in a* $4 \times 4 \times 4$ *supercell. In the upper figure, the longest axis is* $\boldsymbol{r} + \boldsymbol{r'}$*, while in the lower one is* $\boldsymbol{r} - \boldsymbol{r'}$*. The pure system's force constants (cyan squares) are also depicted for comparison.*

axis. The comparison between this case and d) are depicted in Fig. (1.4), for P-doped silicon in a $4 \times 4 \times 4$ supercell.

1.2.3 Configuration potential

One of the main reasons to construct the potential in reciprocal space is that a lattice translation becomes a phase factor. Starting from the lattice-translation property in Eq. (1.56), we apply the Fourier-transform definition. The specific choice of phase signs is discussed in Section (1.3.7).

$$V_{\boldsymbol{qq'}}(\boldsymbol{T}) = \frac{1}{N} \sum_{\boldsymbol{RR'}} e^{-i\boldsymbol{q}\cdot\boldsymbol{R}} V(\boldsymbol{R}, \boldsymbol{R'}, \boldsymbol{T}) e^{i\boldsymbol{q'}\cdot\boldsymbol{R'}} \tag{1.63}$$

$$= e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} \frac{1}{N} \sum_{\boldsymbol{RR'}} e^{-i\boldsymbol{q}\cdot(\boldsymbol{R}-\boldsymbol{T})} V(\boldsymbol{R}-\boldsymbol{T}, \boldsymbol{R'}-\boldsymbol{T}, \boldsymbol{0}) e^{i\boldsymbol{q'}\cdot(\boldsymbol{R'}-\boldsymbol{T})} \tag{1.64}$$

$$= e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} V_{\boldsymbol{qq'}}(\boldsymbol{0}). \tag{1.65}$$

For a certain configuration, denoted as α, the potential can be expressed as

$$V_{\alpha,\boldsymbol{qj}}^{\boldsymbol{q'}j'} = V_{\boldsymbol{qj}}^{\boldsymbol{q'}j'}(\boldsymbol{0}) \sum_{\boldsymbol{T}\in C_\alpha} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}}, \tag{1.66}$$

where C_α is the set of sites hosting a defect in the configuration α. Many materials have multiple equivalent sites for the same species. In terms of symmetry, there is a symmetry operation that transforms one site into the others. Eq. (1.66) can accommodate this possibility by summing over each subset C_α^i in the sublattice, with

$$V_{\alpha,\boldsymbol{qj}}^{\boldsymbol{q'}j'} = \sum_i V_{i,\boldsymbol{qj}}^{\boldsymbol{q'}j'}(\boldsymbol{\tau_i}) \sum_{\boldsymbol{T}\in C_\alpha^i} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}}. \tag{1.67}$$

The potential is not simply translated to the atomic coordinate of the equivalent site $\boldsymbol{\tau_i}$; it must also be rotated according to the symmetry. This explains why the potential depends explicitly on the equivalent site i. We can generalize even more, by adding non-equivalent sites and sampling C_α^i with a multinomial distribution, with the probabilities given by the relative concentrations. For this second case, we must perform an additional phonon calculation in the supercell, since the new potentials are not defined directly through symmetry.
To construct the potential at equivalent sites, we apply the symmetry in real space and then transform back to the reciprocal basis. The symmetry is applied to the Cartesian basis $(\beta, \boldsymbol{\tau})$, where the first index is the Cartesian direction and the second denotes the atomic position. Defining $(S, \boldsymbol{t})$ as the symmetry rotation plus the fractional translation that transforms the position $\boldsymbol{\tau}$ into an equivalent site $\bar{\boldsymbol{\tau}}$, the lattice vectors are defined as $\bar{\boldsymbol{R}} = S\boldsymbol{R}$. It is important to note that the fractional translation acts only on the atomic position, bringing $\boldsymbol{R} + \boldsymbol{\tau}$ into $\bar{\boldsymbol{R}} + \bar{\boldsymbol{\tau}}$. The force constants, being rank-two tensors, transform as

$$V_{\bar{\beta}}^{\bar{\beta}'}(\bar{\boldsymbol{\tau}}, \bar{\boldsymbol{\tau}}', \bar{\boldsymbol{R}}, \bar{\boldsymbol{R}}') = \sum_{\beta\beta'} S^{-1}_{\bar{\beta}\beta} V_{\beta}^{\beta'}(\boldsymbol{\tau}, \boldsymbol{\tau}', \boldsymbol{R}, \boldsymbol{R}') (S^{-1}_{\beta'\bar{\beta}'})^T. \tag{1.68}$$

The potential on the equivalent site therefore becomes a function only of the symmetry operation, once expressed in reciprocal space and in the phonon basis $(\beta, \boldsymbol{\tau}) \to j$.

1.2.4 Sum rule and detailed balance

Detailed balance requires the transition probabilities for $|\boldsymbol{q}j\rangle \to |\boldsymbol{q'}j'\rangle$ and $|\boldsymbol{q'}j'\rangle \to |\boldsymbol{q}j\rangle$ to be equal. For a time-independent perturbation, this follows from the Hermiticity of V: $\langle \boldsymbol{q}j | V | \boldsymbol{q'}j' \rangle = \langle \boldsymbol{q'}j' | V | \boldsymbol{q}j \rangle^*$. The required Hermiticity is inherited from derivative-exchange symmetry of the force constants. If the coordinates of both Φ and Φ_0 can be exchanged, then $V_{\eta'\eta}(\boldsymbol{R'}, \boldsymbol{R}) = V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})$. Explicitly,

$$\langle \boldsymbol{q'}j' | V | \boldsymbol{q}j \rangle^* = \sum_{\boldsymbol{R}\boldsymbol{R'}\eta\eta'} \langle \boldsymbol{q'}j' | \boldsymbol{R'}\eta' \rangle^* V_{\eta'\eta}(\boldsymbol{R'}, \boldsymbol{R}) \langle \boldsymbol{R}\eta | \boldsymbol{q}j \rangle^* \tag{1.69}$$

$$= \sum_{\boldsymbol{R}\boldsymbol{R'}\eta\eta'} \langle \boldsymbol{q}j | \boldsymbol{R}\eta \rangle V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) \langle \boldsymbol{R'}\eta' | \boldsymbol{q'}j' \rangle = \langle \boldsymbol{q}j | V | \boldsymbol{q'}j' \rangle . \tag{1.70}$$

There are two main mechanisms that can break the exchange symmetry: the centering procedure and the sum rule. From the various centering methods we should always choose the one that treats the indices $(\boldsymbol{R}, \eta)$ and $(\boldsymbol{R'}, \eta')$ symmetrically. The sum rule, instead, is applied to force the acoustic phonons to be zero at Γ and it is a consequence of the translational symmetry of the crystal. Indeed, the crystal should gain no energy after a bulk translation, encoded as a displacement of all the atoms at $\boldsymbol{q} = \boldsymbol{0}$. We should be careful when applying the sum rule, since it breaks the exchange symmetry of the force constants if not applied to both of the indices.

1.2.5 New ASR

There are various kinds of acoustic sum rules reported in the literature. The simplest consists of shifting a subset of the force constants so that a uniform translation costs no energy. Using translational symmetry, the oscillation energy for the unit cell at the origin is

$$\sum_{\boldsymbol{R}ij\alpha\beta} u\begin{pmatrix} i \\ \alpha \\ 0 \end{pmatrix} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} u\begin{pmatrix} j \\ \beta \\ \boldsymbol{R} \end{pmatrix}.$$

The force acting on the site (i, α) is then

$$F\begin{pmatrix} i \\ \alpha \\ 0 \end{pmatrix} = \sum_{\boldsymbol{R}j\beta} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} u\begin{pmatrix} j \\ \beta \\ \boldsymbol{R} \end{pmatrix}.$$

The crystal should gain no energy after a translation of size d of the entire bulk in the direction β'. The forces, being the derivative of the zero function, should also be zero. If we substitute the displacement $u_{j\beta\boldsymbol{R}} = d\delta_{\beta\beta'}$ we obtain

$$F\begin{pmatrix} i \\ \alpha \\ 0 \end{pmatrix} = d\sum_{\boldsymbol{R}j\beta} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} \delta_{\beta\beta'} = d\sum_{\boldsymbol{R}j} \Phi\begin{pmatrix} i & j \\ \alpha & \beta' \\ 0 & \boldsymbol{R} \end{pmatrix} = 0.$$

The condition must be true for each distance d, each direction of shift β and each atom (i, α) inside the unit cell. Hence:

$$\sum_{\boldsymbol{R}j} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} = 0 \qquad \forall\ i,\ \alpha,\ \beta.$$

Our force constants, indicated by $\tilde{\Phi}$, do not satisfy this property exactly. One simple solution is to subtract the residual from the largest element, conventionally the on-site term at $\boldsymbol{R} = 0$. Distributing a constant correction over all force constants would alter their exponential decay. We therefore define

$$\Phi\begin{pmatrix} i & i \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} := \tilde{\Phi}\begin{pmatrix} i & i \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} - \sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix}$$

leaving the rest as it is: $\Phi(\boldsymbol{R}) := \tilde{\Phi}(\boldsymbol{R}) \quad \forall\, \boldsymbol{R} \neq 0$. The condition is then satisfied:

$$\sum_{\boldsymbol{R}j} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} = \sum_{j\neq i \vee \boldsymbol{R}\neq 0} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} + \tilde{\Phi}\begin{pmatrix} i & i \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} - \sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} = 0. \tag{1.71}$$

Using these conventions, the derivative exchange does not hold anymore:

$$\Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} - \Phi\begin{pmatrix} j & i \\ \beta & \alpha \\ 0 & 0 \end{pmatrix} = -\sum_{\boldsymbol{R}j'} \tilde{\Phi}\begin{pmatrix} i & j' \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} + \sum_{\boldsymbol{R}j'} \tilde{\Phi}\begin{pmatrix} j & j' \\ \beta & \alpha \\ 0 & \boldsymbol{R} \end{pmatrix} \neq 0.$$

If preserving detailed balance is more important, we can symmetrize the subtracted expression by defining

$$\Phi\begin{pmatrix} i & i \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} := \tilde{\Phi}\begin{pmatrix} i & i \\ \alpha & \beta \\ 0 & 0 \end{pmatrix} - \frac{1}{2}\sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} - \frac{1}{2}\sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \beta & \alpha \\ 0 & \boldsymbol{R} \end{pmatrix}.$$

The sum rule is then approximately satisfied, while derivative-exchange symmetry is preserved by construction. The residual sum-rule violation is

$$\sum_{\boldsymbol{R}j} \Phi\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} = \sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} - \frac{1}{2}\sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \alpha & \beta \\ 0 & \boldsymbol{R} \end{pmatrix} - \frac{1}{2}\sum_{\boldsymbol{R}j} \tilde{\Phi}\begin{pmatrix} i & j \\ \beta & \alpha \\ 0 & \boldsymbol{R} \end{pmatrix} \tag{1.72}$$

The first two terms cancel partially; hence, this transformation guarantees at least half the correction of a standard sum rule. Generally, due to other internal symmetries, the sum rule is completely satisfied, as you can see in Fig. (1.5) Both requirements can also be respected by imposing a weaker sum rule. For example, one can require the energy to remain unchanged under a uniform translation:

$$E(\alpha) = \sum_{\boldsymbol{R}ij} \Phi\begin{pmatrix} i & j \\ \alpha & \alpha \\ 0 & \boldsymbol{R} \end{pmatrix} = 0 \qquad \forall\, \alpha, \tag{1.73}$$

but this gives less information on the shape of the potential, hence the acoustic phonons are farther from zero at Γ.

1.2.6 Long range interactions

A defect breaks translational invariance. Chemical changes, structural relaxation, and mass differences are short-ranged and can therefore be described by a localized perturbation extracted from a periodic supercell. Electrostatic effects, by contrast, may be long-ranged and converge slowly with supercell size. They can produce interpolation artefacts and spurious interactions between periodic images of the defect. The importance of these effects depends on both the material and the defect type. The following cases require particular care:

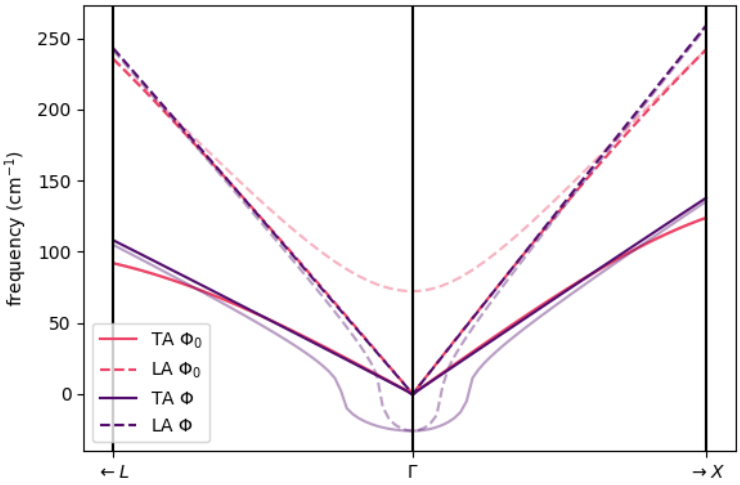


Figure 1.5: *Acoustic phonons in Si (Φ_0) and P-doped Si (Φ) before and after applying the symmetrized sum rule. The acoustic phonons are zero at Γ and the exchange symmetry is respected.*

- **Polar materials**: as discussed in Section (1.1.7), some insulators exhibit a polar behavior, where the displacement induced by an optical phonon near Γ generates dipoles and a long-range dipole-dipole interaction, which scales as $1/r^3$ with the distance r. Symmetry can suppress the first order polarity, but higher-order multipole moments can still generate polynomially decaying interactions.

- **Charged defects**: The ionic potential of a charged defect decays as $1/\epsilon r$, where ϵ is the macroscopic and static dielectric matrix of the material. Even if the charge is localized within a small region near the defect and the charge neutrality is imposed by a uniform background, the lattice of point charges generates a long-range Coulomb potential [23], that can be separated from the short range part [24, 25]. In the absence of a localized charge, higher order multipole moments generate a polynomially decaying potential [26].

- **Shallow defects**: the wavefunction of a bound state near the band edges extends over many unit cells. The bound electron does not see the atomic details of the defect core and feels a hydrogen-like screened Coulomb potential, while its mass is renormalized to the effective mass of the nearest band edge [27]. Together with central-cell corrections [28], this model correctly predicts the binding energy of shallow donors in Silicon. The concern in this case is that the supercell may alias the long tail of the wavefunction, creating an artificial band-like dispersion of the defect state. At finite temperature, a portion of these states may be thermally excited: the ionized impurity becomes a charged defect and the particle gets adsorbed into the band edges. These free carriers contribute to screen the Coulomb potential [29].

- **Kohn anomalies**: like for the polar materials, this feature is not related to the presence of the defect, but it can emerge due to the presence of defects. Since it is a feature of metallic systems, doping can introduce this effect. Low-energy phonons that connect two points near the Fermi Energy give rise to a strong response in the electronic system. For a free electron gas, the Fermi surface is spherical and $\boldsymbol{q} = 2\boldsymbol{k_F}$, where $\boldsymbol{k_F}$ is the Fermi wavevector [30]. This anomaly in the electronic susceptibility χ due to electron-phonon interaction is seen in real space as Friedel oscillations of wavevector $2\boldsymbol{k_F}$ [31]. Asymptotic analytical expressions for the long range part of force constants due to these oscillations have been derived in [32], by using the free electron gas susceptibility from Lindhard theory [33]. Calculations with high smearing corresponding to high fictitious temperatures effectively suppress Kohn anomalies and restore the short-range character of IFCs; if needed, the long range part can be deduced from electron-phonon matrix elements [34].

1.2.7 Dipole-dipole constants corrections

We now isolate every long-range contribution to the IFCs so that we can Fourier transform the short-range part and restore the long-range contribution in reciprocal space. As for charged-defect formation energies, the periodic supercell contribution must be separated from the isolated one:

$$\Delta\Phi^{SR}(\boldsymbol{Q}, \boldsymbol{Q'}) = \Delta\Phi(\boldsymbol{Q}, \boldsymbol{Q'}) - \Delta\Phi^{pLR}(\boldsymbol{Q}, \boldsymbol{Q'}) \tag{1.74}$$

on the $\boldsymbol{Q}$ grid used for the IFC calculation. We then Fourier transform $\Delta\Phi^{SR}$ to real space, center it on the defect as discussed in Section (1.2.2), and transform it back onto the desired $\boldsymbol{q}$ grid before restoring the isolated long-range part:

$$\Delta\Phi(\boldsymbol{q}, \boldsymbol{q'}) = \Delta\Phi^{SR}(\boldsymbol{q}, \boldsymbol{q'}) + \Delta\Phi^{iLR}(\boldsymbol{q}, \boldsymbol{q'}). \tag{1.75}$$

The electrostatic Green's function of a point charge, solving the anisotropic Poisson equation

$$\nabla \cdot \epsilon^{\infty} \nabla G(\boldsymbol{r}) = -4\pi\delta(\boldsymbol{r}), \tag{1.76}$$

is the screened Coulomb potential $G_{\epsilon}(\boldsymbol{r}) = (\epsilon^{\infty} r)^{-1}$. The reciprocal space counterpart is, including the Thomas-Fermi screening

$$G_{\epsilon,k_s}(\boldsymbol{Q}) = \frac{4\pi}{\boldsymbol{Q} \cdot \epsilon^{\infty} \cdot \boldsymbol{Q} + k_s^2}. \tag{1.77}$$

The screening effect for doped insulators can be inferred from the free carrier concentration due to dopants. The electrostatic contribution to IFC comes from the curvature of the Hartree functional $E^H = \frac{1}{2}\rho G \rho$, where ρ is the charge density. Expanding ρ in the atomic displacements, we get

$$\rho(\boldsymbol{Q}) = \rho^{(0)}(\boldsymbol{Q}) + \sum_{\eta \boldsymbol{R}} u_{\eta}(\boldsymbol{R}) \rho^{(1)}_{\eta \boldsymbol{R}}(\boldsymbol{Q}) + \dots \tag{1.78}$$

Although the expansion of the Hartree term alone also contains a term $\rho^{(0)}W\rho^{(2)}$, this term is not needed to construct the second derivative of the total ground-state energy. This is a consequence of the variational character of the Kohn-Sham energy functional, or equivalently of the $2n+1$ theorem of density-functional perturbation theory: the second-order variation of the total energy is completely determined by the first-order electronic response, together with the explicit second derivatives of the external and ion-ion terms. The second-order density multiplies the first functional derivative of the total energy and therefore vanishes at self-consistency. The universal long-range electrostatic part is thus obtained from the first-order charge response to an atomic displacement. At long wavelength this response is fixed by the Born effective charge (BEC),

$$\rho^{(1)}_{\eta\boldsymbol{R}}(\boldsymbol{Q}) = -ie^{-i\boldsymbol{Q}\cdot\boldsymbol{R}}\sum_{\gamma} Q_\gamma Z^*_{\gamma\alpha}(\boldsymbol{R}) + O(Q^2). \tag{1.79}$$

Combining two such dipolar responses with the screened Coulomb kernel $G(\boldsymbol{Q})$ gives the leading nonanalytic contribution

$$\rho^{(1)}G\rho^{(1)} \sim Q\,\frac{1}{Q^2}\,Q, \tag{1.80}$$

which is the reciprocal-space counterpart of the dipole-dipole energy. The long range IFC contribution is then

$$\Phi^{LR}_{\eta\eta'}(\boldsymbol{R},\boldsymbol{R'}) = \sum_{\boldsymbol{Q}} \rho^{(1)}_{\eta\boldsymbol{R}}(-\boldsymbol{Q})G(\boldsymbol{Q})\rho^{(1)}_{\eta'\boldsymbol{R'}}(\boldsymbol{Q}), \tag{1.81}$$

which becomes the well-known dipole-dipole interaction kernel in [12]:

$$\Phi^{LR}_{\alpha\boldsymbol{\tau},\alpha'\boldsymbol{\tau'}}(\boldsymbol{R},\boldsymbol{R'}) = \sum_{\gamma\delta} Z^*_{\boldsymbol{\tau},\alpha\gamma}(\boldsymbol{R})K_{\gamma\delta}(\boldsymbol{R}+\boldsymbol{\tau}-\boldsymbol{R'}-\boldsymbol{\tau'})Z^*_{\boldsymbol{\tau'},\delta\alpha'}(\boldsymbol{R'}), \tag{1.82}$$

where multiplication by $\boldsymbol{Q}$ becomes a derivative: $2K_{\gamma\delta}(\boldsymbol{r}) = -\partial_\gamma\partial_\delta G(\boldsymbol{r}) \sim r^{-3}$. Here we used $\eta = (\alpha,\boldsymbol{\tau})$ to distinguish between the cartesian and the atomic indices. The kernel has to be separated into

$$\frac{1}{r} = \frac{\mathrm{erfc}(\Lambda r)}{r} + \frac{\mathrm{erf}(\Lambda r)}{r}, \tag{1.83}$$

where Λ is a cutoff parameter that makes the first term short ranged in real space, while the second is smooth and converges rapidly in reciprocal space. This allows to Fourier transform separately the short and long range part of the kernel, as in Ewald summation techniques: $K(\boldsymbol{q})$ can be split into a convergent sum of reciprocal lattice vectors, the Fourier transform of the short range part and a limiting contribution. The kernel retains its translation symmetry, the supercell's effects come exclusively from the BEC:

$$\Delta Z^*_{\boldsymbol{\tau},\alpha\beta}(\boldsymbol{R}) = Z^*_{\boldsymbol{\tau},\alpha\beta}(\boldsymbol{R}) - Z^{*,0}_{\boldsymbol{\tau},\alpha\beta} \tag{1.84}$$

and possibly from the dielectric tensor ϵ^{∞}. For dilute defects, because the dielectric response is a macroscopically averaged quantity, we can assume that the long-range tail due to dipole-dipole interactions is not affected by the response $\Delta\epsilon$ of the locally perturbed environment. The leading contribution for long range interactions is then

$$\Delta\Phi^X(\boldsymbol{R},\boldsymbol{R'}) = \Delta Z^X(\boldsymbol{R})K^X(\boldsymbol{R}-\boldsymbol{R'})Z^0 + \tag{1.85}$$

$$Z^0K^X(\boldsymbol{R}-\boldsymbol{R'})\Delta Z^X(\boldsymbol{R'}) + \tag{1.86}$$

$$\Delta Z^X(\boldsymbol{R})K^X(\boldsymbol{R}-\boldsymbol{R'})\Delta Z^X(\boldsymbol{R'}), \tag{1.87}$$

where X is either the periodic or the isolated long-range contribution. The isolated kernel is the usual Coulomb kernel $K^{iLR} = K$, while the periodic one is obtained by summing over the supercell's lattice vectors $\boldsymbol{T}$:

$$K^{pLR}(\boldsymbol{r}) = \sum_{\boldsymbol{T}} K(\boldsymbol{r}+\boldsymbol{T}). \tag{1.88}$$

In the same way, if the BEC do not converge to the pristine ones within the supercell, they must follow

$$Z^{pLR}(\boldsymbol{R}) = \sum_{\boldsymbol{T}} Z^{iLR}(\boldsymbol{R}+\boldsymbol{T}), \tag{1.89}$$

from which we can fit the isolated contribution from the one given by the supercell calculation. In reciprocal space, the Fourier transform becomes a convolution:

$$\sum_{\boldsymbol{R},\boldsymbol{R'}} e^{-i\boldsymbol{q}\cdot\boldsymbol{R}}A(\boldsymbol{R})K(\boldsymbol{R}-\boldsymbol{R'})A'(\boldsymbol{R'})e^{i\boldsymbol{q'}\cdot\boldsymbol{R'}} \tag{1.90}$$

$$= \sum_{\boldsymbol{Q}} A(\boldsymbol{q}-\boldsymbol{Q})K(\boldsymbol{Q})A'(\boldsymbol{Q}-\boldsymbol{q'}), \tag{1.91}$$

where A and A' are either the BEC or their difference. Since $Z^0(\boldsymbol{q}) = Z^0\delta_{\boldsymbol{q},\boldsymbol{0}}$, Eq. (1.87) is transformed into

$$\Delta\Phi^X(\boldsymbol{q},\boldsymbol{q'}) = \Delta Z^X(\boldsymbol{q}-\boldsymbol{q'})K^X(\boldsymbol{q'})Z^0 + \tag{1.92}$$

$$Z^0K^X(\boldsymbol{q})\Delta Z^X(\boldsymbol{q}-\boldsymbol{q'}) + \tag{1.93}$$

$$\sum_{\boldsymbol{Q}} \Delta Z^X(\boldsymbol{q}-\boldsymbol{Q})K^X(\boldsymbol{Q})\Delta Z^X(\boldsymbol{Q}-\boldsymbol{q'}). \tag{1.94}$$

The last term vanishes in the isolated case, since $\Delta Z^{iLR}(\boldsymbol{R})$ decays within the supercell. At large separations $\boldsymbol{R}-\boldsymbol{R'}$, the dipole is therefore the one generated by the pristine system.

1.3 Green's function formalism

1.3.1 Dyson expansion

Particle-defect scattering is conveniently described using multiple-scattering theory, particularly when the perturbing potential is too strong for low-order perturbation theory

to be reliable. The formalism recasts the perturbative expansion as an operator series. Its central quantity, the self-energy, shifts the particle energies through its real part and gives the spectral broadening and lifetime through its imaginary part. When the perturbation series converges, this formulation is equivalent to ordinary quantum-mechanical perturbation theory. Consider, for example, the first two orders of the energy correction

$$E_n^{(1)} + E_n^{(2)} = \langle n| V |n\rangle + \sum_{m\neq n} \frac{\langle n| V |m\rangle \langle m| V |n\rangle}{E_n^{(0)} - E_m^{(0)}}. \tag{1.95}$$

Introducing the free propagator, also known as the retarded lattice Green function or resolvent, as

$$g(E) = \sum_m \frac{|m\rangle \langle m|}{E - E_m^{(0)} + i0^+} \tag{1.96}$$

we can rewrite the first two orders — leaving the $E = E_n^{(0)}$ dependence implicit — as

$$\langle n| (V + V g_n V) |n\rangle , \tag{1.97}$$

where $g_n = \bar{P}_n g \bar{P}_n$ is the free propagator with the n-th state projected out by $\bar{P}_n = 1 - |n\rangle \langle n|$. The third order

$$E_n^{(3)} = \sum_{m\neq n, l\neq n} \frac{\langle n| V |m\rangle \langle m| V |l\rangle \langle l| V |n\rangle}{(E_n^{(0)} - E_m^{(0)})(E_n^{(0)} - E_l^{(0)})} \tag{1.98}$$

$$- \langle n| V |n\rangle \sum_{m\neq n} \frac{\langle n| V |m\rangle \langle m| V |n\rangle}{(E_n^{(0)} - E_m^{(0)})^2} \tag{1.99}$$

can be rewritten as $\langle n| V g_n V g_n V |n\rangle - \langle n| V |n\rangle \langle n| V g_n^2 V |n\rangle$. The first term continues the pattern in Eq. (1.97); the second corrects it. To identify its origin, we define the self-energy as

$$\Sigma_n(E_n) = \sum_{i=1}^{\infty} E_n^{(i)} = E_n - E_n^{(0)}, \tag{1.100}$$

where E_n is the energy of state n to infinite order. The Born series gives

$$\Sigma_n(E_n^{(0)}) = \langle n| (V + V g_n V + V g_n V g_n V + ...) |n\rangle = \langle n| V(1 - g_n V)^{-1} |n\rangle . \tag{1.101}$$

The last equality is the Born series: the geometric expansion is valid when the spectral radius of $g_n V$ is smaller than one. The inverse $(1 - g_n V)^{-1}$ can nevertheless be defined more generally, except at energies for which $g_n V$ has an eigenvalue equal to one. Such singularities signal that the perturbation creates a state whose energy cannot be obtained from a convergent weak-scattering expansion. By expanding Eq. (1.100) around $E_n^{(0)}$:

$$\delta E_n = \Sigma_n(E_n) = \Sigma_n(E_n^{(0)}) + \delta E_n \frac{\partial \Sigma_n}{\partial E}\bigg|_{E_n^{(0)}} + ..., \tag{1.102}$$

The energy derivative produces a renormalization correction: Σ must be evaluated at the unknown interacting energy E_n. The equation can be solved iteratively, starting from $\Sigma_n(E_n^{(0)})$. If the potential has no dynamical dependence on E, the energy derivative of Eq. (1.101) is

$$\left.\frac{\partial \Sigma_n}{\partial E}\right|_{E_n^{(0)}} = -\langle n|\, V g_n^2 V\, |n\rangle + \mathcal{O}(V^3), \tag{1.103}$$

since $\partial_E g_n = -g_n^2$. The lowest order correction is then

$$\delta E_n \left.\frac{\partial \Sigma_n}{\partial E}\right|_{E_n^{(0)}} = (E_n^{(1)} + \mathcal{O}(V^2))(-\langle n|\, V g_n^2 V\, |n\rangle + \mathcal{O}(V^3)) \tag{1.104}$$

$$= -\langle n|\, V\, |n\rangle \langle n|\, V g_n^2 V\, |n\rangle\,, \tag{1.105}$$

which is exactly the second term appearing in the third order of perturbation theory. The equation that determines the complex energy is

$$E_n = E_n^{(0)} + \mathrm{Re}\,\Sigma_n(E_n) + i\,\mathrm{Im}\,\Sigma_n(E_n). \tag{1.106}$$

The imaginary part determines the lifetime. Since the wavefunction evolves as $e^{-iE_n t}$, its probability decays as $e^{2\,\mathrm{Im}\,\Sigma_n(E_n)t}$; the scattering rate is

$$\tau_n^{-1} = -2\,\mathrm{Im}\,\Sigma_n(E_n). \tag{1.107}$$

For weak scattering, we expand the self-energy around the real energy solution $E_n^* = E_n^{(0)} + \mathrm{Re}\,\Sigma_n(E_n^*)$ up to linear order in the imaginary part:

$$\Sigma_n(E_n) = \Sigma_n(E_n^*) + (E_n - E_n^*)\partial_E \Sigma_n(E_n^*) \tag{1.108}$$

Substituting this expansion into Eq. (1.106) gives

$$E_n = E_n^* + (E_n - E_n^*)\partial_E \Sigma_n(E_n^*) + i\,\mathrm{Im}\,\Sigma_n(E_n^*), \tag{1.109}$$

which is solved for

$$E_n = E_n^* + \frac{i\,\mathrm{Im}\,\Sigma_n(E_n^*)}{1 - \partial_E \Sigma_n(E_n^*)}. \tag{1.110}$$

Within this approximation, the scattering rate is

$$\tau_n^{-1} = -2Z\,\mathrm{Im}\,\Sigma_n(E_n^*) \tag{1.111}$$

where $Z = \mathrm{Re}(1 - \partial_E \Sigma_n(E_n^*))^{-1}$ is the renormalization factor. Throughout this work, we neglect self-energy renormalization. This approximation is valid when a small parameter c controls the self-energy and its derivative. In that case,

$$\mathrm{Im}\,\Sigma_n(E_n^*) = \mathrm{Im}\,\Sigma_n(E_n^{(0)}) + \mathcal{O}(c^2), \qquad Z = 1 + \mathcal{O}(c), \tag{1.112}$$

and the scattering rate becomes

$$\tau_n^{-1} = -2\,\mathrm{Im}\,\Sigma_n(E_n^{(0)}) + \mathcal{O}(c^2). \tag{1.113}$$

For defect scattering, the small parameter is the defect concentration c.

1.3.2 T-matrix and self-energy

The self-energy operator Σ_n, defined in Eq. (1.101), is constructed to exclude the state n from the intermediate states. We now define a new operator, called T-matrix, as:

$$T = V(1 - gV)^{-1}. \tag{1.114}$$

This operator contains all scattering events involving the defect at any order, including the intermediate state n. We now focus on the relation between the two operators, in the momentum basis in which $g_{\boldsymbol{qq'}} = g_{\boldsymbol{q}}\delta_{\boldsymbol{qq'}}$ is diagonal. The T-matrix represents the infinite series of irreducible scattering events contained in $\Sigma_{\boldsymbol{q}}$, followed by the free propagation $g_{\boldsymbol{q}}$:

$$T_{\boldsymbol{q}} = \Sigma_{\boldsymbol{q}} + \Sigma_{\boldsymbol{q}} g_{\boldsymbol{q}} \Sigma_{\boldsymbol{q}} + ... = \Sigma_{\boldsymbol{q}}(1 - g_{\boldsymbol{q}}\Sigma_{\boldsymbol{q}})^{-1}. \tag{1.115}$$

The same rearrangement applies to the case in which Σ is allowed to have non-equal initial and final states, for which the previous equation is rewritten in matrix form as

$$T = \Sigma(1 - g\Sigma)^{-1}. \tag{1.116}$$

The full resolvent, defined as $G = (E - H_0 - V)^{-1}$ can be extracted from both these operators as

$$G = g + gTg = g + g\Sigma G = (1 - g\Sigma)^{-1} g, \tag{1.117}$$

since $H_0 + V$ can be substituted with $E_0 + \Sigma$, as shown in the previous section. The resolvent G and the T-matrix T therefore retain the information encoded in the perturbation series. They provide the starting point for calculating the macroscopic properties of interest.

1.3.3 Localized states

We now focus on the series in Eq. (1.114), specifically on the energy value where the series diverges. Let's consider the simplest case: a single defect with a real-valued perturbing potential V. The operators reduce to scalars, and it is then immediate that

$$1 - g(E)V = 0 \tag{1.118}$$

is fulfilled only when the resolvent is real. This condition is met when E lies outside the continuum spectrum, since the density of states (DOS), defined as

$$\rho(E) = \sum_m \delta(E - E_m) = -\frac{1}{\pi} \operatorname{Im} \left(\sum_m g_{mm} \right) = -\frac{1}{\pi} \operatorname{Im}(\operatorname{Tr} g), \tag{1.119}$$

vanishes when g is real. Outside the continuum, the resolvent is known to decay as $g \sim (E - E_0)^{-1}$, where E_0 is the center of the continuum, so that the pole condition translates into $E = E_0 + V$. The full resolvent of the perturbed Hamiltonian

$$G(E) = (E - H_0 - V)^{-1} \tag{1.120}$$

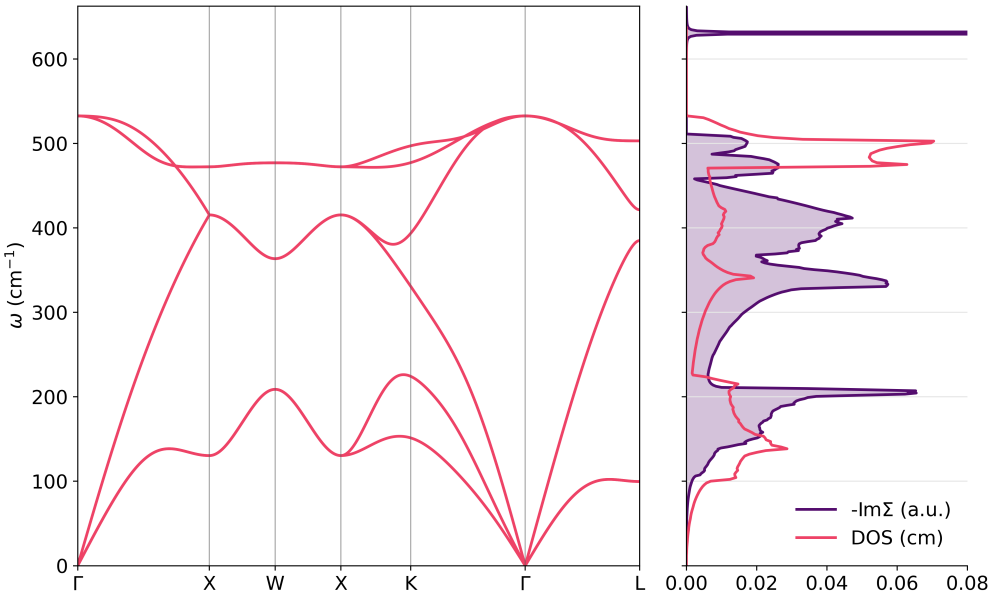


Figure 1.6: *Energy width for single mass defect in Silicon, with* $\Delta m/m = -0.6$*. The mass defect is maximally localized, hence the width is expected to be constant in reciprocal space, for cubic crystals. The lines up to* ~ 500 *are inside the DOS of the system, while the line at* ~ 630 *is the localized state, outside the bands. For phonons,* $E = \omega^2$ *and the equation to solve is* $\omega^2 = \omega_0^2 + 0.6\omega^2$*, where* ω_0 *is the center of the band.*

is divergent once the condition in Eq. (1.118) is met, since $G^{-1} = g^{-1} - V$ and then

$$G = g + gVg + ... = (1 - gV)^{-1}g. \tag{1.121}$$

This new resolvent has a pole exactly at $E = E_0 + V$, hence this must be an isolated eigenvalue of $H_0 + V$ outside the continuum of H_0. We now show that this state is spatially localized. In a site-localized basis $|R\rangle$, suppose that the defect is at the origin and that only one matrix element is non-zero: $V(\boldsymbol{R}, \boldsymbol{R'}) = V_{\mathbf{00}}\delta_{\boldsymbol{R}\mathbf{0}}\delta_{\boldsymbol{RR'}}$. The argument extends to any potential with finite support or sufficiently rapid decay. Substituting this definition of V into Eq. (1.121), the full resolvent will be

$$G_{\boldsymbol{RR'}} = g_{\boldsymbol{RR'}} + V_{\mathbf{00}}g_{\boldsymbol{R}\mathbf{0}}G_{\mathbf{0}\boldsymbol{R'}} \tag{1.122}$$

$$G_{\mathbf{0}\boldsymbol{R'}} = g_{\mathbf{0}\boldsymbol{R'}} + V_{\mathbf{00}}g_{\mathbf{00}}G_{\mathbf{0}\boldsymbol{R'}} = g_{\mathbf{0}\boldsymbol{R'}}(1 - V_{\mathbf{00}}g_{\mathbf{00}})^{-1}. \tag{1.123}$$

Combining the two equations gives the diagonal part of G:

$$G_{\boldsymbol{RR}} = g_{\boldsymbol{RR}} + \frac{V_{\mathbf{00}}|g_{\boldsymbol{R}\mathbf{0}}|^2}{1 - V_{\mathbf{00}}g_{\mathbf{00}}}. \tag{1.124}$$

The state is said to be localized when the local density of states (LDOS), defined as

$$\rho(\boldsymbol{R}, E) = -\frac{1}{\pi}\operatorname{Im} G_{\boldsymbol{RR}} \tag{1.125}$$

shows an exponential decay. The LDOS is the density of states at energy E and distance $\boldsymbol{R}$ from the defect. Outside the band, $g_{\boldsymbol{RR}}$ is real; in the second term of Eq (1.124), the denominator gives a delta function at $E_0 + V_{\mathbf{00}}$, so that

$$\rho(\boldsymbol{R}, E) \sim |g_{\boldsymbol{R0}}|^2 \delta(E - E_0 - V_{\mathbf{00}}). \tag{1.126}$$

This equation shows how the LDOS has the same decay as the lattice resolvent $g_{\boldsymbol{R0}}$. The Green function in the localized basis can be seen as the Fourier transform of $g_{\boldsymbol{qq'}}$, which is diagonal in the momentum basis:

$$g_{\boldsymbol{R0}}(E) \sim \int d\boldsymbol{q}\, \frac{e^{i\boldsymbol{q}\cdot\boldsymbol{R}}}{E - \epsilon_{\boldsymbol{q}}}. \tag{1.127}$$

In a periodic lattice, the spectrum consists of dispersive bands $\epsilon_{\boldsymbol{q}j}$; we omit the branch index j for brevity. Outside a band, $E = \epsilon_{\boldsymbol{q}}$ has no solution for real $\boldsymbol{q}$, but it can have solutions at complex wavevectors. Let $\boldsymbol{\kappa}$ be the smallest positive imaginary shift that reaches such a solution. Shifting the integration contour by $i\boldsymbol{\kappa}$ leaves the analytic integrand unchanged; the vertical contour segments vanish in the thermodynamic limit. Consequently,

$$g_{\boldsymbol{R0}}(E) \sim \int d\boldsymbol{q}\, \frac{e^{i(\boldsymbol{q}+i\boldsymbol{\kappa})\cdot\boldsymbol{R}}}{E - \epsilon_{\boldsymbol{q}+i\boldsymbol{\kappa}}} \sim e^{-\boldsymbol{\kappa}\cdot\boldsymbol{R}} \tag{1.128}$$

since the remaining integral is bounded. This contour argument shows why states outside the bands are exponentially localized.

1.3.4 Resonant states

Even when $g(E)V = 1$ has no real solution inside a band, the condition can be nearly satisfied. Allowing E to be complex gives generalized solutions known as resonances. Like localized states, resonances are associated with poles of the resolvent; unlike them, they lie within the continuum and acquire a finite width from the imaginary part of their energy. For a moderate perturbation, they are expected near the condition

$$\operatorname{Re} g(E)V = 1. \tag{1.129}$$

From Eq. (1.124), the on-site element of the resolvent reduces to

$$G_{\mathbf{00}} = \frac{g_{\mathbf{00}}}{1 - g_{\mathbf{00}} V_{\mathbf{00}}}, \tag{1.130}$$

while the LDOS becomes

$$\rho(\mathbf{0}, E) = \frac{\rho_0(\mathbf{0}, E)}{(1 - V_{\mathbf{00}} \operatorname{Re} g_{\mathbf{00}}(E))^2 + (V_{\mathbf{00}} \pi \rho_0(\mathbf{0}, E))^2}, \tag{1.131}$$

where ρ_0 is the LDOS of the lattice. The resulting LDOS is maximal when the condition of Eq. (1.129) is satisfied. The profile resembles a Lorentzian, with a width proportional to $V_{\mathbf{00}}\rho_0(E)$. If resonant states are present, we would expect the eigenvalues of gV — where they exist — to be close to 1 in modulus. We then must use the non-perturbative expansion of Eq. (1.121), since the Born series is divergent. Fig. (1.7) shows a real system in which this condition happens.

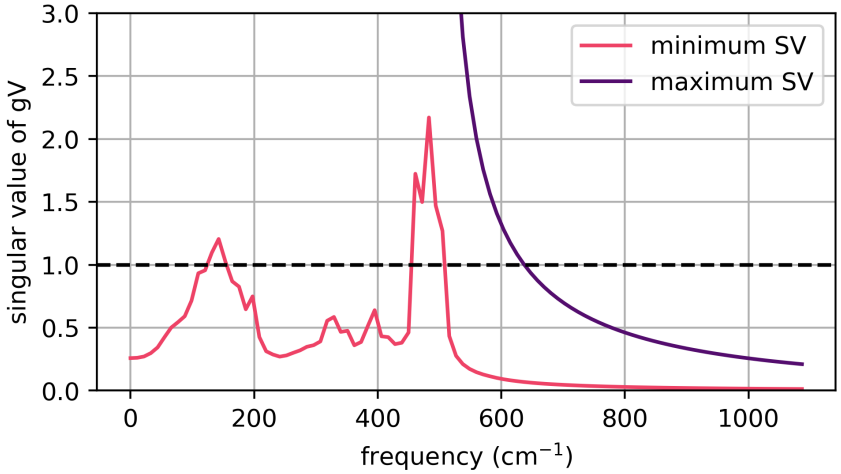


Figure 1.7: *Singular values of gV as a function of energy, for B-doped Silicon. They can be viewed as the modulus of the eigenvalues of gV, when they exist. For certain frequencies, the singular values are close to 1 inside the bands, we then expect a resonant state in the proximity of these intervals. The purple line marks a localized state at $\sim 630\text{cm}^{-1}$, at the same energy as in Fig* (1.6)*. It is caused by the same mass part of the perturbation, since $\Delta m/m$ is around 0.6 for this defect.*

1.3.5 Configurational Average

A phonon propagating through a disordered crystal scatters from localized defect potentials at randomly distributed sites. We restrict the discussion to phonon-defect scattering. Two features make the problem challenging:

- phonons inside a crystal have a dispersion relation and are fully delocalized. They are described as displacements in the unit cell enveloped with a phase that can span many unit cells, according to the Bloch equation $\langle \boldsymbol{R}\tau\alpha|\boldsymbol{q}j\rangle = e_{\boldsymbol{q}j}(\tau,\alpha)e^{i\boldsymbol{q}\cdot\boldsymbol{R}}$, where $e_{\boldsymbol{q}j}$ is the j-th eigenvector of the dynamical matrix of the system at $\boldsymbol{q}$. On the other hand, the potential is localized, hence it has no structure in the phonon basis. As a consequence, there will always be a tradeoff between real and reciprocal bases, in which the potential and the Green function, respectively, are conveniently represented.

- Defect positions must be sampled from their probability distribution over the entire lattice. In principle, this requires an average over large supercells, each with force constants calculated using DFPT or finite differences. In the infinite-supercell limit, a single representative configuration would be sufficient. In practice, calculations are limited to supercells not much larger than the one used to obtain the single-defect force-constant differences.

To show how these features enter the formalism, consider the configuration potential V^α of Eq. (1.66). We restrict the discussion to a single defect species in a lattice with equivalent sites. For a fixed defect configuration α, the full resolvent follows from $G^\alpha = g + gV^\alpha G^\alpha$. For one defect in N cells, the scattering matrix T is independent of the defect position; therefore,

$$\langle G \rangle = (1-c)g + cG^{(1)} = (1-c)g + cgTg, \tag{1.132}$$

where $c = 1/N$ is the defect concentration. All scattering processes are contained in the single-defect scattering matrix $T = V(1-gV)^{-1}$. For multiple defects, the average $\langle G \rangle = g + g\langle VG \rangle$ cannot be evaluated by simply replacing V with its average, because the random defect positions produce phase factors. This positional disorder can be represented by an indicator function: $\rho_\alpha(\boldsymbol{T}) = 1$ when site $\boldsymbol{T}$ is occupied by a defect in configuration α, and zero otherwise. Suppressing phonon indices, the potential becomes

$$V^\alpha = \sum_{\boldsymbol{T}} V(\boldsymbol{T})\rho_\alpha(\boldsymbol{T}), \tag{1.133}$$

where the sum extends over all N sites of the lattice. The configurational average must then be performed over the indicator functions appearing in the series. The simplest approximation assumes independent site occupations, $\langle \rho(\boldsymbol{T}_1)\dots\rho(\boldsymbol{T}_n)\rangle = c^n$, where $c = \langle \rho(\boldsymbol{T})\rangle$. The unrestricted sums over defect positions give momentum conservation for each scattering:

$$\sum_{\boldsymbol{T}} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} = N\delta_{\boldsymbol{q}\boldsymbol{q'}} \tag{1.134}$$

Under this assumption, a phonon initially in $|\boldsymbol{q}j\rangle$ can scatter only into states $|\boldsymbol{q}j'\rangle$. These are vertical transitions in a band diagram because they conserve crystal momentum. For well-separated bands, such transitions are usually suppressed: at second order, the Green's function (equivalently, the energy denominator in perturbation theory) enforces energy conservation between $\omega_{\boldsymbol{q}j}$ and $\omega_{\boldsymbol{q}j'}$. The self-energy defined by $\langle G\rangle = g + g\Sigma\langle G\rangle$ then reduces to $\Sigma = \langle V\rangle = cV$, with V the potential centred at any equivalent site. This approximation neglects multiple scattering from the same site.

1.3.6 Higher order diagrammatic expansions

Scattering amplitudes in the Born series have a diagrammatic representation in terms of Feynman diagrams. A horizontal line denotes the lattice propagator, whereas a dashed vertical or oblique line denotes scattering from a defect. Several dashed lines attached to one point represent repeated scattering from that same defect. More importantly, a phonon can scatter from one defect, then another, and subsequently return to the first. Such correlated processes depend jointly on several defect positions and cannot be factorized. The self-energy, in this diagrammatic view, is the sum of every irreducible event, for which no lattice propagator is put in between without any scattering; while

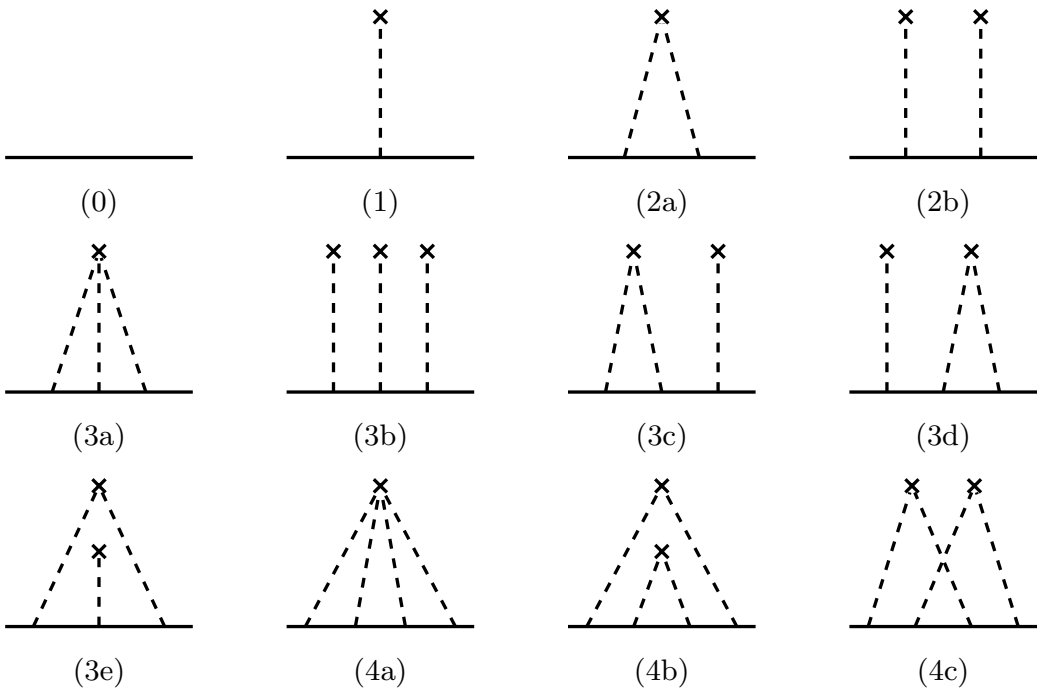


Figure 1.8: *Diagrams up to third order in the scattering potential, together with selected fourth-order diagrams.*

the T-matrix contains all scattering diagrams. Examples are shown in Fig. (1.8). In the previous approximation, known also as Virtual Crystal Approximation (VCA), the self-energy is the average betweeen diagrams (0) and (1): this representation allows us to visually explain why $\Sigma = cV$. The T-matrix embeds instead all the single scattering across many defects: it's composed of (2b) and (3b), etc. In the single defect approximation, the diagrams (2a), (3a), (4a) and so on are allowed for the self-energy, while the T-matrix also includes (3c) and (3d). This approximation is also known as Average T-matrix Approximation (ATA). Better approximations should feature other diagrams in the expansion, among the most successful we should mention the self-consistent ones. The most widely used is the Coherent Potential Approximation (CPA). Starting from ATA, we can substitute the full propagator instead of the lattice one: for example the diagram (3e) can be viewed as a replica of (2a) where the propagator at (1) is used instead of the usual horizontal line. A new propagator is found at each step, this cycle is repeated until self-consistency. These methods is exact up to third order, but it leaves out correlated diagrams, such as (4c): it's impossible to compose this diagram through the use of g, G and V if the diagram itself is not present inside the series of G. Better approximation should incorporate this diagram natively, or at least make a good approximation of its scattering amplitude. These methods are called non-local,

since the self-energy has its own dispersion in reciprocal space. We'll briefly discuss some of these methods, such as Itinerant CPA. Dynamical Cluster Approximation (DCA) is instead discussed in further details, implemented and public available at [35]. It is instructive to see how all the diagrams come out by unrolling the series at each order. Starting from the first, $\langle\rho(\boldsymbol{T})\rangle = c$, the average of Eq. (1.133) becomes

$$cV(\mathbf{0})\sum_{\boldsymbol{T}} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} = cV(\mathbf{0})N\delta_{\boldsymbol{qq'}}. \tag{1.135}$$

The average enforces translation invariance and thus momentum conservation. The first order diagonal self-energy is then $\Sigma^{(1)}_{\boldsymbol{qj}} = cN\,\langle \boldsymbol{qj}|V^{\boldsymbol{qj}}_{\boldsymbol{qj}}|\boldsymbol{qj}\rangle$. For the second order, $\langle\rho(\boldsymbol{T_1})\rho(\boldsymbol{T_2})\rangle = c\delta_{12} + c^2(1-\delta_{12})$. Grouping the delta term, we get

$$c(1-c)\sum_{\boldsymbol{T}} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} e^{i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} = c(1-c)N \tag{1.136}$$

$$c^2\sum_{\boldsymbol{T}} e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}}\sum_{\boldsymbol{T'}} e^{i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T'}} = c^2N^2\delta_{\boldsymbol{qq'}}, \tag{1.137}$$

The first term is associated to (2a), since there can be an intermediate $\boldsymbol{q'}$ between the two dashed scattering lines. The second represents the diagram (2b), for which momentum is conserved on both single scattering events. At third order, there are three momentum transfers, denoted $\boldsymbol{q_1}$, $\boldsymbol{q_2}$, and $\boldsymbol{q_3}$. The phase factors encode momentum conservation at each scattering process. It is implicit that the total momentum is always conserved: $\boldsymbol{q_1}+\boldsymbol{q_2}+\boldsymbol{q_3} = \mathbf{0}$. This law comes naturally by taking only the (momentum) diagonal part of the self-energy, for which the initial and final momentum of the propagating particle are the same. The summation over the position can be decomposed in five parts, using the following identity:

$$1 = \delta_{12}\delta_{23} + (1-\delta_{12})(1-\delta_{23})(1-\delta_{31}) + \tag{1.138}$$

$$\delta_{12}(1-\delta_{23}) + \delta_{23}(1-\delta_{31}) + \delta_{31}(1-\delta_{12}), \tag{1.139}$$

corresponding to the diagrams (3a) to (3e), respectively. We include only the index in the delta, so that δ_{ij} is meant to give $\boldsymbol{T_i} = \boldsymbol{T_j}$. Splitting the summation over defect position into these five parts make the calculation of the average of the indicator functions trivial: since $\langle\rho(\boldsymbol{T})^n\rangle = \langle\rho(\boldsymbol{T})\rangle = c$, the whole average $\langle\rho(\boldsymbol{T_1})...\rho(\boldsymbol{T_n})\rangle$ gives a factor c for each different position in the product. Defining $A_{lmn} = e^{i(\boldsymbol{q_1}\cdot\boldsymbol{T_l}+\boldsymbol{q_2}\cdot\boldsymbol{T_m}+\boldsymbol{q_3}\cdot\boldsymbol{T_n})}$ the sum becomes:

$$c\sum_{1} A_{111} + c^3\left(\sum_{123} A_{123} - \sum_{12}(A_{112}+A_{122}+A_{121}) + 2\sum_{1} A_{111}\right) + \tag{1.140}$$

$$c^2\left(\sum_{12}(A_{112}+A_{122}+A_{121}) - 3\sum_{1} A_{111}\right), \tag{1.141}$$

rearranging terms and summing we get to the final expression:

$$(c - 3c^2 + 2c^3)N\delta_{\boldsymbol{q_1}+\boldsymbol{q_2}+\boldsymbol{q_3}} + c^3N^3\delta_{\boldsymbol{q_1}}\delta_{\boldsymbol{q_2}}\delta_{\boldsymbol{q_3}} + \tag{1.142}$$

$$c^2(1-c)N^2(\delta_{\boldsymbol{q_1}+\boldsymbol{q_2}}\delta_{\boldsymbol{q_3}} + \delta_{\boldsymbol{q_1}}\delta_{\boldsymbol{q_2}+\boldsymbol{q_3}} + \delta_{\boldsymbol{q_1}+\boldsymbol{q_3}}\delta_{\boldsymbol{q_2}}). \tag{1.143}$$

Apart from the lengthy derivation, we can clearly see how the diagrams combine already all the information on momentum, by simply stating the following principle: *the set of scatterings across each defect conserves momentum globally*. For example, the conservation in (3c) is $\boldsymbol{q_1} + \boldsymbol{q_2} = \boldsymbol{0} \wedge \boldsymbol{q_3} = \boldsymbol{0}$, corresponding to the third piece in our calculation. Since we had to sum each part of the space spanned by the defect position individually, each diagrams got a higher order correction on the coefficient. For example, (3c), (3d) and (3e) got $c^2(1-c)$, which is the product of the coefficients of (2a) and (1), the diagrams composing them. The coefficient $c(1-3c+2c^2)$ belonging to (3a) is probably the least obvious: its bare strength should be c, but the following diagrams (3b)-(3e) are equivalent to (3a) if the defects coincide. Its strength must be reduced by the sum of all the other diagrams: $3c^2(1-c)+c^3$. There are closed formula derived from set theory, named cumulant expansions in this context [36]. One last aspect to be aware of, is that these coefficients should change accordingly to the approximation being used. Within the ATA, the diagram (3e) is not included, so that the previous coefficient becomes $c(1-2c+c^2) = c(1-c)^2$. In next chapters we'll show how the terms of the kind (na) acquire the coefficients $c(1-c)^{n-1}$. Up to now, we could already derive the self-energy in this approximation. We want only multiple scattering on one defect, we then need to sum all the diagrams composing the single particle T-matrix, multiplied each by its coefficient:

$$\Sigma^{ATA} = cV + c(1-c)VgV + \ldots = cV\left[1-(1-c)gV\right]^{-1}. \tag{1.144}$$

Without the overcounting corrections, the expression would have been

$$\Sigma^{ATA} = cV + cVgV + \ldots = cV(1-gV)^{-1} = cT, \tag{1.145}$$

the self-energy becomes the single particle T-matrix rescaled by the concentration.

1.3.7 Real and reciprocal space

This section shows that the real- and reciprocal-space representations yield the same Born series. Every expression in one representation has a corresponding expression in the other. As stated before, there are advantages and drawbacks for both representations: the potential is better represented in real space, while the lattice Green's function is already diagonal in reciprocal space. It is advisable to combine the two whenever possible. We start, as before with a single defect system and express the T-matrix as:

$$T = V + VgV + ... \tag{1.146}$$

The perturbation V is expressed in matrix form in real space as:

$$V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) = D_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}, \boldsymbol{0}) - D^0_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}). \tag{1.147}$$

The lattice vectors $\boldsymbol{R}$ and $\boldsymbol{R'}$ are assumed to have been centred using the procedure described in Section (1.2.2). We now consider the second-order term of the Born series.

The (complex) energy shift is obtained by taking the matrix element of the operator in the phonon state of interest:

$$T^{(2)}_{\boldsymbol{q}j} = \langle \boldsymbol{q}j | \, VgV \, | \boldsymbol{q}j \rangle \, . \tag{1.148}$$

We must express both g and V in a common representation. Although the reciprocal-space representation of V has not yet been derived, the Green's function is diagonal in reciprocal space: $g^{\boldsymbol{q'}j'}_{\boldsymbol{q}j}(\omega) = (\omega^2 - \omega^2_{\boldsymbol{q}j})^{-1}\delta_{\boldsymbol{q}\boldsymbol{q'}}\delta_{jj'}$. The basis transformation between cartesian real space and phonon reciprocal space is:

$$\langle \boldsymbol{R}\eta | \boldsymbol{q}j \rangle = \frac{1}{\sqrt{N}} e_{\boldsymbol{q}j}(\eta) e^{i\boldsymbol{q}\cdot\boldsymbol{R}} \tag{1.149}$$

where $e_{\boldsymbol{q}j}(\eta)$ is the phonon polarization vector. N is both the number of unit cells and the number of $\boldsymbol{q}$ points, as required by the discrete Fourier transform. The Green's function in real space is then:

$$g_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}, \omega) = \sum_{\boldsymbol{q}j} \frac{\langle \boldsymbol{R}\eta | \boldsymbol{q}j \rangle \langle \boldsymbol{q}j | \boldsymbol{R'}\eta' \rangle}{\omega^2 - \omega^2_{\boldsymbol{q}j}} = \frac{1}{N} \sum_{\boldsymbol{q}j} \frac{e^{i\boldsymbol{q}\cdot\boldsymbol{R}} e_{\boldsymbol{q}j}(\eta) e^*_{\boldsymbol{q}j}(\eta') e^{-i\boldsymbol{q}\cdot\boldsymbol{R'}}}{\omega^2 - \omega^2_{\boldsymbol{q}j}}. \tag{1.150}$$

The second order now reads:

$$T^{(2)}_{\boldsymbol{q}j} = \sum_{\boldsymbol{R}\boldsymbol{R'}\eta\eta'} \langle \boldsymbol{q}j | \boldsymbol{R}\eta \rangle \, [VgV]_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) \, \langle \boldsymbol{R'}\eta' | \boldsymbol{q}j \rangle \, . \tag{1.151}$$

Combining the indexes $(\boldsymbol{R}, \eta)$ into a single one, the term VgV under square brackets is simply given by matrix multiplication. We can also start from the reciprocal space representation to get the same result. The second order term is given by:

$$T^{(2)}_{\boldsymbol{q}j}(\omega) = \sum_{\boldsymbol{q'}j'} V^{\boldsymbol{q'}j'}_{\boldsymbol{q}j} g_{\boldsymbol{q'}j'} V^{\boldsymbol{q}j}_{\boldsymbol{q'}j'} = \sum_{\boldsymbol{q'}j'} \frac{\left| V^{\boldsymbol{q'}j'}_{\boldsymbol{q}j} \right|^2}{\omega^2 - \omega^2_{\boldsymbol{q'}j'}}. \tag{1.152}$$

The modulus squared follows from the Hermiticity of the perturbation, $V^{\boldsymbol{q}j}_{\boldsymbol{q'}j'} = (V^{\boldsymbol{q'}j'}_{\boldsymbol{q}j})^*$, which is the condition relevant to detailed balance. Section (1.2.4) discusses how this property is maintained. Adding a small imaginary shift $i\epsilon$ to regularize the poles makes the imaginary part of the Green's function proportional to a delta function; the resulting expression resembles Fermi's golden rule for transition rates. This simplification applies only at second order: higher orders require sums over intermediate states and thus matrix multiplications. By comparing this expression with Eq. (1.150) and Eq. (1.151) we conclude that the potential in reciprocal space in the phonon basis is given by:

$$V^{\boldsymbol{q'}j'}_{\boldsymbol{q}j} := \langle \boldsymbol{q}j | V | \boldsymbol{q'}j' \rangle = \sum_{\boldsymbol{R}\boldsymbol{R'}\eta\eta'} \langle \boldsymbol{q}j | \boldsymbol{R}\eta \rangle \, V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) \, \langle \boldsymbol{R'}\eta' | \boldsymbol{q'}j' \rangle \tag{1.153}$$

$$= \frac{1}{N} \sum_{\boldsymbol{R}\boldsymbol{R'}\eta\eta'} e^*_{\boldsymbol{q}j}(\eta) e^{-i\boldsymbol{q}\cdot\boldsymbol{R}} V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'}) e^{i\boldsymbol{q'}\cdot\boldsymbol{R'}} e_{\boldsymbol{q'}j'}(\eta'). \tag{1.154}$$

The potential must first be Fourier transformed, since the rotation to the phonon basis is $\boldsymbol{q}$-dependent. The choice of signs in the phases is then fixed in this specific way to restore the equivalence of the two representations. In reciprocal space, as discussed in the previous Section (1.2.3), the presence of multiple defects builds an additional phase factor. The information on disorder comes exclusively from this factor, while each term of the Born series remains the same as in the single defect picture. Unfortunately, the phase factor is different for each order in the series, hence it cannot be factorized out and we should still treat each order separately.

1.3.8 Normalization of the potential

For a single defect, an nth-order matrix element of the Born series contains an overall factor $1/N$, where N is the number of unit cells. This does not imply that scattering vanishes as the system grows, because the dimensions of g and V grow simultaneously. In real space, V has compact support and acts on a finite set of coordinates; consequently, its nonzero eigenvalues and its trace remain of order unity. The factor $1/N$ arises from the external brackets in Eq. (1.151). In reciprocal space, V is delocalized: its matrix elements scale as N^{-1}, while the number of states scales as N. The trace, which is invariant under the Fourier transformation, remains of order unity. Each of the $n-1$ internal sums contributes a factor N, which cancels one factor N^{-1} from a potential matrix element. The same overall $1/N$ scaling is therefore retained. Two limiting behaviours are possible:

- The eigenvalues of V spread out and become negligibly small as the system size increases. The scattering then disappears completely in the thermodynamic limit.
- A small cluster of eigenvalues remains finite, while the rest is zero. The scattering is then dominated by this cluster, in the thermodynamic limit each element of the Green's function approaches a finite value and the non-zero eigenvalues of gV are the same as the ones in which g and V are restricted to the compact support of the potential. Interactions with the defect can cause the formation of states with spectral weight only inside the support. These states are hence localized.

The number of cells N is not the size of the supercell in which the potential was originally calculated; it is the number of $\boldsymbol{q}$ points used to represent the Green's function. Centering shifts the coordinates relative to the defect and pads the real-space potential matrix with zeros. The added eigenvalues are therefore zero, while the original eigenvalues remain unchanged. For a single defect, the scattering contribution can still vanish because of the overall $1/N$ factor. A finite contribution requires a divergent T-matrix, corresponding to a potential strong enough to produce localized states or resonances. In the ATA, there are cN defects, but only repeated scattering from the same defect is retained. Since each defect contributes a factor $1/N$, the total scattering strength is independent of system size and phonons can decay into the continuum. More generally, the disorder phase factor supplies exactly one global factor of N for each scattering channel. If α sites are involved, there are $\alpha - 1$ independent

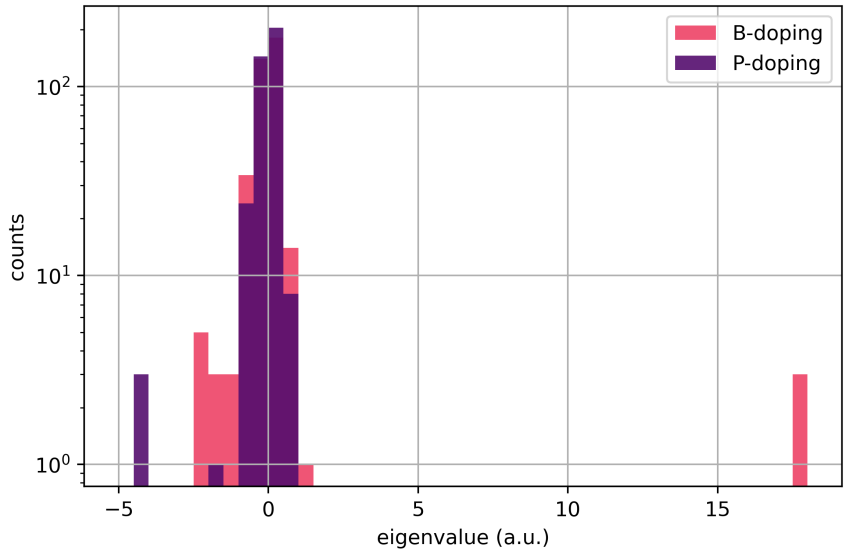


Figure 1.9: *Distribution of the eigenvalues of V for P- and B-doped silicon. The outliers can produce localized states. The eigenvalues are positive for B and negative for P, following the sign of the mass perturbation.*

momentum-conservation conditions. Equations (1.136) and (1.137) illustrate that the remaining factor of N follows from imposing equal initial and final momenta.

1.4 Disorder approximations

1.4.1 Virtual Crystal Approximation (VCA)

The disorder-averaged one-particle Green's function contains the information relevant to single-particle excitations, including spectral functions, lifetimes, and line shifts. Configurational averaging also generates correlations between propagators. For example, the two-particle Green's function

$$\left\langle G^{(2)}(1,2;1',2')\right\rangle = \left\langle G(1,1')G(2,2') + G(1,2')G(2,1')\right\rangle \tag{1.155}$$

$$\neq \left\langle G(1,1')\right\rangle \left\langle G(2,2')\right\rangle + \left\langle G(1,2')\right\rangle \left\langle G(2,1')\right\rangle \tag{1.156}$$

cannot be factorized into a Wick's sum because of the averaging operation. The single particle Green's function is plagued by the same problem: the average of

$$\langle G\rangle = \langle g + gVG\rangle = g + g\langle VG\rangle = g + g\langle V\rangle g + g\langle VgV\rangle g + \ldots \tag{1.157}$$

cannot be easily decoupled. We've already seen how the potential can be factorized into the single scatterer and the configurational part in reciprocal space in Eq. (1.66).

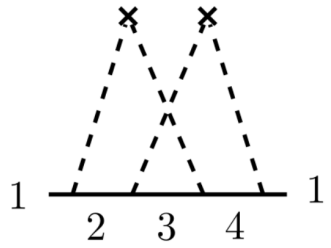


Figure 1.10: *Example of a correlated diagram, where the momentum conservation links the degrees of freedom of the two scatterers and makes it impossible to separate them into independent contributions.*

As a result, only a sum of phases is involved in the averaging process: the configurational term alone is capable of generating all the correlations and it is the only term that requires an approximation. The main challenge in the average is that a phonon can visit a site, scatter elsewhere, and return to the original site. In Sec. (1.3.6) we briefly showed how the averaging process produces a sum of diagrams, where each term is composed of a multiplicative coefficient expressed as a function of the concentration and the momentum conservation at vertices. We can now understand why the correlated diagrams, i.e. the ones with crossing lines, are the most difficult to evaluate. Referring to Fig. (1.10), the contribution of the diagram is

$$\mathcal{O}(c^2)g(1)V(1,2)g(2)V(2,3)g(3)V(3,4)g(4)V(4,1)g(1)\delta(2-1+4-3), \tag{1.158}$$

where V is the single scatterer part of the potential. We wrote only one momentum conservation, since the other gives the same information if the diagram itself conserves momentum globally. In any case, the conservation mixes all the four degrees of freedom of the particle, making it impossible to decouple the contribution of the two scatterers. The first attempt to decouple the correlated contributions might be something similar to a random phase approximation, in which

$$\langle p(1,2)...p(n,1)\rangle \simeq \langle p(1,2)\rangle ... \langle p(n,1)\rangle\,, \tag{1.159}$$

where $p(1,2) = \sum_{\boldsymbol{T}} e^{i(\boldsymbol{q_1}-\boldsymbol{q_2})\cdot\boldsymbol{T}}$ is the configurational part of the potential. Since $\langle p(1,2)\rangle = Nc\delta(1-2)$, the approximation neglects any momentum exchange with the crystal:

$$\langle G(1)\rangle \simeq g(1) + g(1)^2 cV(1,1) + ... \tag{1.160}$$

This approximation is called the virtual crystal approximation (VCA) and is known to show divergences in the power series near critical points. Indeed, if the band energy has a quadratic form, the integration volume $d^3q = q^2dq = \sqrt{E}dE$ is not sufficient to

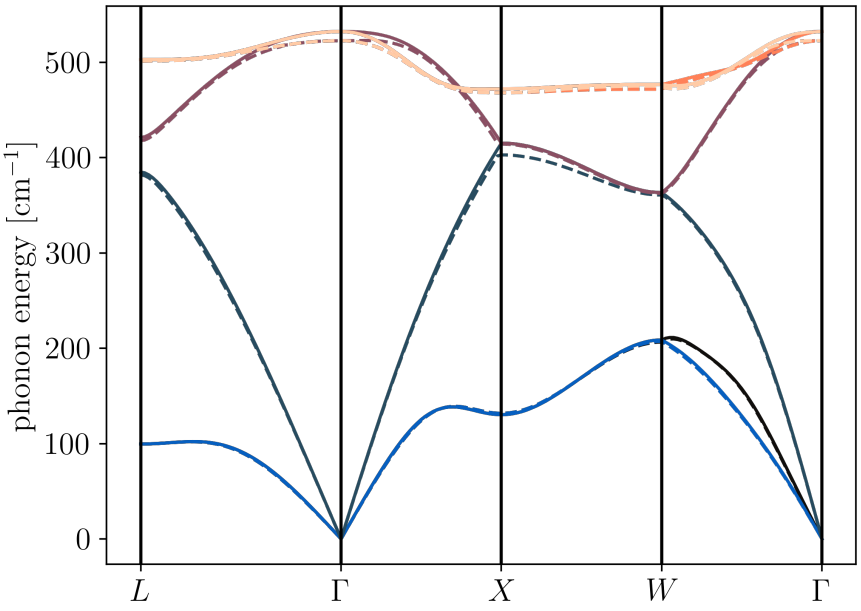


Figure 1.11: *First order lineshifts in 1% P-doped Si, calculated with the VCA. The dashed lines represent the new phonon energies; we can see that in the region $X-W$ the first order is already able to lift the band degeneracy.*

compensate for the divergence of g^n:

$$\rho(E) = -\frac{1}{\pi}\mathrm{Im}\int \frac{d^3q}{(2\pi)^3} G(E,\boldsymbol{q}) \tag{1.161}$$

$$= -\frac{1}{\pi}\mathrm{Im}\sum_n c^{n-1}\int \frac{dE_q}{(2\pi)^3}\frac{V(E_k)^{n-1}\sqrt{E_q}}{(E-E_q+i\eta)^n}. \tag{1.162}$$

Knowing that V has a finite value at the critical point, the integral converges for $n=1$ and diverges for all the other orders. The VCA, as do most disorder approximations, transforms the expression into a Dyson equation, which is analytic at critical points:

$$\langle G\rangle = g + g\Sigma\langle G\rangle \simeq g + gcV\langle G\rangle\,, \tag{1.163}$$

then $\Sigma_{\text{VCA}} = cV$ is the VCA self-energy and

$$\left\langle G_{jj'}(\omega,\boldsymbol{q})\right\rangle = (\omega^2 - \omega^2_{\boldsymbol{q}j} - cV^{\boldsymbol{q}j'}_{\boldsymbol{q}j} + i\eta)^{-1}. \tag{1.164}$$

Here the inverse operator acts on the full (j,j') space. The scattering rates are given by the imaginary part of the self-energy, but since the potential comes from real force constants, it is Hermitian in reciprocal space and then $\mathrm{Im}\,\Sigma_{\text{VCA}} = 0$. Its role is to shift the band energies by a $\boldsymbol{q}$–dependent term, but it cannot give rise to a finite lifetime. This approximation is valid in the limit of small scattering strength, since Eq. (1.159) holds exactly for $n=1$ and yields more severe suppressions as n increases. In this limit, V^n decreases rapidly and gives less weight to the suppressed terms.

1.4.2 Average T-matrix Approximation (ATA)

For many materials, the single-particle perturbation series diverges near impurity states or resonances. The weak-scattering VCA is therefore inadequate in this regime. The ATA instead solves the single-impurity problem exactly, through the T-matrix $t = V + Vgt$, and approximates the self-energy of a dilute system as a concentration-weighted form of this full single-impurity scattering. Indeed, when the average distance between impurities is large, we can expect a fairly localized phonon packet to have a small probability to find two different impurities before losing its spatial coherence, due to mode-dependent scattering. For a single mode, which is completely delocalized and it has intuitively similar amplitudes for each defective site: the phonon itself does not seem to be affected by the average impurity distance, or by the concentration. This paradox highlights the role of the averaging process over the phase factors, which is responsible for the concentration dependence in each scattering term. For a fixed configuration, the phonon has the same amplitude for scattering from the same defect twice or from two different defects. The number of single defects is d, while the number of pairs of defects is d^2. However, the phase factor of the former in Eq. (1.136) cancels out constructively, while the latter in Eq. (1.137) suppresses the scattering $\boldsymbol{q} \neq \boldsymbol{q'}$. As a result, the final amplitudes are c and c^2, respectively. This is the reason why the single scattering term is dominant at low concentrations, and why we can expect the self-energy to be proportional to the T-matrix of a single impurity, with a prefactor given by the concentration.
The ATA starts by treating exactly the multiple scattering on a single defect, by expanding the Green's function as:

$$\langle G \rangle = g + g \langle t \rangle g + g \left\langle t g' t \right\rangle g + ... \tag{1.165}$$

In real space, the new operator g' can be seen as the propagator that pushes the phonon away from the defect in which it has already fully scattered in the previous expansion order. Formally, it is $g' = g - g(\mathbf{0})$, where $g(\mathbf{0}) = g(\boldsymbol{T}, \boldsymbol{T}) = g(\boldsymbol{T} - \boldsymbol{T})$ and $\boldsymbol{T}$ is the defect position. Within the ATA, the decoupling now involves the single T-matrix t, instead of the potential V as in Eq. (1.159):

$$\langle t(1,2)...t(n,1) \rangle \simeq \langle t(1,2) \rangle ... \langle t(n,1) \rangle . \tag{1.166}$$

The approximation excludes the possibility of a phonon fully scattering from a site, then from another and finally returning to the first, since the phases are averaged out independently. It is already a significant improvement with respect to the VCA, since it treats exactly the single impurity scattering. The Green's function is then

$$\langle G \rangle = g + g[\langle t \rangle (1 - g' \langle t \rangle)^{-1}] g = g + g \langle T \rangle g, \tag{1.167}$$

where $\langle T \rangle$ is the average T-matrix of the entire scattering process. The self-energy is found by inverting Eq. (1.116), giving:

$$\Sigma_{\text{ATA}} = \langle T \rangle (1 + g \langle T \rangle)^{-1} \tag{1.168}$$
$$= \langle t \rangle (1 - g' \langle t \rangle)^{-1} (1 + g \langle t \rangle (1 - g' \langle t \rangle)^{-1})^{-1} \tag{1.169}$$
$$= \langle t \rangle [(1 + g \langle t \rangle (1 - g' \langle t \rangle)^{-1})(1 - g' \langle t \rangle)]^{-1} \tag{1.170}$$
$$= \langle t \rangle [1 - g' \langle t \rangle + g \langle t \rangle]^{-1} \tag{1.171}$$
$$= \langle t \rangle [1 + g(\mathbf{0}) \langle t \rangle]^{-1}. \tag{1.172}$$

The average of the single-defect T-matrix is simply rescaled by the concentration. We shall use the defect Green's function $g(\mathbf{0})$ in the expansion, since only one defect is involved. The self-energy now becomes:

$$\Sigma_{\text{ATA}} = cV(1 - g(\mathbf{0})V)^{-1}(1 + g(\mathbf{0})cV(1 - g(\mathbf{0})V)^{-1})^{-1} \tag{1.173}$$
$$= cV(1 - g(\mathbf{0})V)^{-1}(1 - g(\mathbf{0})V)(1 - g(\mathbf{0})V + cg(\mathbf{0})V)^{-1} \tag{1.174}$$
$$= cV(1 - (1 - c)g(\mathbf{0})V)^{-1}. \tag{1.175}$$

The solution is equal to Eq. (1.144), apart from the new $g(\mathbf{0})$ instead of g. The full potential also involves force-constant changes and is non-zero in the whole region around the defect: the phonon must be able to scatter from $\boldsymbol{R}$ to $\boldsymbol{R'}$ inside the support of the potential and still be considered as scattered by a single defect. For this reason, the true Green's function should have two indices, although remembering that we are dealing with one single defect. The ATA correction slightly shifts the poles of t by $(1 - c)$; this is also called the overcounting correction. By using the full g in Eq. (1.165) we would have had $\Sigma_{\text{ATA}} = ct$, similar to Eq. (1.145). There is another clever way to obtain the overcounting corrections, directly from Eq. (1.165). With reference to Fig. (1.8), after having summed up the single defect diagrams (1), (2a), (3a), etc., we should subtract all the multi-defect diagrams at same order that fall into the single defect ones after substituting g with $g(\mathbf{0})$:

$$\langle G \rangle = g + g \langle t \rangle g + g \langle tg't \rangle g + ... \tag{1.176}$$
$$= g + g \langle t \rangle g + g \langle tgt \rangle g - g \langle tg(\mathbf{0})t \rangle g + ... \tag{1.177}$$

The last term represents the correction of two defect diagrams as (2b), (3c), (3d), etc., that become the single defect diagram (2a) if the phonon does not propagate to the next defect. For this reason, they must be taken once and should be then subtracted from the self-energy. Following this recipe,

$$\Sigma = \langle t \rangle - \Sigma g(\mathbf{0})\Sigma - \Sigma g(\mathbf{0})\Sigma g(\mathbf{0})\Sigma - ... \tag{1.178}$$
$$= \langle t \rangle (1 + \langle t \rangle g(\mathbf{0}))^{-1}. \tag{1.179}$$

This last equation is also found in Eq. (1.172) and the same algebra can be repeated to arrive at the same conclusions.

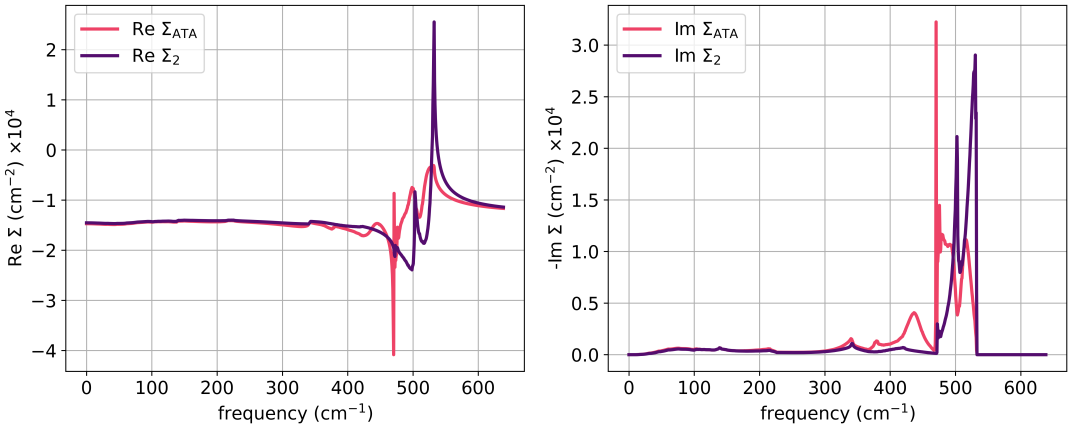


Figure 1.12: *Comparison between ATA self-energy and the one obtained by summing diagrams up to the second order for 1% P-doped Si. New features emerge in the optical range around 470 cm^{-1}, in the proximity of a resonance due probably to mass disorder. In the higher frequency range the scattering is suppressed in the ATA, where the Born series is diverging.*

The ATA self-energy now acquires an imaginary part, due to the presence of the bare propagator. It is written in reciprocal space as:

$$\Sigma_{jj'}(\omega, \boldsymbol{q}) = \sum_{\boldsymbol{q}'j''} c V_{\boldsymbol{q}j}^{\boldsymbol{q}'j''} (A^{-1})_{\boldsymbol{q}'j''}^{\boldsymbol{q}j'}. \tag{1.180}$$

It is the momentum-diagonal part of the matrix product between the single defect potential V and the inverse of the matrix

$$A_{\boldsymbol{q}j}^{\boldsymbol{q}'j'} = 1 - (1-c) g_j(\omega, \boldsymbol{q}) V_{\boldsymbol{q}j}^{\boldsymbol{q}'j'}. \tag{1.181}$$

The computational cost of this direct formula makes it difficult to apply, since a good description of g requires a large $\boldsymbol{q}$-point grid and the inversion scales as the cube of the matrix dimension. Moreover, we could not take advantage of symmetries to reduce the grid into its irreducible zone, since we would need the whole matrix to perform the inversion. Mixed real-reciprocal-space methods are discussed in the next section; they are able to reduce the degrees of freedom of the system without losing accuracy.

1.4.3 Coherent Potential Approximation (CPA)

The coherent potential approximation (CPA) is a widely used self-consistent method for treating disorder. At each iteration, the propagator obtained from the current self-energy is inserted back into the scattering expansion. The procedure is repeated until

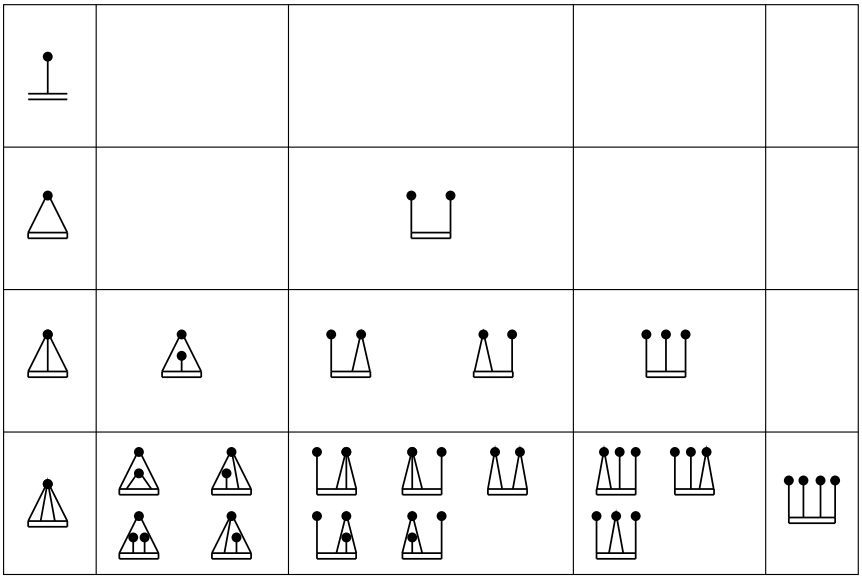


Figure 1.13: *Diagrams of the CPA expansion up to fourth order, adapted from [37]. The first column is the bare self-energy of Eq.* (1.183)*, while the other columns are the overcounting correction that must be subtracted from the bare self-energy. Each row is a different expansion order; the first two columns are the irreducible diagrams of the self-energy, while the following columns have one more irreducible diagram moving to the right.*

the propagator and self-energy converge:

$$G = g(1 - \Sigma g)^{-1} \tag{1.182}$$

$$\Sigma = cV(1 - GV)^{-1}. \tag{1.183}$$

This approach does not remove overcounting, even after introducing the ATA correction, since there are now more diagrams to subtract in the overcounting correction. Historically, the CPA introduces an effective medium with its own propagator $\mathcal{G}$, obtained by the interaction of the particle with the medium:

$$\mathcal{G} = g + g\Sigma\mathcal{G}. \tag{1.184}$$

Starting from the effective medium, the full propagator feels the true configuration potential V^α at each site. Thus, at a defective site the particle feels the potential $V - \Sigma$, and $-\Sigma$ otherwise.

$$\langle G \rangle = \mathcal{G} + \mathcal{G} \langle (V_\alpha - \Sigma) G \rangle . \tag{1.185}$$

If $\mathcal{G}$ is able to reproduce the entire system scattering, then

$$\langle G \rangle = \mathcal{G} + \mathcal{G} \langle T \rangle \mathcal{G} \tag{1.186}$$

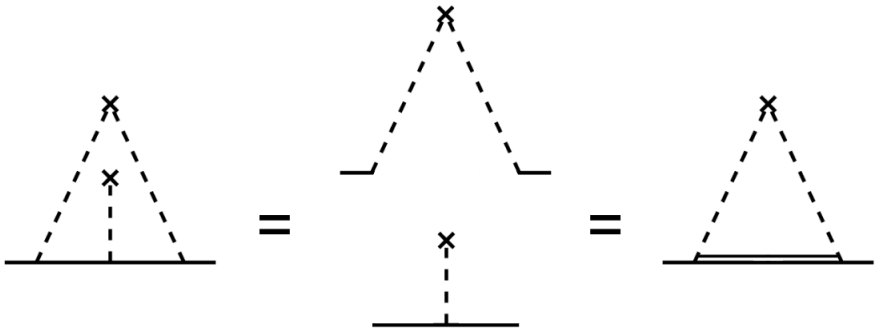


Figure 1.14: *Diagram (3e) of Fig.* (1.8)*, in which the first order propagator* $G^{(1)}$ *in the center-bottom figure is first calculated and then plugged into the diagram as the full double line propagator.*

and then $\langle T \rangle$ must be zero. For a certain configuration α, the T-matrix consists in summing over all sites the single site T-matrix t_α:

$$\langle T_\alpha \rangle = \left\langle \sum_i t_{\alpha,i} + \sum_{i,j \neq i} t_{\alpha,i} \mathcal{G} t_{\alpha,j} + \dots \right\rangle = 0. \tag{1.187}$$

The approximation consists in requiring the configurational average of each single-site T-matrix to vanish:

$$\langle t_\alpha \rangle = \frac{-\Sigma(1-c)}{1+\Sigma\mathcal{G}} + \frac{(V-\Sigma)c}{1-(V-\Sigma)\mathcal{G}} = 0. \tag{1.188}$$

By multiplying by $1-(V-\Sigma)\mathcal{G}$ we obtain:

$$\Sigma = cV(1-(1-c)\mathcal{G}(1+\Sigma\mathcal{G})^{-1}V)^{-1}. \tag{1.189}$$

After some algebra, the self-energy can be isolated and written as:

$$\Sigma = \frac{cV}{1-(V-\Sigma)\mathcal{G}}, \tag{1.190}$$

where the quantities V and $\mathcal{G}$ are single-site quantities, since they come from the single site T-matrix. In particular, V is the potential given by force constants, centered at the site; $\mathcal{G} = \mathcal{G}(\mathbf{0})$ is to be intended as the local propagator, in which the phonon starts and ends at the same site. We could also start from the self-energy expansion of Eq. (1.183) and remove the overcounting by hand, as in the ATA. The source of overcounting comes from the self-consistency, since the full propagator of Eq. (1.182) should be instead:

$$G = g + g\Sigma g + g\Sigma g' \Sigma g + \dots \tag{1.191}$$

where $g' = g - g(\mathbf{0})$, since the case in which the particle scatters twice on the same defect in the third term is already included in the second term. With reference to

Fig. (1.13), the first column represents the bare self-energy, while the other are the overcounting correction that must be subtracted. We introduce a new propagator $\gamma = G + G\Sigma G + \cdots = G(1 - \Sigma G)^{-1}$ that, once plugged into the self-energy $\Sigma[\gamma]$, gives all the diagrams with any number of insertions. These diagrams are summarized by summing the first two columns of Fig. (1.13), where the first has no insertion and the second one or more insertions. The second column is then $\sigma_2 = -\Sigma[\gamma] + \Sigma[G]$, where we put a minus sign since we're removing the overcounting. The third column has two irreducible diagrams, hence $\sigma_3 = -\Sigma[\gamma]G\Sigma[\gamma]$. By summing all the columns, the self-energy becomes:

$$\Sigma[G] = cV(1 - GV)^{-1} - \frac{\Sigma[\gamma]}{1 - \Sigma[\gamma]G} + \Sigma[G], \tag{1.192}$$

which gives, by isolating $\Sigma[\gamma]$:

$$\Sigma[\gamma] = cV(1 - (1 - c)GV)^{-1}. \tag{1.193}$$

Now we would need the explicit expression of Σ as a function of the full propagator. Since we know the functional dependence of Σ we can make the following substitution:

$$\gamma \to G \tag{1.194}$$

$$G \to G(1 + \Sigma G)^{-1} \tag{1.195}$$

which gives the same expression of Eq. (1.189), by substituting G with the medium propagator $\mathcal{G}$. This series of substitutions is not harmful for two reasons: the self-energy has been set since the beginning as a functional of a generic propagator, we can then plug the one we chose to represent the system afterwards. The second reason is that the medium propagator is supposed to converge to the full one, so they can be considered as interchangeable in the limit of convergence.

The central feature of the CPA is the locality of the condition in Eq. (1.188), which defines a local self-energy $\Sigma(\omega)$. This approximation is well suited to the mass-disorder case, but the force constant disorder is usually spreaded across multiple sites. The ATA self-energy is also supposed to be local, but it has the advantage of being a function of the single site T-matrix only. The single site problem can be solved by plugging one single defect in a large supercell and calculating the whole non-local effects due to the presence of the defect, encapsulated in the single site T-matrix. The same is not possible in the previous CPA formulation, since the single site T-matrix in Eq. (1.188) is really a site quantity. Placing a defect has the effect of creating a cluster of size equal at least to the support of the potential, in which the self-consistent equation must be applied to retain a non-local self-energy $\Sigma(\boldsymbol{q}, \omega)$. Various non-local formulations of the CPA have been proposed in the literature, such as the Itinerant CPA (ICPA), and the Dynamical Cluster Approximation (DCA). The ICPA works by extending the space in which the Hamiltonian acts with a new disorder field, making it easier to perform averages. This new space includes all possible configurations of the system, including the virtual crystal state, in which each site is prepared in a superposition of

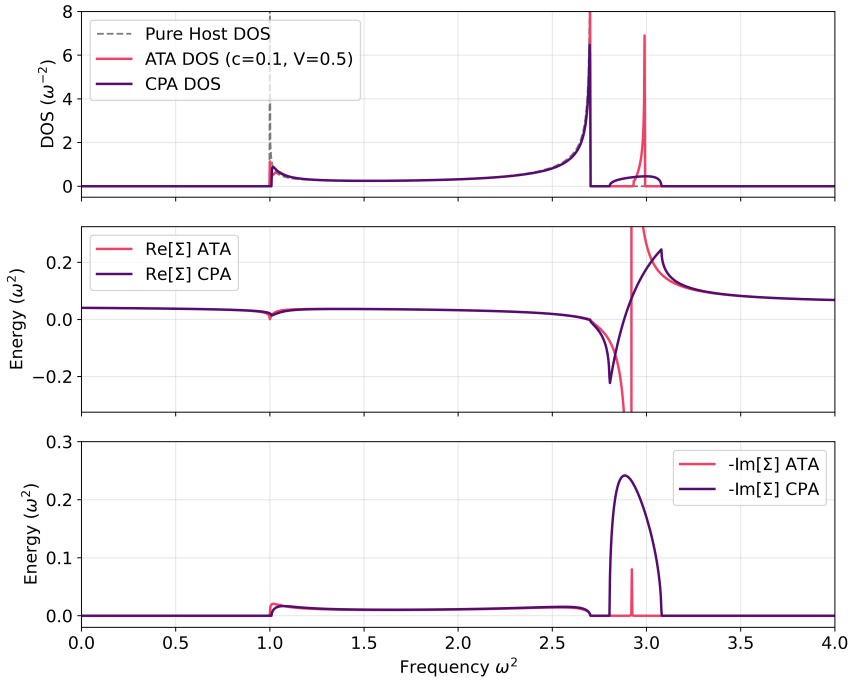


Figure 1.15: *Comparison between ATA and CPA for a 1D linear chain of exponentially decaying force constants and a scalar potential. The ATA is known to produce a peaked DOS in the proximity of the localized state, around 3 in arbitrary units. The CPA has a broader self-energy and shapes the impurity state into a band. The reason for this behaviour is that in the CPA the impurity is no longer isolated and can interact with the impurities contained in the effective medium, behaving like isolated atoms brought in contact to form a material with its own unique band structure.*

being host or defective according to the concentration. The rest of the space represents the fluctuations with respect to the virtual crystal state. The approximation consists in treating exactly only one fluctuation, leaving the others as a mean field. For an alloy of A and B atoms, the force constants Φ_{ij} take the value Φ_{ij}^{AA}, Φ_{ij}^{AB} or Φ_{ij}^{BB} according to the configuration of sites i and j. We could in principle exactly know the AA and BB force constants by performing a DFT calculation of phonons for the species A and B. Similarly, for the cross term AB we could use the force constants in a supercell with a certain number of A and B atoms, but the value would depend on the concentration of the two species and their spatial configuration. Moreover, the force constant between two equal sites is modified by the presence of the defective species. Since our potential already contains this information, we opted for cluster theories — such as the DCA — to grasp the non-local effects of the disorder.

1.4.4 Dynamical cluster approximation (DCA)

The dynamical cluster approximation (DCA) retains the self-consistency of the CPA but replaces its single site with a cluster embedded in an effective medium. The disorder average is evaluated explicitly over a subset of defect configurations. The cluster is a supercell of N_c cells, in which the configuration potential V_α is constructed by translating the centered single-defect potential to each defective site. As shown in Eqs. (1.66) and (1.67), this construction is most convenient in reciprocal space. Each lattice vector is decomposed into $\boldsymbol{R}$, which locates a cell within the cluster, and $\tilde{\boldsymbol{R}}$, which locates the cluster itself. In the unit-cell Brillouin zone, the corresponding momentum is $\boldsymbol{Q}+\tilde{\boldsymbol{q}}$: $\boldsymbol{Q}$ labels the centers of N_c momentum patches, whereas $\tilde{\boldsymbol{q}}$ spans the patch $p(\boldsymbol{Q})$. This construction yields a $\boldsymbol{Q}$-dependent generalization of the local CPA propagator $\mathcal{G}(\boldsymbol{0})$. The correlated diagrams, such as the one in Fig. (1.10) are treated explicitly within the cluster; we therefore expect a self-energy with a range set by the cluster size in real space. In terms of amplitudes, we are approximating the scattering in the figure explicitly written in Eq. (1.158) with a coarse-grained version of the momentum conservation:

$$\delta(\boldsymbol{q_2}-\boldsymbol{q_1}+\boldsymbol{q_4}-\boldsymbol{q_3}) \approx \delta(\boldsymbol{Q_2}-\boldsymbol{Q_1}+\boldsymbol{Q_4}-\boldsymbol{Q_3}). \tag{1.196}$$

For this reason, the cluster must be chosen large enough to capture the relevant correlations, so that the self-energy can be close to zero outside the maximum distance cutoff imposed by the cluster size. The DCA is formally exact in the limit of infinite cluster size, which is equivalent to solving the problem in a system with randomly distributed defects. The generalized propagator $\mathcal{G}$ is set to be:

$$\mathcal{G}_{jj'}(\boldsymbol{Q},\omega) = \int_{p(\boldsymbol{Q})} (\omega^2 - \omega^2_{\boldsymbol{q}j} - \Sigma_{jj'}(\boldsymbol{Q},\omega))^{-1} d\boldsymbol{q} \tag{1.197}$$

The self-energy in this approximation is a $\boldsymbol{Q}$-dependent constant function within each patch. The average Green's function is a function of a new propagator, called the cluster excluded propagator $\mathcal{G}_0$, defined by removing the medium contribution from the coarse-grained effective-medium propagator $\mathcal{G}$:

$$\mathcal{G}_{0,jj'}(\boldsymbol{Q},\omega)^{-1} = \mathcal{G}_{jj'}(\boldsymbol{Q},\omega)^{-1} + \Sigma_{jj'}(\boldsymbol{Q},\omega). \tag{1.198}$$

The effective medium propagator $\mathcal{G}$ describes a particle feeling the average effect of the disorder at each site, while the cluster excluded $\mathcal{G}_0$ feels the effect of the medium only outside the cluster, since we want to count it inside it by performing an explicit average over the configurations. The average Green's function is:

$$\left\langle G_{jj'}(\boldsymbol{Q},\omega)\right\rangle = \left\langle (\mathcal{G}_{0,jj'}(\boldsymbol{Q},\omega)^{-1}\delta_{\boldsymbol{Q}\boldsymbol{Q'}} - V^{\boldsymbol{Q'}j'}_{\alpha,\boldsymbol{Q}j})^{-1}\right\rangle_\alpha \delta_{\boldsymbol{Q}\boldsymbol{Q'}}, \tag{1.199}$$

where we impose the translational invariance by taking the diagonal elements in $\boldsymbol{Q}$ space. This operation is necessary since we're taking the average only over a small subset of the possible configurations. This subset must be chosen large enough so that the symmetry-breaking off-diagonal elements are negligible. The self-consistency

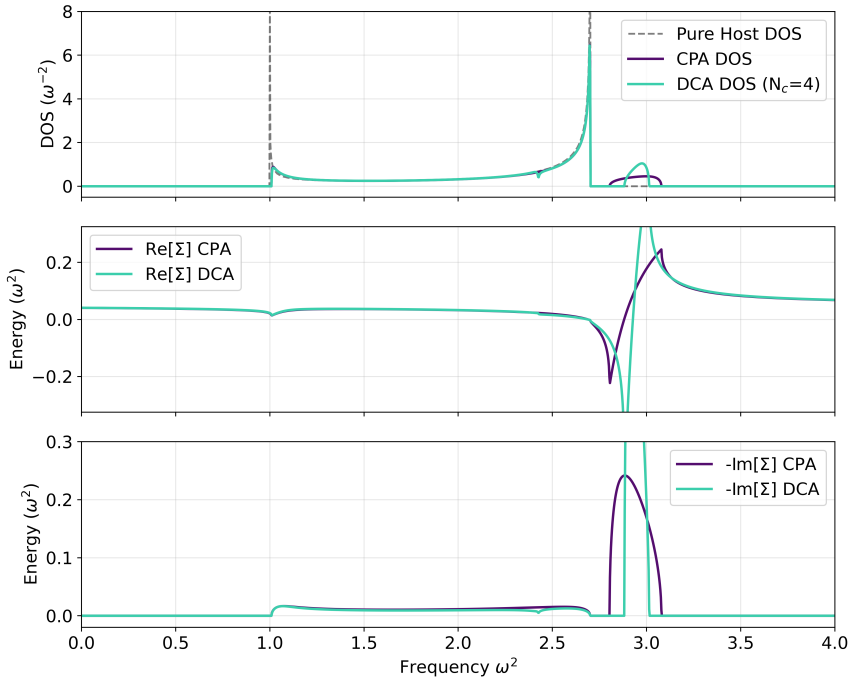


Figure 1.16: *Comparison between DCA and CPA for the same model of Fig.* (1.15). *On the two bottom panels, we plotted the self-energy for* $\boldsymbol{Q} = \mathbf{0}$.

condition can be imposed on the Green's function or the self-energy. In the first case, it translates into $\langle G \rangle = \mathcal{G}$, while in the second we extract the new self-energy as

$$\Sigma_{jj'}(\boldsymbol{Q}, \omega) = \mathcal{G}_{0,jj'}(\boldsymbol{Q}, \omega)^{-1} - \left\langle G_{jj'}(\boldsymbol{Q}, \omega) \right\rangle^{-1} \tag{1.200}$$

and terminate the iteration when $\Sigma_{in} = \Sigma_{out}$ within a chosen tolerance.
We recover the CPA in the limit $N_c = 1$, with a local potential $V_{\alpha,jj'} = V_{jj'}$ with probability c. The average Green's function becomes:

$$\mathcal{G} = \langle G \rangle = \left\langle (\mathcal{G}^{-1} - V_{\alpha,jj'} + \Sigma)^{-1} \right\rangle_{\alpha} \tag{1.201}$$

$$= c(\mathcal{G}^{-1} - V + \Sigma)^{-1} + (1-c)(\mathcal{G}^{-1} + \Sigma)^{-1} \tag{1.202}$$

which is indeed equivalent to Eq. (1.190). We can also recover the ATA in the limit of small concentration, by taking only configurations with zero or one defect in the cluster and by averaging as before. To obtain a non-local result, we should make the average in the entire BZ, by taking the cluster size equal to the grid size. In this case, the measure of $p(\boldsymbol{Q})$ is trivially zero, the coarsening procedure is suppressed entirely: $\mathcal{G}$ becomes the full propagator G and $\mathcal{G}_0$ becomes the bare propagator g. The average Green's function

is then:

$$\langle G \rangle = \left\langle (g^{-1} - V_\alpha)^{-1} \right\rangle_\alpha \tag{1.203}$$
$$= c(g^{-1} - V)^{-1} + (1-c)g \tag{1.204}$$
$$= g + gcV(1 - gV)^{-1} g \tag{1.205}$$
$$= g + g \langle T \rangle g, \tag{1.206}$$

which gives the ATA self-energy of Eq. (1.175) by using Eq. (1.168).
The correlations have a characteristic decay over a certain correlation length, that should be smaller than the cluster size to capture all the relevant effects. The resulting self-energy is a translationally invariant, fast-decaying function in real space: these two properties allow us to perform Fourier interpolation with the usual algorithm used for force constants, adapted for complex variables. In other words, the matrix elements of the self-energy can be expressed through N_c basis functions, with accuracy given by the correlation length. By using a single index η for cartesian and atomic indices:

$$\Sigma_\eta(\boldsymbol{q}, \omega) = \sum_{\boldsymbol{Q}} c_{\eta \boldsymbol{Q}}(\boldsymbol{q}) \Sigma_\eta(\boldsymbol{Q}, \omega), \tag{1.207}$$
$$c_{\eta \boldsymbol{Q}}(\boldsymbol{q}) = \sum_{\boldsymbol{R}_\eta} e^{i(\boldsymbol{Q}-\boldsymbol{q})\cdot \boldsymbol{R}_\eta} \tag{1.208}$$

where $\boldsymbol{R}_\eta = \boldsymbol{R}_{\tau\tau'} = \boldsymbol{R} + \boldsymbol{T}_{\tau\tau'}$ and $\boldsymbol{T}_{\tau\tau'}$ is the translation vector of the cluster chosen to minimize $||\boldsymbol{R} + \boldsymbol{T} + \boldsymbol{\tau} - \boldsymbol{\tau'}||$. The continuous self-energy can then be used in Eq. (1.197) to obtain a better estimate of the coarsened Green's function, while the patch average of the self-energy $\bar{\Sigma}$ replaces the coarsened self-energy in Eq. (1.198). These patch averages can be seen as integrals over the entire BZ of an indicator function that specifies which patch is integrated, as:

$$\phi_{\boldsymbol{Q}}(\boldsymbol{q}) = \begin{cases} 1 & \text{if } \boldsymbol{q} \in p(\boldsymbol{Q}) \\ 0 & \text{otherwise} \end{cases} \tag{1.209}$$

This function has the following properties:

$$\phi_{\boldsymbol{Q}}(\boldsymbol{q}) = \phi_{\boldsymbol{0}}(\boldsymbol{q} - \boldsymbol{Q}) := \phi(\boldsymbol{q} - \boldsymbol{Q}) \tag{1.210}$$
$$\phi(\boldsymbol{q} - \boldsymbol{Q}) = \phi(\boldsymbol{Q} - \boldsymbol{q}). \tag{1.211}$$

The coarsening procedure becomes a convolution with this new indicator function, written as:

$$\mathcal{G}_\eta(\boldsymbol{Q}, \omega) = \int \phi(\boldsymbol{q} - \boldsymbol{Q}) G_\eta(\boldsymbol{q}, \omega) d\boldsymbol{q} \tag{1.212}$$
$$\bar{\Sigma}_\eta(\boldsymbol{Q}, \omega) = \int \phi(\boldsymbol{q} - \boldsymbol{Q}) \Sigma_\eta(\boldsymbol{q}, \omega) d\boldsymbol{q} \tag{1.213}$$

By using Eq. (1.207) we can convolve the elements of the basis once, saving useful computational time:

$$\bar{\Sigma}_\eta(\boldsymbol{Q},\omega) = \sum_{\boldsymbol{Q'}} \left(\int \phi(\boldsymbol{q}-\boldsymbol{Q}) c_{\eta\boldsymbol{Q'}}(\boldsymbol{q}) d\boldsymbol{q} \right) \Sigma_\eta(\boldsymbol{Q'},\omega) \tag{1.214}$$

$$:= \sum_{\boldsymbol{Q'}} P^\eta_{\boldsymbol{Q}\boldsymbol{Q'}} \Sigma_\eta(\boldsymbol{Q'},\omega), \tag{1.215}$$

where P, defined as the matrix inside the braces, is the operator that performs the convolution. Unfortunately, the average over a patch of an interpolated function is not equal to the function at the center of the patch, in other words $\bar{\Sigma} \neq \Sigma$. If we use $\bar{\Sigma}$ to estimate the cluster excluded propagator, the cluster solver will output a new self-energy $\bar{\Sigma}_{out}$ that reflects this mismatch and introduces spurious oscillations across cycles. At convergence, the output of the cluster solver $\Sigma(\boldsymbol{Q})$ should be equal to the convolution $\bar{\Sigma}(\boldsymbol{Q})$ of a deconvoluted lattice self-energy $\Sigma'(\boldsymbol{q})$, defined implicitly as:

$$\Sigma_\eta(\boldsymbol{q},\omega) = \sum_{\boldsymbol{q'}} P^\eta_{\boldsymbol{q}\boldsymbol{q'}} \Sigma'_\eta(\boldsymbol{q'},\omega). \tag{1.216}$$

The convolution operator has been extended to accept vectors in the entire BZ, by using the function $\phi(\boldsymbol{q}-\boldsymbol{q'})$ as a kernel. There are several algorithms to perform the deconvolution in the literature; the first authors of this DCA extension [38] used the Richardson–Lucy algorithm. It is best suited to preserve the causality of the self-energy, since it conserves the sign of strictly positive and negative functions. The main issue in our case is that the self-energy is a matrix and causality is a global property of the matrix. In principle, we could invert the relation in Eq. (1.216) if the $\boldsymbol{q}$-grid is not too dense to make the operation impractical and especially if the convolution operator is invertible. This operator can be written as:

$$P_{\boldsymbol{q}\boldsymbol{q'}} = \int \phi(\boldsymbol{k}-\boldsymbol{q}) \sum_{\boldsymbol{R}} e^{i(\boldsymbol{q'}-\boldsymbol{k})\cdot\boldsymbol{R}} d\boldsymbol{k} \tag{1.217}$$

$$= \sum_{\boldsymbol{R}} e^{i\boldsymbol{q'}\cdot\boldsymbol{R}} \int \phi(\boldsymbol{k}-\boldsymbol{q}) e^{-i\boldsymbol{k}\cdot\boldsymbol{R}} d\boldsymbol{k} \tag{1.218}$$

$$= \sum_{\boldsymbol{R}} e^{i(\boldsymbol{q'}-\boldsymbol{q})\cdot\boldsymbol{R}} \int \phi(\boldsymbol{k}) e^{-i\boldsymbol{k}\cdot\boldsymbol{R}} d\boldsymbol{k} \tag{1.219}$$

$$= UWU^{-1}, \tag{1.220}$$

where U is the phase matrix and $W(\boldsymbol{R})$ is a diagonal matrix which eigenvalues are defined as the Fourier transform of the indicator function:

$$W(\boldsymbol{R}) = \int \phi(\boldsymbol{k}) e^{-i\boldsymbol{k}\cdot\boldsymbol{R}} d\boldsymbol{k}. \tag{1.221}$$

The patch is a box of size $\Delta k_x \times \Delta k_y \times \Delta k_z$ in crystal coordinates, so the eigenvalues of W are given by the product of three sinc functions:

$$W(\boldsymbol{R}) = \prod_{i=x,y,z} \int_{-\Delta k_i/2}^{\Delta k_i/2} e^{-ik_i R_i} dk_i \tag{1.222}$$

$$= \prod_{i=x,y,z} \Delta k_i \,\mathrm{sinc}(\Delta k_i R_i/2). \tag{1.223}$$

Defining the cluster grid as composed of $n_x \times n_y \times n_z = N_c$ points, $\Delta k_i = b_i/n_i$ is a submultiple of the reciprocal lattice vector b_i, while $R_i = m_i a_i$ is a multiple of the direct lattice vector a_i. The sine function evaluates to zero if m_i is a multiple of n_i. The self-energy is Fourier interpolated on the same real grid composed of N_c points, but through the minimal image method we chose the lattice vector $\boldsymbol{R}$ which minimizes the distance between the two atoms, given a certain matrix element. If the BZ is not extremely stretched, this method picks usually m_i contained in $[-n_i/2, n_i/2]$, so that the eigenvalues of W are non-zero and the convolution operator is invertible. The subset of $\boldsymbol{R}$ outside this interval yields a small self-energy, since the correlation length is chosen to be smaller than the cluster size. The convolution operator projects out these small contributions: in real space, Eq. (1.216) becomes $\Sigma(\boldsymbol{R}) = W(\boldsymbol{R})\Sigma'(\boldsymbol{R}) = \boldsymbol{0}$ for large $\boldsymbol{R}$. We can still perform the inversion on the non-zero subspace and force the remaining eigenvalues to be zero, which is equivalent to neglecting the small contributions coming from outside the cluster. We call this new operator $\tilde{W}$ and define $P = U\tilde{W}U^{-1}$ to completely avoid these divergences. Finally, we can use $\Sigma' = P^{-1}\tilde{\Sigma}$ to obtain a deconvoluted cluster self-energy:

$$\Sigma'_\eta(\boldsymbol{Q}, \omega) = \sum_{\boldsymbol{Q'}} (P^{-1})^\eta_{\boldsymbol{Q}\boldsymbol{Q'}} \Sigma_\eta(\boldsymbol{Q'}, \omega). \tag{1.224}$$

The two main equations of this improved DCA method are:

$$\Sigma'_\eta(\boldsymbol{q}, \omega) = \sum_{\boldsymbol{Q}} c_{\eta\boldsymbol{Q}}(\boldsymbol{q}) \Sigma'_\eta(\boldsymbol{Q}, \omega) \tag{1.225}$$

$$\mathcal{G}_\eta(\boldsymbol{Q}, \omega) = \int \phi(\boldsymbol{q} - \boldsymbol{Q})(\omega^2 - D_\eta(\boldsymbol{q}) - \Sigma'_\eta(\boldsymbol{q}, \omega))^{-1} d\boldsymbol{q} \tag{1.226}$$

$$\mathcal{G}_{0,\eta}(\boldsymbol{Q}, \omega)^{-1} = \mathcal{G}_\eta(\boldsymbol{Q}, \omega)^{-1} + \Sigma_\eta(\boldsymbol{Q}, \omega). \tag{1.227}$$

The last equation holds since

$$\bar{\Sigma}_\eta(\boldsymbol{Q}, \omega) = \int \phi(\boldsymbol{q} - \boldsymbol{Q}) \Sigma'_\eta(\boldsymbol{q}, \omega) d\boldsymbol{q} \tag{1.228}$$

$$= \sum_{\boldsymbol{Q'}} P^\eta_{\boldsymbol{Q}\boldsymbol{Q'}} \Sigma'_\eta(\boldsymbol{Q'}, \omega) \tag{1.229}$$

$$= \sum_{\boldsymbol{Q'}\boldsymbol{Q''}} P^\eta_{\boldsymbol{Q}\boldsymbol{Q'}} (P^{-1})^\eta_{\boldsymbol{Q'}\boldsymbol{Q''}} \Sigma_\eta(\boldsymbol{Q''}, \omega) \tag{1.230}$$

$$= \Sigma_\eta(\boldsymbol{Q}, \omega). \tag{1.231}$$

The property of causality is guaranteed for retarded quantities by the analyticity in the upper half-plane and by a strictly negative imaginary part of the self-energy

$$\langle v|\,\mathrm{Im}\,\Sigma\,|v\rangle := \frac{1}{2i}\,\langle v|\,\Sigma - \Sigma^{\dagger}\,|v\rangle < 0 \quad \forall\,|v\rangle \tag{1.232}$$

Each DCA step has been proven to preserve the causality of the self-energy. The only possible source of non-causality is the deconvolution step. The convolution operator $W(\boldsymbol{R})$ is a real function, so it preserves the sign of the imaginary part of the self-energy. In real space,

$$\mathrm{Im}\,\Sigma'(\boldsymbol{R}) = \mathrm{Im}[W^{-1}(\boldsymbol{R})\Sigma(\boldsymbol{R})] = W^{-1}(\boldsymbol{R})\,\mathrm{Im}\,\Sigma(\boldsymbol{R}) < 0 \tag{1.233}$$

if all the eigenvalues of W are positive, which is true in the region where $\Delta k_i R_i < 2\pi$. We could simply set the eigenvalues to zero outside this region, but it was already shown that the inequality is respected in the borders of the cluster, in which the self-energy approaches zero. The Fourier interpolation can be also a source of non-causality, since the oscillations in between the points where the cluster self-energy in reciprocal space approaches zero can change sign. In this region, the self-energy can be safely set to zero, since in both cases it's supposed to be negligible. These issues are never observed in practice, after having achieved full convergence of the DCA cycle.

2 Numerical implementations

2.1 Tetrahedron Methods

The Green functions, defined as $g(\omega) = (\omega - H)^{-1}$, have poles near the eigenvalues of the Hamiltonian H by construction. These divergences must be handled with care when performing numerical calculations. The usual way to treat these divergencies is to calculate them near the poles, by shifting the poles along the real axis by a small amount $i\eta$. The new function, also called retarded Green's function, is defined as $g^R(\omega) = g(\omega + i\eta)$. Near an eigenvalue ω_0, the real and imaginary part can be split as:

$$(\omega - \omega_0 + i\eta)^{-1} = \frac{\omega - \omega_0}{(\omega - \omega_0)^2 + \eta^2} - i\frac{\eta}{(\omega - \omega_0)^2 + \eta^2} \tag{2.1}$$

$$= \mathcal{P}\left(\frac{1}{\omega - \omega_0}\right) - i\pi\delta(\omega - \omega_0). \tag{2.2}$$

Both the principal value $\mathcal{P}$ and the delta function δ are defined as distributions in the limit $\eta \to 0$. Integrated quantities over the degrees of freedom of the Green's function, such as the density of states $\rho(\omega)$, yield a well-defined and finite result:

$$\rho(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\int d\boldsymbol{q}\, g(\omega + i\eta, \boldsymbol{q}) \in \mathbb{R}. \tag{2.3}$$

There are two main approaches to perform numerical calculations of the Green's function: the first is to keep a finite η, for which the principal value is well defined and the delta becomes a Lorentzian function. Other functions, such as the Gaussian, are widely used to approximate the imaginary part. These methods introduce an artificial broadening of the peaks in the density of states: the result is a smooth and analytical function, unable to mimic all the features of the system, such as non-analyticities at critical points. The second approach is to replace at each point $\boldsymbol{q}$ the Green's function with its integrated values around $\boldsymbol{q}$:

$$g(\omega, \boldsymbol{q}) \to \frac{1}{\mu(S_{\boldsymbol{q}})}\int_{S_{\boldsymbol{q}}} d\boldsymbol{q}' g(\omega, \boldsymbol{q}'), \tag{2.4}$$

where $S_{\boldsymbol{q}}$ is a small volume around $\boldsymbol{q}$ and $\mu(S_{\boldsymbol{q}})$ is the measure of $S_{\boldsymbol{q}}$. This new quantity, being an integral of a distribution, is well defined and finite. The Hamiltonian is usually defined on a discrete grid of $\boldsymbol{q}$ points. The aim of the integration methods is to find a valid interpolation of the Hamiltonian between the grid points and a geometry for which the integration of the interpolated Green's function is analytically solvable.



Figure 2.1: *Tessellation of a torus with a tetrahedral grid of* 15^3 *(left) and* 30^3 *(right) points. Blue areas are triangles, red areas are quadrilaterals.*

A linear interpolation of a function in a three-dimensional space is fixed by four conditions: one for $f(\boldsymbol{q_0})$ and three for the gradient $\nabla f(\boldsymbol{q_0})$; for this reason, the best suited integration geometry for linear interpolants is a tetrahedron. Denoting by f_i the value of the function at the vertex i of the tetrahedron, the linear interpolation is defined as:

$$f(\boldsymbol{q}) = (1 - x_1 - x_2 - x_3)f_0 + x_1 f_1 + x_2 f_2 + x_3 f_3, \tag{2.5}$$

$$\boldsymbol{q} = (1 - x_1 - x_2 - x_3)\boldsymbol{q_0} + x_1 \boldsymbol{q_1} + x_2 \boldsymbol{q_2} + x_3 \boldsymbol{q_3}, \tag{2.6}$$

with $x_i \in [0, 1]$ and $x_1 + x_2 + x_3 \leq 1$. If $\omega(\boldsymbol{q})$ is the dispersion of a single eigenvalue of the Hamiltonian, the integral we want to solve analytically is:

$$\int d\boldsymbol{q}\,(\omega - \omega(\boldsymbol{q}))^{-1} = \tag{2.7}$$

$$\int_0^1 dx_1 \int_0^{1-x_1} dx_2 \int_0^{1-x_1-x_2} dx_3\,(\omega - \omega(x_1, x_2, x_3))^{-1}. \tag{2.8}$$

The imaginary part is a delta function; it is therefore proportional to the measure of the surface with energy ω. The linear tetrahedron method divides the full Brillouin zone into a set of tetrahedra, then computes the area of the plane defined implicitly by $\omega = \omega(\boldsymbol{q})$, bisecting each tetrahedron. The result is a polygon with three or four vertices, as shown in Fig. (**??**), whose area can be calculated analytically. Better interpolations, such as the quadratic one, require 6 more conditions, which can be fixed by using the value of the function at neighbouring points. The issue is that the measure of a quadratic surface defined by $\omega = \omega_2(\boldsymbol{q})$ is not known in terms of elementary functions, so the integral cannot be solved analytically. A way to improve the linear method is to use a polynomial interpolation of order n to find the plane which minimizes the distance between the plane and the surface $\omega = \omega_n(\boldsymbol{q})$, then calculate the area of the plane as in the linear method, as in [39]. The minimization is performed by using least squares fitting, which is a linear operation in the coefficients defining the polynomial. The same new energies $\tilde{\omega}_i$ can be used to calculate the real part of the Green's function. Since we can integrate each variable separately and since the integral of the inverse of a linear function is a set of rational functions and logarithms, the result is still analytical. The closed formula can be found in [40], while in [41] the authors derive the imaginary

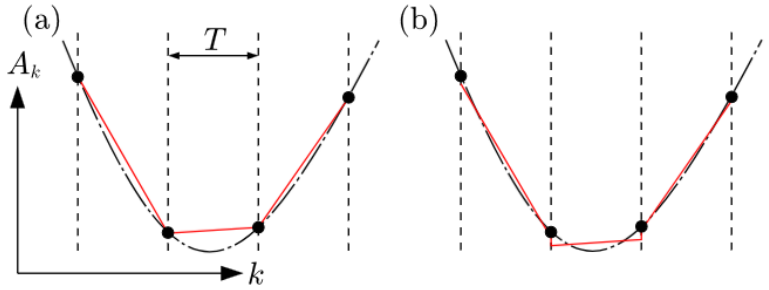


Figure 2.2: *Comparison between the linear (a) and the piecewise linear (b) interpolation of a function A_k, from [39]. The black points indicate the values at the grid points, the dashed lines indicate the boundaries of the tetrahedra. The linear interpolation overestimates convex functions, while the piecewise interpolation obtained by fitting a plane to a high order polynomial interpolant cures this issue.*

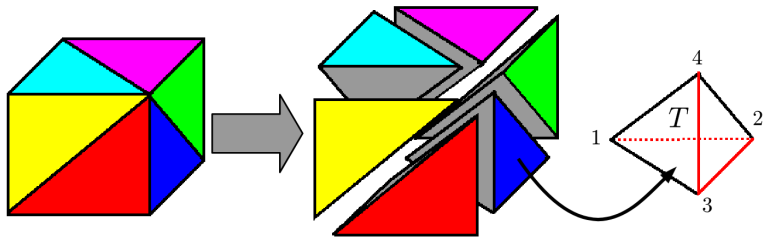


Figure 2.3: *Division of a cube into six tetrahedra, from [39]. The $\boldsymbol{q}$-points sit in the vertices of the cube.*

part starting from the volume integral and then taking the derivative to get the closed formula for the polygon area. In both cases, the contribution can be split into four parts, one for each vertex of the tetrahedron. The integration volume is first split into cubes, with vertices at the points of a uniformly spaced grid, then each cube is split into six tetrahedra by cutting the diagonal of the cube, as shown in Fig. (2.3). Each point then gets contributions from the 24 neighbouring tetrahedra: the integral over the Brillouin zone of a function $A(\boldsymbol{q})$ can be rewritten as:

$$\int_{BZ} d\boldsymbol{q}\, A(\boldsymbol{q}) g(\boldsymbol{q}, \omega) \simeq \sum_{\boldsymbol{q}} w_{\boldsymbol{q}}(\omega) A_{\boldsymbol{q}}, \tag{2.9}$$

where $w_{\boldsymbol{q}}$ is the tetrahedron weight at point $\boldsymbol{q}$. The closed formulas present degenerate cases in which the reference energy coincides with one of the energies at the vertices, or when two vertices have the same energy. The reason is that the logarithmic functions can propagate the machine precision error ϵ. In cases in which $\omega_{ij} := (\omega_i - \omega_j)/(\omega_i +$

$\omega) \to 0$, the real part of the weight is [42]:

$$\frac{\mathcal{O}(\omega_{ij}^2) + \mathcal{O}(\epsilon)}{\omega_{ij}^2}, \tag{2.10}$$

that should converge to a finite value, but diverges due to the presence of ϵ. For this reason, two energies must be considered equal if their difference is smaller than a certain threshold, that must be higher than machine precision. The use of the $\mathrm{Log1p}(x) = x\ln(1+x)/(1+x-1)$ also helps to reduce the error, since for small arguments the precision is $\mathcal{O}(x\epsilon)$ instead of $\mathcal{O}(\epsilon)$. The degeneracy can be also lifted by hand, as in [42]: the authors introduce a small bias ϵ_{tet} to the renormalized energies, so that ω_{ij} cannot be lower than ϵ_{tet}. The integral converges when $\epsilon_{tet}^2 \gg \epsilon$.
The full resolvent G acquires a complex argument due to the presence of the self-energy. In the complex plane, the comparison operators are no longer defined, so the integral must be solved algebraically for both the real and the imaginary part. The degeneracy lifting introduced before needs to order the energies and can no longer be applied, while the use of limit formulas for degenerate cases is still possible, as long as the logarithms are evaluated with the same branch cut. Unfortunately, the recipe for the imaginary part follows a purely geometric construction, that fixes the position of the plane with respect to the tetrahedron. This involves the use of comparisons or the Heaviside function, as in [43], to discriminate the multiple ways in which the plane can intersect the tetrahedron. We could in principle apply Kramers-Kronig relations to get the imaginary part from the real part, but this procedure involves an integral over the whole frequency range, while the real part is still non-zero even far away from the continuum. Moreover, the weights must be calculated for each frequency, even if we could need to explore only a small frequency range. An algebraic formulation for complex resolvents of complex values is given in [44] and [45]. Since the branch cut of the logarithm is fixed to be in the negative real axis, the complex energies at the four vertices of the tetrahedron must not intersect the branch cut; but as long as the self-energy is causal, the eigenvalues will have a negative imaginary part. On the other hand, even small causality violations make the algorithm ill conditioned in the evaluation of the resolvent, hence it must be enforced by hand on the eigenvalues of G^{-1}.

2.1.1 Non-diagonal operators

When the self-energy is not diagonal in the basis of the Hamiltonian, the inversion of the operator $G^{-1}_{jj'}(\boldsymbol{q},\omega) = \omega - \omega_{\boldsymbol{q}j}\delta_{jj'} - \Sigma_{jj'}(\boldsymbol{q},\omega)$ introduces non-linearity. Apart from small matrix cases in which the inversion has a closed form, generally the barycentric coordinates used in the interpolation of $\omega_{\boldsymbol{q}j}$ and Σ become mixed, making an analytical solution unfeasible. We could in principle rotate G into the basis in which G^{-1} is diagonal, but the existence of this rotation is not guaranteed, since Σ is not hermitian. Moreover, if Σ respects the symmetries of the system, the degenerate subspaces of H are still invariant under the action of Σ. The algebraic multiplicity of the subspace is given

by the underlying irreducible representation, which can be reduced by the presence of Σ only if the degeneracy is accidental. If the numerical diagonalization succeds, we would get right and left eigenvectors R and L. The resolvent is represented by the weight:

$$W_{jj'} = \sum_k \frac{R_{jk} L^*_{j'k} w_k}{\langle L_k | R_k \rangle}, \tag{2.11}$$

where w_k is the tetrahedron weight associated to the k-th eigenvalue. The dependence of each quantity by $\boldsymbol{q}$ and ω is implicit. The denominator represents the overlap between the two basis sets and ensure biorthogonality. Since standard LAPACK routines such as ZGEEV do not sort the complex eigenvalues, the branch order between two nearby $\boldsymbol{q}$ points can be scrambled, giving tetrahedra with incorrect vertices. The solution of sorting the real part of the new eigenvalue works unless the real contribution of the self-energy is large enough to make two eigenvalues cross each other.
The last topic to be discussed is the symmetry enforcing. The point-like quantity G keeps the symmetries of the Hamiltonian and the self-energy, while the integrated quantity depends also on the symmetries of the neighbouring points. Since it's not guaranteed that two symmetrically equivalent points have equivalent neighbours, the weights can break the symmetries of the system. This mismatch is even more pronounced in the improved tetrahedron method, in which 20 neighbouring points are used to fit the plane for the imaginary contribution. This issue, however, becomes less and less relevant as the grid size is increased. Symmetry can also be enforced by merging the degeneracy at each point, according to the irreducible representations of the point group; since the weights are scalar quantities like the energies. Also matrices like W can be symmetrized by generating each matrix in the set of equivalent $\boldsymbol{q}$ points, also called the star of $\boldsymbol{q}$ and then averaging. This procedure is described in the next sections. Since the calculation of weights is usually one of the most expensive steps in a numerical code, we could also calculate them in the irreducible BZ and generate the other weights by symmetry. The computational gain in calculating only the IBZ weights allows us to use a finer grid, that usually mitigates the symmetry breaking issue and improves the convergence of the integrals.

The code we wrote builds the grid and the irreducible BZ and maps each point to a representative point in the IBZ. For each tetrahedron, we check if one of the vertices hits a representative point, saving the mapping between points and tetrahedra. For real arguments, during the initialization of the routine, the vertices are sorted once for each tetrahedron, saving also the mapping between the original and the sorted vertices. The energies are also renormalized with the full band average to improve convergence of the real part. For each reference energies, the weights are calculated for each tetrahedron and then summed together for each point, using the previously defined mapping. The fitting procedure is equivalent to multiplications of the weights with predefined matrices.

2.1.2 Three-phonon surfaces and self-energy interpolation

The tetrahedron method can be generalized to more complex surfaces, wherever we need to approximate a surface integral given by the conservation of energy restrictions. For example, in the anharmonic scattering the three phonon vertex must conserve energy and momentum (up to a reciprocal lattice vector). The energy conservation in Eq. (1.51) can be restated as:

$$\omega_{\boldsymbol{q}j} = \omega_{\boldsymbol{q''}j''} \pm \omega_{\boldsymbol{q'}j'} \tag{2.12}$$

for generic triplets of momenta $\boldsymbol{q}+\boldsymbol{q'}+\boldsymbol{q''} = \boldsymbol{0}$. For linewidth calculation, $\boldsymbol{q}$ is kept fixed: the right hand side of the previous equation depends then on a single variable $\boldsymbol{q'}$, since the other is fixed by momentum conservation. The computational time grows quadratically with the number of atoms in the unit cell, but only linearly with the number of internal $\boldsymbol{q'}$ points. Another way to reduce the computational time for large internal grids of $\boldsymbol{q'}$ is to sample an equispaced interval of frequencies ω up to the maximum frequency of the system; then the weights can be interpolated with any polynomial. This approach fails for low dimensional systems, where divergencies remain even in integrated quantities such as the density of states. On the other hand, the self-energy is usually a smoother function of ω and then it can be interpolated with fewer points. The self-energy, obtained by simply substituting $\omega_{\boldsymbol{q}j}$ with ω, has still to be calculated for each $\boldsymbol{q}$ point in the external grid, therefore $\delta(\omega - \omega_{-\boldsymbol{q}-\boldsymbol{q'}j} \mp \omega_{\boldsymbol{q'}j'})$ must be computed $N_\omega N_{\boldsymbol{q}}$ times; this approach is then used only to calculate the off-shell self-energy $\Sigma_{\boldsymbol{q}j}(\omega)$. In the defect case, we can instead gain in computational efficiency, since the costly part consists in the inversion of $(1 - g(\omega)V(\omega))$. If the inversion is performed in real space, the total calculation cost scales as N_ω instead of $3N_{at}N_{\boldsymbol{q}}$. The cost of Fourier transform is common to both approaches and it is usually negligible with respect to the inversion.

2.2 Numerical implementation of ATA

The Average T-matrix Approximation (ATA) has been widely used as an analytic method to study mass disorder in alloys [46] [47]. Extensions for force-constant disorder are not trivial, even when modeling the interatomic interaction [48], since both the lattice and defective force constants must respect the acoustic sum rule (ASR) to ensure the translational invariance of the system. Many other authors have included force constants disorder from first principles [49] [50] [51]. Here we summarize the fundamental aspects and discuss numerical issues in the implementation. We start by defining the lattice resolvent $g_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R'})$ in the real-space Cartesian representation as in Eq. (1.150).

The potential can be split into the mass and force constant perturbations, defined as:

$$V_{M,\eta\eta'}(\boldsymbol{R},\boldsymbol{R'},\boldsymbol{T},\omega) = -\delta_{\boldsymbol{RT}}\delta_{\boldsymbol{R'T}}\delta_{\eta\eta'}\frac{M_\eta(\boldsymbol{R})-M^0_\eta(\boldsymbol{R})}{M^0_\eta(\boldsymbol{R})}\omega^2 \tag{2.13}$$

$$V_{K,\eta\eta'}(\boldsymbol{R},\boldsymbol{R'},\boldsymbol{T}) = \frac{\Phi_{\eta\eta'}(\boldsymbol{R},\boldsymbol{R'},\boldsymbol{T})-\Phi^0_{\eta\eta'}(\boldsymbol{R},\boldsymbol{R'})}{\sqrt{M^0_\eta(\boldsymbol{R})M^0_{\eta'}(\boldsymbol{R'})}}, \tag{2.14}$$

where $M_\eta(\boldsymbol{R})$ is the mass of the atom with coordinate τ and direction β, with $\eta=(\beta,\tau)$, in the cell $\boldsymbol{R}$. The T-matrix is defined in matrix form as:

$$T = V(1-gV)^{-1} = (1-Vg)^{-1}V. \tag{2.15}$$

2.2.1 Optical Theorem and scattering rate

The scattering rate $1/\tau_{\boldsymbol{q}j}$ is proportional to the imaginary part of the T-matrix element in the phonon basis. Since $\Sigma_{\boldsymbol{q}j}=\Delta\omega^2_{qj}=2\omega_{\boldsymbol{q}j}\Delta\omega_{\boldsymbol{q}j}$, the scattering rate is given by:

$$\frac{1}{\tau_{\boldsymbol{q}j}} = -\frac{2}{2\omega_{\boldsymbol{q}j}}\operatorname{Im}\Sigma_{\boldsymbol{q}j}(\omega) = -\frac{c}{\omega_{\boldsymbol{q}j}}\operatorname{Im}\langle\boldsymbol{q}j|T|\boldsymbol{q}j\rangle\,. \tag{2.16}$$

The optical theorem also follows from Fermi's golden rule for scattering-matrix elements $T^{\boldsymbol{q'}j'}_{\boldsymbol{q}j}$:

$$\frac{1}{\tau_{\boldsymbol{q}j}} \sim \sum_{\boldsymbol{q'}j'}|T^{\boldsymbol{q'}j'}_{\boldsymbol{q}j}|^2\delta(\omega_{\boldsymbol{q}j}-\omega_{\boldsymbol{q'}j'}). \tag{2.17}$$

Starting with a Hermitian Hamiltonian, such as our dynamical matrix D, the time-evolution operator must be unitary, hence the matrix $S=1+iT$, which is the infinite time evolution operator $U(-\infty,\infty)$, must be unitary. This leads to the optical theorem:

$$S^\dagger S = (1-iT^\dagger)(1+iT) = 1+i(T-T^\dagger)+T^\dagger T = 1, \tag{2.18}$$

then the modulus $|T|^2=T^\dagger T$ is related to the imaginary part of T as:

$$T^\dagger T = i(T^\dagger - T) = -2\operatorname{Im}T. \tag{2.19}$$

The sum of the scattering matrix elements $|T^{\boldsymbol{q'}j'}_{\boldsymbol{q}j}|^2$ represents $T^\dagger T$, which is linked to the imaginary part of the zero-momentum $T^{\boldsymbol{q}j'}_{\boldsymbol{q}j}$, equivalent to $\langle\boldsymbol{q}j|T|\boldsymbol{q}j'\rangle$. If the T-matrix is diagonal, then we could give a scattering rate $\tau_{\boldsymbol{q}j}$ for each phonon mode, while off-diagonal elements mix different phonon modes and the scattering rate is ill-defined.

2.2.2 Mass disorder and Tamura model

For mass disorder, the previous formalism is equivalent to the widely used Tamura model introduced in [52], where the author starts with an interaction Hamiltonian acting on the kinetic-energy term:

$$V = \frac{1}{2}\sum_{\boldsymbol{R}\eta}(M_\eta(\boldsymbol{R}) - \bar{M}_\eta(\boldsymbol{R}))\dot{u}^2_\eta(\boldsymbol{R}). \tag{2.20}$$

The perturnation in Eq. (2.13) already gives the Tamura results at the second order in the perturbation expansion. For small mass perturbations, we can also express the potential $V(\boldsymbol{R}, \boldsymbol{R}', \boldsymbol{0})$ in terms of a dynamical-matrix difference as:

$$\begin{cases} \left(\frac{1}{m+\Delta m} - \frac{1}{m}\right)\Phi_0 \simeq -\frac{\Delta m}{m}D_0 & \text{if } \boldsymbol{R} = \boldsymbol{R}' = \boldsymbol{0} \\ \left(\frac{1}{\sqrt{m(m+\Delta m)}} - \frac{1}{m}\right)\Phi_0 \simeq -\frac{\Delta m}{2m}D_0 & \text{if } \boldsymbol{R} \neq \boldsymbol{R}' \wedge \boldsymbol{0} \in \{\boldsymbol{R}, \boldsymbol{R}'\} \\ 0 & \text{otherwise} \end{cases} \tag{2.21}$$

In terms of deltas, it can be rewritten as:

$$V_{\eta\eta'}(\boldsymbol{R}, \boldsymbol{R}') = -\frac{\Delta m}{2m}D_0(\boldsymbol{R}, \boldsymbol{R'})\delta_{\eta\eta'}(\delta_{\boldsymbol{R0}} + \delta_{\boldsymbol{R'0}}). \tag{2.22}$$

The Fourier transform of the potential is:

$$V_{\eta\eta'}(\boldsymbol{q}, \boldsymbol{q'}) = -\frac{\Delta m}{2m}\delta_{\eta\eta'}(D_{0,\eta\eta'}(\boldsymbol{q}, -\boldsymbol{q}) + D_{0,\eta\eta'}(\boldsymbol{q'}, -\boldsymbol{q'})). \tag{2.23}$$

In the phonon basis, we multiply by $\langle\eta|\boldsymbol{q}j\rangle = e_{\boldsymbol{q}j}(\eta)$ and get:

$$V^{\boldsymbol{q'}j'}_{\boldsymbol{q}j} = -\frac{\Delta m}{2m}(\omega^2_{\boldsymbol{q}j} + \omega^2_{\boldsymbol{q'}j'})\sum_\eta e_{\boldsymbol{q}j}(\eta)e^*_{\boldsymbol{q'}j'}(\eta). \tag{2.24}$$

The dynamical matrices produce two different eigenvalues $\omega^2_{\boldsymbol{q}j}$ and $\omega^2_{\boldsymbol{q'}j'}$. Considering the perturbation up to second order, we need to calculate VgV, which introduces the energy conservation $\delta(\omega^2_{\boldsymbol{q}j} - \omega^2_{\boldsymbol{q'}j'})$ through the imaginary part of the lattice resolvent. The two energies are then the same and the scattering rate becomes:

$$\operatorname{Im} T^{(2)}_{M,\boldsymbol{q}j} = \left(\frac{\Delta m}{m}\right)^2 \omega^4_{\boldsymbol{q}j}\sum_{\boldsymbol{q'}j'}\delta(\omega^2_{\boldsymbol{q}j} - \omega^2_{\boldsymbol{q'}j'})\left|\sum_\eta e_{\boldsymbol{q}j}(\eta)e_{\boldsymbol{q'}j'}(\eta)\right|^2. \tag{2.25}$$

This formula coincides with the Tamura model with the substitution

$$\delta(\omega^2_{\boldsymbol{q}j} - \omega^2_{\boldsymbol{q'}j'}) = \frac{1}{2\omega_{\boldsymbol{q}j}}\delta(\omega_{\boldsymbol{q}j} - \omega_{\boldsymbol{q'}j'}). \tag{2.26}$$

While Tamura starts with the VCA crystal as the unperturbed system, we instead add the perturbation on a perfect lattice. These two approaches have different definitions of the mass-perturbation term, as discussed in [53].

2.2.3 Aliasing in Fourier transforms

The core of the ATA is to find a reliable and computationally efficient way to invert the matrix $1 - gV$. In the first-principles papers cited above, this condition is reached by working in periodic real space, while defining the lattice resolvent as a Fourier transform from a dense reciprocal grid to real space. This is subject to aliasing, since the two grids do not coincide. In particular, if $\boldsymbol{\mathcal{T}}$ is a supercell translation vectors, the Fourier transform in Eq. (1.150) reads:

$$g_{\eta\eta'}(\boldsymbol{R}+\boldsymbol{\mathcal{T}},\boldsymbol{R'},\omega) = \frac{1}{N}\sum_{\boldsymbol{q}} e^{i\boldsymbol{q}\cdot(\boldsymbol{R}-\boldsymbol{R'})}\frac{e_{\boldsymbol{q}j}(\eta)e^{*}_{\boldsymbol{q}j}(\eta')}{\omega^2-\omega^2_{\boldsymbol{q}j}}e^{i\boldsymbol{q}\cdot\boldsymbol{\mathcal{T}}} \tag{2.27}$$

$$\neq g_{\eta\eta'}(\boldsymbol{R},\boldsymbol{R'},\omega). \tag{2.28}$$

The only way to avoid aliasing is to use the same grid size for the definition of g and V. On the other hand, the matrix to invert becomes very large if we want to resolve features of the resolvent g. This problem can be partially overcome by noting that in this large real space the potential V acts on a small support: the set of $(\boldsymbol{r},\boldsymbol{r'})$ which minimize a chosen distance as prescribed in Section (1.2.2). In a large real space grid, in the new coordinates the potential is unknown but small, hence can be fixed to zero. We define the potential in this support as V_S, while the resolvent is defined by the submatrices g_{SS}, $g_{S\bar{S}}$, $g_{\bar{S}S}$ and $g_{\bar{S}\bar{S}}$, where $\bar{S}$ is the complement of S. Matrix operations such as VgV can be rewritten as:

$$VgV = \begin{bmatrix} V_S & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} g_{SS} & g_{S\bar{S}} \\ g_{\bar{S}S} & g_{\bar{S}\bar{S}} \end{bmatrix}\begin{bmatrix} V_S & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} V_S g_{SS} V_S & 0 \\ 0 & 0 \end{bmatrix}. \tag{2.29}$$

The main result is that the scattering depends only on the resolvent projected on the support of the potential g_{SS}, which is indeed a small matrix. The cardinality of the support is usually way larger than the supercell size on which it is generated by the centering procedure, but it is still small enough to be computationally manageable. The code performs then the inversion of the matrix $1 - g_{SS}V_S$, which is then transformed back to the reciprocal space by an additional centering procedure. The subspace S is defined as the set of $\boldsymbol{R}$ in which the potential $V(\boldsymbol{R},\boldsymbol{R'})$ is non-zero for at least one $\boldsymbol{R'}$, because we need a square matrix to perform the inversion, see Fig. (2.4). A large amount of space is then wasted by putting a zero potential. For extremely stretched cells this can be an issue and probably sparse inversion techniques should be used. For our test cases, the support was small enough to be handled by dense matrix LAPACK inversions. We should be cautious in translating the same argument to the resolvent g, since it is not a fast decaying function. By neglecting the displacement mixing and for a linear dispersion $\omega(q) = vq$, we get

$$g(\boldsymbol{R}) = \int \frac{d^3\boldsymbol{q}}{(2\pi)^3}\frac{e^{i\boldsymbol{q}\cdot\boldsymbol{R}}}{\omega^2 - v^2q^2} \sim \frac{1}{R}. \tag{2.30}$$

The algorithm is implemented through the following steps:

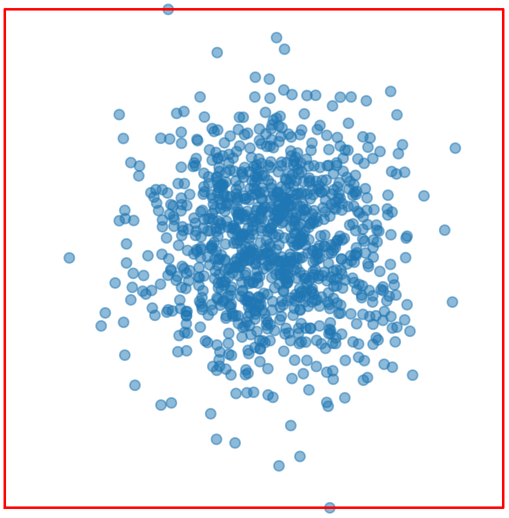


Figure 2.4: *Schematic representation of the support (red) of the potential V (blue dots) in the real space. The red area must be squared to perform the inversion of the matrix $1 - gV$. The potential, being mostly zero at the edges of the support is a sparse matrix in this subspace.*

- We generate the set S from the lattice vectors defining the support of the potential. The centering procedure gives a set of $\boldsymbol{R}$ and a set of $\boldsymbol{R'}(\boldsymbol{R})$ for each $\boldsymbol{R}$ in the support. We insert each lattice vector appearing at least once by cycling along each $\boldsymbol{R}$ and $\boldsymbol{R'}(\boldsymbol{R})$

- We generate the square potential matrix as in Fig. (2.4)

- The Green function is translationally invariant; we must generate a second set S' containing all the possible differences from two lattice vectors in S. The differences that fall outside the full real space grid, reciprocal of the q-points grid, are brought inside it by applying the periodicity of the lattice

- We transform in cartesian coordinates the tetrahedra weights by applying the inverse of the eigenvectors of the unperturbed dynamical matrix, then apply an FFT in the whole space with the size of the $\boldsymbol{q}$-point grid

- We restrict the green function to the subspace S'. The majority of the elements of g are thrown away, but still FFT on the whole space is faster than picking the elements we need

- The resolvent is transformed into the matrix $g(\boldsymbol{R}, \boldsymbol{R'})$ by mapping back each vector in S' to the corresponding couple which generated the specific difference $S \otimes S$

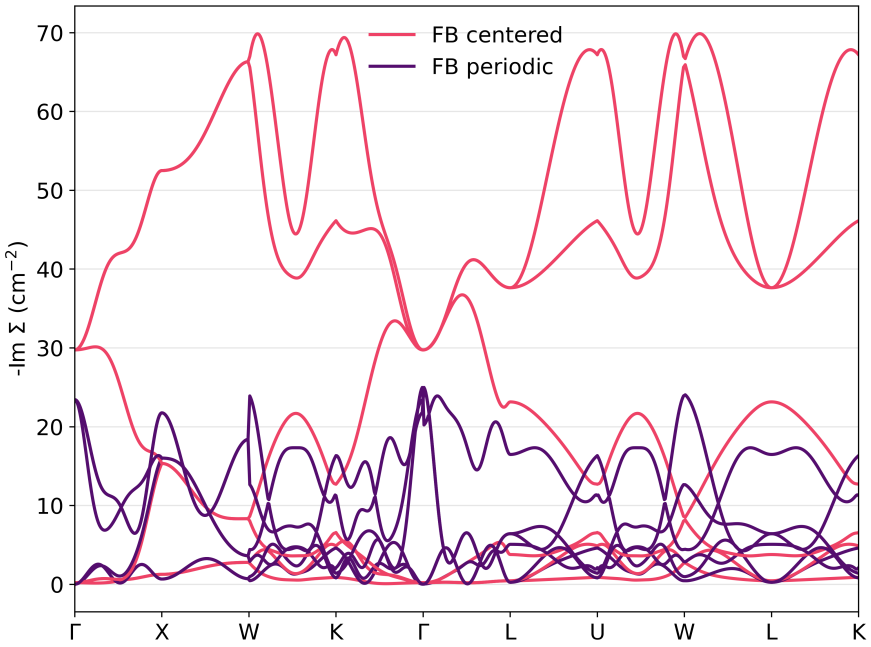


Figure 2.5: *Discrepancy between the centered and periodic ATA, in P-doped Silicon with* $c = 10^{-4}$ *and* $N_q = 18 \times 18 \times 18$*. The periodic self-energy is calculated in the supercell of the defective force constants*

- We calculate the T-matrix by applying the ATA formula $V(1 - gV)^{-1}$ with these matrices
- Since T is supposed to be translationally invariant, we brought it back from $S \otimes S$ to S'. If we need the self-energy in a grid commensurate with the inner q-points grid, we could directly FFT the matrix $T_{\eta\eta'}(\boldsymbol{R})$. Otherwise, if the outer q-points follow a dense path, the T-matrix should be already in the centered form: it should decay rapidly with the Frobenius distance $||\boldsymbol{\tau} + \boldsymbol{R} - \boldsymbol{\tau'}||$ and should not create aliasing.

The results for a test case between the aliased (periodic) and our method is shown in Fig. (2.5).

2.2.4 Self-energy normalization

The concentration c is defined as the probability of a certain site to be occupied by a defect. The normalization constant N is not the number of sites in the supercell of the fine q-points grid, but simply the number of cells. The total number of occupied sites in the supercell is then $cn_{eq}N$. If the single-site T-matrix of different equivalent sites is the same, we would expect these two contributions to add up, making the effective concentration cn_{eq}. First we check that the scattering strength is independent on the

choice of the lattice vector $\boldsymbol{T}$. In reciprocal space, the translated potential acquires a phase, as in Eq. (1.63). Defining the phase matrix

$$U_{qj}^{\boldsymbol{q'}j'}(\boldsymbol{T}) = \delta_{\boldsymbol{qq'}}\delta_{jj'}e^{-i\boldsymbol{q}\cdot\boldsymbol{T}}, \tag{2.31}$$

the potential can be written in matrix form as $V(\boldsymbol{T}) = U(\boldsymbol{T})V(\boldsymbol{0})U^{\dagger}(\boldsymbol{T})$, where $V(\boldsymbol{0})$ is the potential centered at the origin. The inverse of $1 - gV(\boldsymbol{T})$ is then given by:

$$(1 - gV(\boldsymbol{T}))^{-1} = (1 - gU(\boldsymbol{T})V(\boldsymbol{0})U^{\dagger}(\boldsymbol{T}))^{-1} \tag{2.32}$$

$$= U(\boldsymbol{T})(1 - gV(\boldsymbol{0}))^{-1}U^{\dagger}(\boldsymbol{T}). \tag{2.33}$$

The T-matrix acquires the same phase as the potential:

$$T(\boldsymbol{T}) = V(\boldsymbol{T})(1 - gV(\boldsymbol{T}))^{-1} \tag{2.34}$$

$$= U(\boldsymbol{T})V(\boldsymbol{0})(1 - gV(\boldsymbol{0}))^{-1}U^{\dagger}(\boldsymbol{T}) \tag{2.35}$$

$$= U(\boldsymbol{T})T(\boldsymbol{0})U^{\dagger}(\boldsymbol{T}). \tag{2.36}$$

While the scattering amplitude $T_{\boldsymbol{qj}}^{\boldsymbol{q'}j'}(\boldsymbol{T}) = T_{\boldsymbol{qj}}^{\boldsymbol{q'}j'}(\boldsymbol{0})e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}}$ undergoes a phase shift after a translation, the scattering probability $|T|^2$ and the zero-momentum scattering amplitude $T_{\boldsymbol{q}}^{\boldsymbol{q}}$ are left invariant. For two equivalent sites in the same cell $\boldsymbol{0}$ and $\boldsymbol{\tau}$, we can define a set of rotation matrices S which are symmetries of the unperturbed lattice and brings one site to the other. Each symmetry is the composition of one proper rotation (plus fractional translation) with one of the symmetries belonging to the small group that leaves the atomic coordinate unchanged. The rotated potential $V(\boldsymbol{\tau}) = SV(\boldsymbol{0})S^{-1}$ obtained with one of these symmetries is the same as the one we would obtain by a phonon calculation with the defect at τ. The same argument can be brought to real space in cartesian coordinates and so $T(\boldsymbol{\tau}) = ST(\boldsymbol{0})S^{-1}$. The T-matrix would have the same spectrum, but the projection over $|\boldsymbol{q}j\rangle$ would be different. The self-energy formally depends on the average T-matrix $\langle T\rangle$, which is

$$\Sigma = cn_{eq}\langle T\rangle = cn_{eq}\frac{1}{NN_g}\sum_{\boldsymbol{T}g}U(\boldsymbol{T})S(g)T(\boldsymbol{0})S^{-1}(g)U^{\dagger}(\boldsymbol{T}), \tag{2.37}$$

where g is a symmetry operation and $S(g)$ the related matrix. The sum over $\boldsymbol{T}$ gives the momentum conservation $\delta_{\boldsymbol{qq'}}$ and restores the translational invariance:

$$\sum_{\boldsymbol{T}} T_{\boldsymbol{qj}}^{\boldsymbol{q'}j'}e^{-i(\boldsymbol{q}-\boldsymbol{q'})\cdot\boldsymbol{T}} = N\delta_{\boldsymbol{qq'}}T_{\boldsymbol{qj}}^{\boldsymbol{q'}j'}. \tag{2.38}$$

The average over symmetry realizations of the T-matrix produces an operator with the same space group symmetries of the crystal, since:

$$S(g')\sum_{g}S(g)TS^{-1}(g)S^{-1}(g') = \sum_{g}S(g'g)TS^{-1}(g'g) \tag{2.39}$$

$$= \sum_{g''}S(g'')TS^{-1}(g''), \tag{2.40}$$

knowing that the space group is closed under composition. The presence of the self-energy indeed replaces each equivalent host site with an effective medium site. Each new site can be considered as a different atomic species, but since we just changed the species i with another species j in each equivalent site, the new effective medium must have the same symmetries of the original one. The average over symmetries also gives insights on the presence of the term n_{eq} in the self-energy. The correct average would be on every symmetry realization of the sum of equivalent site $T(\boldsymbol{\tau})$, which is entirely given by a specific symmetry operation $S(g(\boldsymbol{\tau}))$. We can then use the closure property of the space group to transform the sum over $gg(\boldsymbol{\tau})$ operations into a sum over g operations, leaving the factor n_{eq} in front of the average.

2.3 Numerical implementation of DCA

The Dynamical Cluster Approximation (DCA) poses significant computational challenges with respect to ATA. The main new implementation steps are the following points:

- The self-energy is obtained by self-consistently solving the Dyson equation. The iterative procedure can drastically change the self-energy from one iteration to the next, causing the iteration to diverge or oscillate.
- A reliable mixing algorithm must be found to ensure the convergence. The choice of mixing should be rooted in the complex plane and should retain the physical properties of the self-energy, such as causality and symmetry.
- The configuration space should be sampled to guarantee that the defect average is the concentration c, but should also give some freedom for each configuration to fluctuate around the average, without dramatically restricting the entire space.
- The coarsening procedure should be consistent with their respective integral definitions.

A necessary condition for convergence is a potential that decays with distance from the defect. Otherwise, the potential would affect the entire cluster, making the correlation length infinite. As a result, the self-energy would be non-local within the cluster, making the interpolation on a finer q-points grid impossible.

2.3.1 Broyden's mixing

We chose to apply a computationally efficient version [54] of the modified Broyden's method [55] to update the self-energy at each iteration. Each mixing algorithm aims to minimize the difference in a vector $|s\rangle$ from one iteration to the next. After solving the cluster problem, we obtain a new self-energy vector $|s_{out}\rangle$; our goal is to correct

the input vector $|s^{(m)}\rangle$ at iteration m into a new vector $|s^{(m+1)}\rangle$ so that $|S^{(m+1)}\rangle := |s_{out}^{(m+1)}\rangle - |s^{(m+1)}\rangle$ is as small as possible. The simplest algorithm allows linear mixing:

$$|s^{(m+1)}\rangle = (1-\alpha)\,|s^{(m)}\rangle + \alpha\,|s_{out}^{(m)}\rangle \tag{2.41}$$

for a certain empirical parameter α. Broyden's method, instead, builds a linear approximation of $|S\rangle$, with

$$|S^{(m+1)}\rangle = |S^{(m)}\rangle + J(|s^{(m+1)}\rangle - |s^{(m)}\rangle) \tag{2.42}$$

where J is the Jacobian of $|S\rangle$. We could also define difference vectors $|\Delta s^{(m)}\rangle = |s^{(m+1)}\rangle - |s^{(m)}\rangle$ and $|\Delta S^{(m)}\rangle = |S^{(m+1)}\rangle - |S^{(m-1)}\rangle$, normalized by the norm of of $|\Delta S\rangle$. With $G = -J^{-1}$, the previous equation can be restated as:

$$|\Delta s^{(m)}\rangle + G^{(m+1)}\,|\Delta S^{(m)}\rangle = 0. \tag{2.43}$$

If we could know the Jacobian or G, we would apply the Newton's step

$$|s^{(m+1)}\rangle = |s^{(m)}\rangle + G\,|S^{(m)}\rangle\,, \tag{2.44}$$

which satisfies the condition $|S^{(m+1)}\rangle = 0$. Broyden's method tries to approximate G by asking it to satisfy the secant equation (2.43) and to minimize the Frobenius norm of ΔG. The solution of this minimization subject to a linear equation $U\,|x\rangle = |y\rangle$ is $U = |y\rangle\langle x|$ if $|x\rangle$ is normalized. The equation is trivially satisfied, while the norm is minimized. To show this, we define U_0 as the outer product; for each U satisfying the secant equation, the difference $W = U - U_0$ satisfies it as well, hence $W\,|x\rangle = 0$. The inner product inducing the Frobenius norm between U_0 and W is:

$$\langle U_0, W\rangle = \mathrm{Tr}[|x\rangle\langle y|\,W] = \mathrm{Tr}[\langle y|\,W\,|x\rangle] = 0 \tag{2.45}$$

by using the cyclic property of the trace. The norm of U is then:

$$\|U\|^2 = \|U_0 + W\|^2 = \|U_0\|^2 + \|W\|^2 \geq \|U_0\|^2, \tag{2.46}$$

since U_0 and W are orthogonal. The issue with Broyden's method is that $\Delta G = U_0$, being an outer product, is a rank-1 matrix. It can then satisfy only the secant equation at the current step, while information from previous iterations is lost. To overcome this issue, the authors at [55] proposed to minimize an error function defined as:

$$E = w_0\left\|G - G^{(m)}\right\|^2 + \sum_{n=1}^{m} w_n||\,|\Delta s^{(n)}\rangle + G\,|\Delta S^{(n)}\rangle\,||^2. \tag{2.47}$$

The condition $\partial E/\partial G = 0$ leads to a closed form solution for G. The authors of [54] found a way to express G in terms of $m \times m$ matrices and m vectors with the size of $|s\rangle$, where m is the memory, or the number of previous iterations to be retained. With an inductive argument, the matrix G is put in terms of the initial guess $G^{(1)} = \alpha I$. Here the

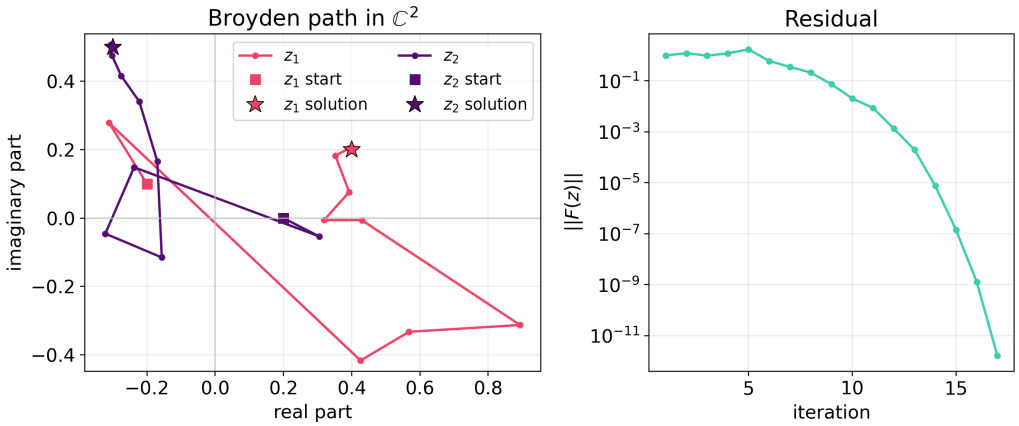


Figure 2.6: *Convergence test for a non-linear coupled equation in $\mathbb{C}^2$, solved by the Broyden's method with memory $m = 6$ and $\alpha = 0.7$. On the left, the trajectory of the guesses is shown in the complex plane, while on the right the norm of the function $f(z) - f(z^*)$ is plotted as a function of the iteration, where z^* is the solution.*

parameter α is equivalent to the mixing parameter in the linear mixing method. The algorithm can be easily adapted in the complex plane, using the complex inner product. The weight w_0 is kept small to obtain a simpler limiting formula for the update of G. For our purpose, the self-energy $\Sigma_{\eta\eta'}(\boldsymbol{Q}, \omega)$ is flattened into a single vector by using the multi-index $\beta = (\eta, \eta', \boldsymbol{Q})$. The vector $|s\rangle$ is then simply $\Sigma_\beta(\omega)$. The scheme has no constraint, such as the causality condition or the space-group symmetry. For the causality, if the imaginary part is far from zero, the mixing should not cause any issue. Small fluctuations around zero can give negative eigenvalues of the imaginary part of the self-energy. These deviations are handled during the tetrahedron integration, once we diagonalize the full denominator $D^0_{\eta\eta'}(\boldsymbol{q}) + \Sigma_{\eta\eta'}(\boldsymbol{q}, \omega)$. The mixing step is fundamentally linear, so it preserves the symmetry of the self-energy if all the previous iterations in memory share the same symmetry. We chose then to impose the point-group symmetry of the cluster self-energy before the mixing step.

2.3.2 Configurational average

The configuration space is sampled by placing a defect in each site with a probability c. The simulated concentration is then the average of the concentration of each configuration. Since the sampling is from a grand canonical ensemble, the average concentration can fluctuate around the target value c. The self-energy is then rescaled by a factor $c/\langle c \rangle$: the error on the concentration scales as $\mathcal{O}(N^{-1/2}_{samples})$. We don't know the scaling of the self-energy with the concentration, but we could expect from the ATA limit that it's almost linear. For low concentrations, when the number of defects is less than two for each sample, there is no defect-defect correlation and each configuration is

equivalent to the others. We can then average the Green's function as $(1-c)G^0 + cG^1$, with the one-defect resolvent obtained by putting the defect in a fixed site and symmetrizing afterwards. The configurations can be sampled once for each frequency and each iteration step, with the advantage of having a faster convergence. For concentrations slightly larger than $1/N_c$, the number of samples with at least two defects can fluctuate and yield a different average. If $N_{samples}$ is large enough, these fluctuations become negligible and we can sample the configurations at each step, to better explore the entire configuration space.

2.3.3 Coarsening and interpolation

The resolvent is calculated by tetrahedral integration, as explained in Section (2.1). These partial integrations are necessary to regularize these divergent functions. The coarsening procedure integrates performs a second integration in the same reciprocal space where the resolvents are defined. To make the DCA numerically converge to ATA, we would need the cluster excluded $\mathcal{G}_0$ to converge to the lattice resolvent g if the coarsening is omitted by setting the cluster size equal to the q-grid size. This is the case if G defined as a point-like quantity as $(\omega^2 - D^0 - \Sigma)^{-1}$, while in our case the resolvent is defined as the integral over a neighbourhood of the point of the linearly interpolated denominator:

$$g_{int}(\boldsymbol{q})^{-1} = \frac{1}{\mu(S(\boldsymbol{q}))}\int_{S(\boldsymbol{q})} d^3q' \, (\omega^2 - D^0(\boldsymbol{q'}))^{-1} \tag{2.48}$$

$$\neq \frac{1}{\mu(S(\boldsymbol{q}))}\int_{S(\boldsymbol{q})} d^3q' \, (\omega^2 - D^0(\boldsymbol{q'}) - \Sigma(\boldsymbol{q'},\omega))^{-1} + \Sigma(\boldsymbol{q},\omega). \tag{2.49}$$

Without coarsening, $g^{-1} = G^{-1} + \Sigma$, but the integral counterparts do not respect the same equation: $g_{int}^{-1} \neq G_{int}^{-1} + \Sigma$. Even if we couldn't avoid this inconsistency, we can mitigate it by increasing the grid size, by effectively reducing the integration region $S(\boldsymbol{q})$. The most effective way to avoid this issue is to restore the DCA cycle without interpolating the self-energy. If we define g with tetrahedron, we can add the self-energy through inversion as:

$$G_{int}(\boldsymbol{q}) = \left[g_{int}(\boldsymbol{q})^{-1} - \Sigma_{DCA}(\boldsymbol{q})\right]^{-1}, \tag{2.50}$$

with $\Sigma_{DCA}(\boldsymbol{q})$ defined as a piecewise constant function in each patch:

$$\Sigma_{DCA}(\boldsymbol{q}) = \sum_{\boldsymbol{Q}} \Sigma(\boldsymbol{Q})\phi(\boldsymbol{q}-\boldsymbol{Q}). \tag{2.51}$$

If we skip the coarsening step by choosing the cluster size N_c equal to the q-grid size N_q, the cluster-excluded Green's function becomes:

$$\mathcal{G}_0(\boldsymbol{Q})^{-1} = G_{int}(\boldsymbol{Q})^{-1} + \Sigma_{DCA}(\boldsymbol{Q}) = g_{int}(\boldsymbol{Q})^{-1} \tag{2.52}$$

and the DCA should converge to the ATA limit, where we are still using tetrahedron integration. The ATA, however, does not make any approximation, since it computes the

non-perturbative formula in the centered real-space, in which the potential is defined in a grid of $\boldsymbol{R}$ of any desiderable size. This is achieved by padding the rest of the grid with zeros; the inversion is done on the support S in which the potential is non-zero just for computational efficiency. The support size, being the result of centering of a potential defined over $N_{\boldsymbol{R}}$ sites, has the linear dimensions increasing by around two times, by taking values from $-\boldsymbol{R}_{max}$ to $\boldsymbol{R}_{max}$. It's often unfeasible to have a cluster grid bigger than this support, so we would expect the tails of the single-defect potential to fold back on the cluster. In particular, if $\boldsymbol{\mathcal{T}}$ is the translation vector of the cluster for which $e^{i\boldsymbol{Q}\cdot\boldsymbol{\mathcal{T}}}=1$, the fourier transform of the potential $V(\boldsymbol{Q},\boldsymbol{Q'})$ would be:

$$\sum_{\boldsymbol{Q}\boldsymbol{Q'}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}} V(\boldsymbol{Q},\boldsymbol{Q'}) e^{-i\boldsymbol{Q'}\cdot\boldsymbol{R'}} = \tag{2.53}$$

$$= \sum_{\boldsymbol{Q}\boldsymbol{Q'}\boldsymbol{R_1}\boldsymbol{R_2}} e^{i\boldsymbol{Q}\cdot(\boldsymbol{R}-\boldsymbol{R_1})} V(\boldsymbol{R_1},\boldsymbol{R_2}) e^{-i\boldsymbol{Q'}\cdot(\boldsymbol{R'}-\boldsymbol{R_2})} \tag{2.54}$$

$$= \sum_{\boldsymbol{\mathcal{T}}\boldsymbol{\mathcal{T}'}} V(\boldsymbol{R}+\boldsymbol{\mathcal{T}},\boldsymbol{R'}+\boldsymbol{\mathcal{T}'}) \neq V(\boldsymbol{R},\boldsymbol{R'}) \tag{2.55}$$

given that

$$\sum_{\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}} = N_c \sum_{\boldsymbol{\mathcal{T}}} \delta_{\boldsymbol{R}+\boldsymbol{\mathcal{T}}}. \tag{2.56}$$

If $\boldsymbol{\mathcal{T}}$ is larger than the support of the potential, there will be one single choice inside the previous sum, and then the Fourier transform is invertible, otherwise the potential itself becomes aliased. The real-to-reciprocal space transform does not introduce any aliasing, because we can enlarge the $\boldsymbol{R}$ grid by padding. The main feature in DCA is to impose translational invariance on the cluster, yielding a self-energy $\Sigma(\boldsymbol{Q})$ defined only on the cluster momenta. In ATA, the translational invariance is applied in the extended real-space; the self-energy is interpolated from the object

$$\Sigma(\boldsymbol{d}) = \sum_{\boldsymbol{R}\boldsymbol{R'}} \Sigma(\boldsymbol{R},\boldsymbol{R'}) \delta_{\boldsymbol{R}-\boldsymbol{R'}-\boldsymbol{d}}, \tag{2.57}$$

for which the Fourier transform in a generic point $\boldsymbol{q}$ reads:

$$\Sigma(\boldsymbol{q}) = \sum_{\boldsymbol{d}} e^{i\boldsymbol{q}\cdot\boldsymbol{d}} \Sigma(\boldsymbol{d}) = \sum_{\boldsymbol{R}\boldsymbol{R'}} e^{i\boldsymbol{q}\cdot(\boldsymbol{R}-\boldsymbol{R'})} \Sigma(\boldsymbol{R},\boldsymbol{R'}) \tag{2.58}$$

as in the operative definition in Eq. (1.63). The Fourier transform of the cluster DCA self-energy already alias into the periodic cluster:

$$\Sigma_{DCA}(\boldsymbol{d}) = \sum_{\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{d}} \Sigma(\boldsymbol{Q}) = \sum_{\boldsymbol{d'}} \left[\sum_{\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot(\boldsymbol{d}-\boldsymbol{d'})} \right] \Sigma(\boldsymbol{d'}) \tag{2.59}$$

$$= \sum_{\boldsymbol{\mathcal{T}}} \Sigma(\boldsymbol{d}+\boldsymbol{\mathcal{T}}) \tag{2.60}$$

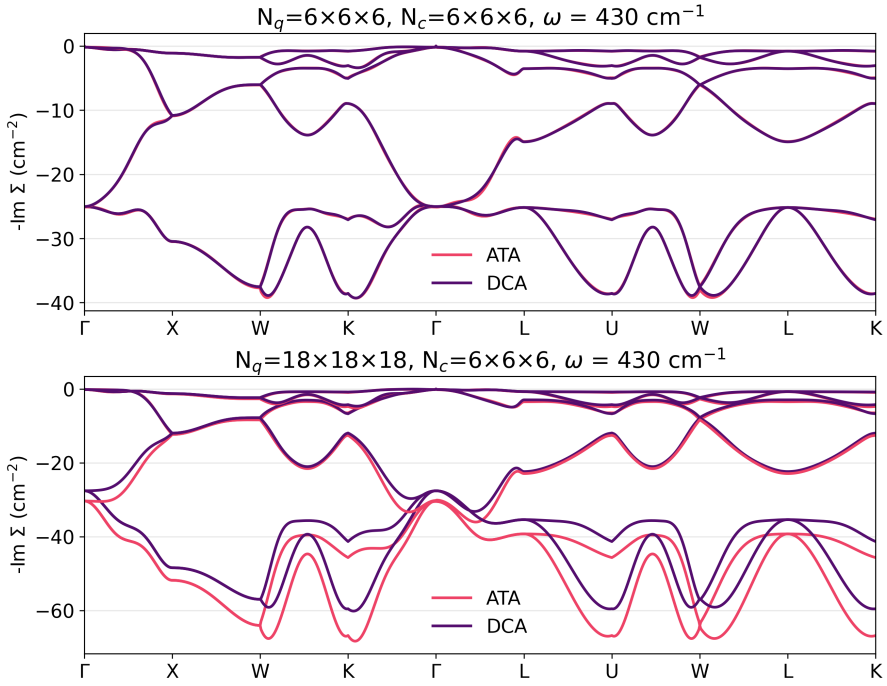


Figure 2.7: *Comparison between ATA and DCA diagonal part of* $\mathrm{Im}\,\Sigma$*, for P-doped Si at* $c = 10^{-4}$*. The potential is obtained from a* $4\times4\times4$ *supercell. The mismatch in the upper panel is due to aliasing, while in the lower panel a more pronounced mismatch is due to the coarsening error. The self-energy increases in the lower panel because the* $6 \times 6 \times 6$ *q-grid is not enough to converge the result.*

is the same for any $\boldsymbol{d} + \boldsymbol{\mathcal{T}}$ and will not yield the same Fourier transform as $\Sigma(\boldsymbol{q})$. The mismatch is visible in the first panel of Fig. (2.7). The other discrepancy between the two methods comes from coarsening: as it can be seen in the second panel of the same figure, it is mainly responsible for the mismatch. By increasing N_q, the ATA lattice resolvent g is able to describe a larger portion of the real defect. The calculation does not scale with the size of the $\boldsymbol{q}$-grid, being performed in the real-space support of the potential. The grid can be easilly taken larger than the support size, increasing it just makes the features of the g sharper. The cluster excluded $\mathcal{G}_0$ comes instead by averaging over each patch, it's no more equal to the lattice resolvent g and has less degrees of freedom. This missing information is then scrambled by a series on non-linear operations inside the clsuter solver, making the self-energy to deviate from the ATA one. In particular, the full resolvent G, written in real space as:

$$G(\boldsymbol{R}) = \left\langle G(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{R}} \right\rangle_{\boldsymbol{q}} = \left\langle \left\langle G(\boldsymbol{Q}+\tilde{\boldsymbol{q}}) e^{i\tilde{\boldsymbol{q}}\cdot\boldsymbol{R}} \right\rangle_{\tilde{\boldsymbol{q}}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}} \right\rangle_{\boldsymbol{Q}} \tag{2.61}$$

loses the phase of the intra-patch momentum $\tilde{\boldsymbol{q}}$ and becomes just:

$$G(\boldsymbol{R}) = \left\langle \langle G(\boldsymbol{Q}+\tilde{\boldsymbol{q}}) \rangle_{\tilde{\boldsymbol{q}}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}} \right\rangle_{\boldsymbol{Q}}. \tag{2.62}$$

This discrepancy grows with the support size S: for a single site perturbation, the self-energy depends only on $G(\mathbf{0})$. Both methods agree in the calculation of this specific matrix element, hence the DCA does not introduce any error. The only way to avoid this issue would be to use the full resolvent G in Eq. (2.61) and solve the cluster problem within S. This option would give the same single particle result as ATA, but it becomes unfeasible for large concentrations, since for each defective sites we should translate the potential and solve the cluster problem in a space that grows linearly with the number of defects (exlcuding the coordinates where overlap occurs).

3 Results and Discussion

3.1 Vacancies and inclusions

Vacancies and inclusions are two very common types of defects in materials. A vacancy is the absence of a host atom, while an inclusion is the presence of an additional atom in an interstitial position. Both are expected to generate a localized perturbation in the system, as reflected in force-constant differences. The effects of this interaction on phonon lifetimes have been studied extensively. The mass perturbation effect, as in analytical models such as Klemens' [56] has been used for vacancies, in the long wavelength limit. Usually, the mass perturbation alone is unable to fully describe the effects of broken bonds in the proximity of the point defect. The authors in [57] found that an effective potential with $\Delta M = 3M$ was suited in the case of vacancies, since the perturbation acts in the kinetic energy T and the average potential $\langle V_{atm} \rangle$ per atom is equal to $\langle T \rangle$. Since the broken bonds affect couples of atoms, the true missing potential is $\langle V \rangle = 2 \langle V_{atm} \rangle$, which is equal to $2 \langle T \rangle$, giving an effective perturbation of $\langle T + V \rangle = 3 \langle T \rangle$ and then consistent with a mass perturbation of $\Delta M = 3M$. Other authors [58, 59] show cases in which this analytical method fails. They identified that the non-perturbative nature of the interaction makes the second-order theory invalid near resonances. Moreover, at low frequencies, where the effect of mass perturbation is expected to be small and the second order to be reasonable, the IFC difference does not scale with frequency and has a major role in this region. The analytical model generally needs ad-hoc mass variances ΔM_{eff}, fitting the results of full potential $V_M + V_K$. In [60] the disorder is treated analytically by inverting the full dynamical matrix to find the resolvent for a large supercell, in which vacancies are modelled as zero-valued matrix elements of the full dynamical matrix. The only potential contribution left out is the relaxation effects, only evident in the a DFPT calculation in a defective supercell. The full thermal conductivity is then calculated through the Callaway model [61].

To treat these two new kinds of defects, we want to stick to the previous methodology, constructing a perturbing potential that can be plugged into ATA equations for small concentrations and DCA for any concentration. The vacancy is easier to treat, since the initial number of atoms remains unchanged: we can see the missing atom as a zero-mass impurity, which does not change the energy of the system while being moved. The host force constants Φ_0 remain unchanged, the defective system is described by the new force constants Φ obtained from DFPT, where the subspace of the missing atom displacement is set to zero. The two force constant matrices have the same dimension and the perturbation is simply $\Phi - \Phi_0$. For an inclusion, we could use the same argument and revert the procedure: we start with the host system, filled with ghost atoms at the position of the inclusion for each unit cell. The matrix Φ_0 this time is enlarged

by $3N_{inc}N_{\boldsymbol{R}}$, where N_{inc} is the number of inclusions in the unit cell. The additional degrees of freedom in all the other unit cells ensure the lattice translation in the host system. The defective system now has N_{inc} inclusions and ghost replicas of the same inclusion in all the other unit cells, so the new force constants matrix Φ has the same dimension as Φ_0. For large systems, the presence of these new degrees of freedom can be computationally expensive. In our ATA implementation, the null subspace shared by both the force cosntants is left out by the support reduction explained in Section (2.2). In particular, we can further reduce the support by excluding the couples $(\boldsymbol{R}\eta, \boldsymbol{R'}\eta')$ for which at least one of the two atoms is a ghost atom. A different approach, valid for any kind of approximation is to project these new degrees of freedom into an effective reduced matrix. We then force the full and reduced dynamical matrices to have the same eigenvalues and eigenvectors. Let's define α and β the degrees of freedom of the host atoms and the inclusion respectively. The eigenvalue problem of the full dynamical matrix can be decomposed as:

$$\begin{pmatrix} \omega_i^2 - D_{\alpha\alpha'} & -D_{\alpha\beta'} \\ -D_{\beta\alpha'} & \omega_i^2 - D_{\beta\beta'} \end{pmatrix} \begin{pmatrix} e^i_{\alpha'} \\ e^i_{\beta'} \end{pmatrix} = 0. \tag{3.1}$$

Since the displacement e_β does not exist in the host system, we isolate it and separate the α degrees of freedom, obtaining:

$$e^i_\beta = (\omega_i^2 - D_{\beta\beta'})^{-1} D_{\beta'\alpha} e^i_\alpha, \tag{3.2}$$

$$(\omega_i^2 - D_{\alpha\alpha'} - D_{\alpha\beta}(\omega_i^2 - D_{\beta\beta'})^{-1} D_{\beta'\alpha'}) e^i_{\alpha'} = 0. \tag{3.3}$$

The potential, being equal to the force constants difference, acquires a new term

$$V_{\alpha\alpha'} \to V_{\alpha\alpha'} + D_{\alpha\beta}(\omega^2 - D_{\beta\beta'})^{-1} D_{\beta\alpha'}, \tag{3.4}$$

which can be easily Fourier transformed, being factorized in the two degrees of freedom $\alpha = (\boldsymbol{R}, \eta)$ and $\alpha' = (\boldsymbol{R'}, \eta')$. The inversion is performd in the small subspace containing the new degrees of freedom. Here we abused of the repeated index notation: the inverse matrix is still indexed with (β, β') and the matrix multiplication is performed after the inversion. The presence of ω in the new term introduces a dynamical behavior, the easiest approximation one can make is to substitute the real solution of the system ω_i with the phonon frequency $\omega_{\boldsymbol{q}j}$, after having performed the Fourier transform.

3.1.1 Vacancies in silicon

Vacancies have been extensively studied in silicon, both experimentally and theoretically. The absence of the atom leaves four reactive dangling bonds, which strongly relax the system. The local point group T_d produces a triply degenerate t_2 electronic state and a low-energy a_1 state. For neutral defects, the four orbitals due to dangling bonds partially occupy the t_2 state with two electrons. Janh-Teller distortion breaks the degeneracy and lowers the symmetry to D_{2d}, with a tetragonal distortion along one of the three $\langle 100 \rangle$ axes. The t_2 state splits into a low-energy b_2 state and a high-energy

e doublet, effectively reducing the total energy of the system. Evidence of the Jahn-Teller distortion has been found in EPR experiments in [62], with an estimated vacancy migration energy of 0.33 eV in p-type silicon. Ab initio calculations confirm the result by placing the nearest-neighbour hopping barrier at 0.4 eV [63]. At room temperature, vacancies are then mobile and can saturate the dangling bonds by clustering with other vacancies, if their concentration is high enough. Various cluster reconstructions have been proposed, exhibiting a different power law reduction for the thermal conductivity [64]. The charge associated to the vacancy depends on the Fermi level. The most common charge states are V_{Si}^{0}, V_{Si}^{2+} and V_{Si}^{2-}, with the neutral state being the most stable one for undoped silicon. Intermediate states +1 and -1 are unfavourable: the vacancy behaves as a negative-U center [65], meaning that the repulsion generated by the presence of a second carrier is highly compensated by the strong lattice relaxation. High formation energies of around 3 to 4 eV are found for the neutral and charged states [66], making the vacancy a rare defect in silicon at room temperature. However, near the silicon melting point of 1685 K, vibrational entropy increases the vacancy concentration by a factor of 10^4, due to highly populated soft modes in the defective system [67]. We made a DFPT calculation of the distorted supercell with point group D_{2d}. We used norm-conserving pseudopotentials [68] and the PBE functional [69] with a 24 Ry energy cutoff in the plane-wave basis set. The host electronic Brillouin is sampled with a $8\times8\times8$ Monkhorst-Pack grid, while the phonons are calculated on a $4\times4\times4$ grid. The system is relaxed with a force threshold of 10^3 Ry and an energy threshold of 10^{-4} Ry. A gaussian smearing of 10^{-3} Ry is introduce to improve the convergence, since defects generally introduce new states in the electronic gap. The electronic convergence threshold for the self-consistent cycle is set to 10^{-12} Ry; for phonons the threshold is 10^8 Ry. The vacancy is then placed in a $4 \times 4 \times 4$ supercell, electronic and phononic states are sampled at Γ point only. The same parameters are used for all the other calculations involving silicon, while the convergence thresholds are valid for all the results of this dissertation. We simulated a concentration of 5% of vacancies to test the DCA implementation, which is more accurate than ATA for large concentrations. We used a $30 \times 30 \times 30$ $\boldsymbol{q}$-grid as the fine grid and a $6 \times 6 \times 6$ cluster grid and 100 samples to perform the resolvent average. Fig. (3.1) shows the diagonal elements of the imaginary self-energy in the phonon basis, along a high symmetry path in the unit cell Brillouin zone. Strong off-shell self-energy is a signature of the interaction between acoustic and optical modes.

3.2 Substitutional doping

3.2.1 Raman spectrum of WSe_2 with Te substitutions

Tungsten diselenide (WSe_2) is a layered material with a hexagonal structure in its stable 2H phase. Within a layer, the Se atoms share the same in-plane coordinates; in the successive layer, the Se atoms sit above the tungsten atoms and this configuration is repeated every two layers. Van der Waals interaction holds the layers together. We

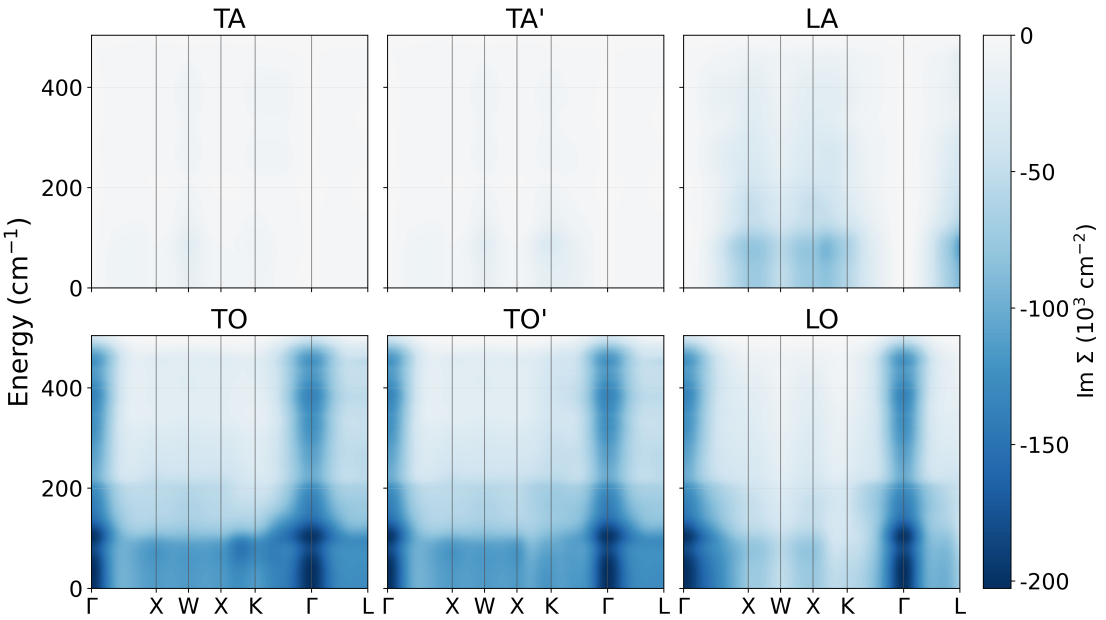


Figure 3.1: *Diagonal elements of the imaginary self-energy in the phonon basis, along a high symmetry path. The self-energy is calculated with DCA for a concentration of* 5% *of vacancies in silicon. Optical modes are strongly suppressed, the acoustic modes near* Γ*, contributing the most to thermal conductivity, are almost unaffected.*

want to focus on the monolayer material, with space group symmetry P$\bar{6}$m2 and point group D_{3h}. We used the Quantum ESPRESSO plane wave code, with norm conserving pseudopotentials [68, 70], [1] and an energy cutoff of 100 Ry. The slab geometry is used to simulate the monolayer, with approximately 15 Å of vacuum; the long-range Coulomb interaction is truncated in the out-of-plane direction [71]. We found that the Perdew-Zunger (PZ) [72] local density approximation (LDA) gives Γ phonons in better agreement with experiments, compared with PBE [69]. Electrons and phonons are calculated in a 8×8 grid in reciprocal space, and in a 5×5 defective supercell at Γ point. The DCA averages 200 cluster grids of size 6×6, while the internal $\boldsymbol{q}$ grid is 30×30. In DOS calculations, we used an external grid of 30×30. The calculated and experimental phonon frequencies at the Γ point are reported in Fig. (3.3). The 2D long-range interaction correction [76] is very important for the correct description of the phonon frequencies. In the 2D limit the LO and TO modes are degenerate at Γ, while the macroscopic electric field associated with the longitudinal mode produces a finite slope of the LO branch near Γ. Moreover, the out-of-plane acoustic mode (ZA), also

[1] The pseudopotentials were generated from SG15 v1.1 input files, using ONCV code version 4.0.1, replacing the PBE choice of functional with PZ. A few parameters have been tuned. Se: the local potential cutoff has been increased to 1.8 a_0, W and Te: the pseudisation radius for the first s channel has been set to 2.4 a_0.

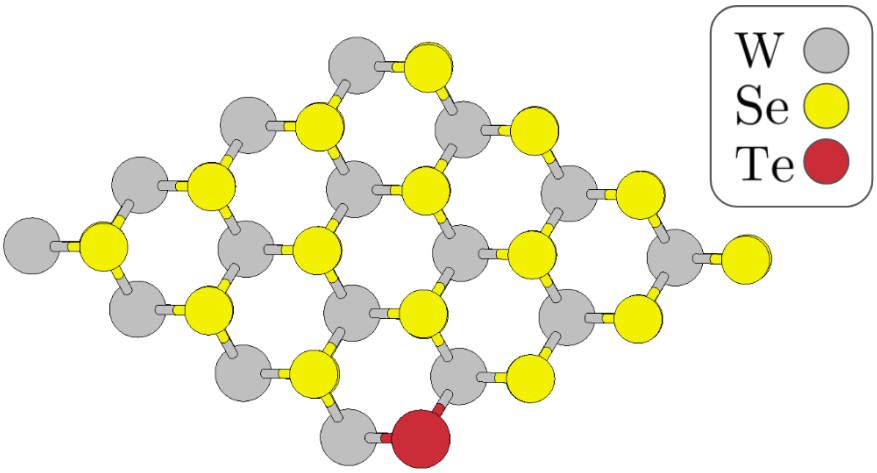


Figure 3.2: *Top view of the WSe_2 monolayer structure of a 4×4 supercell. One of the two stacked Se atoms is substituted with a Tellurium atom in one of the unit cells.*

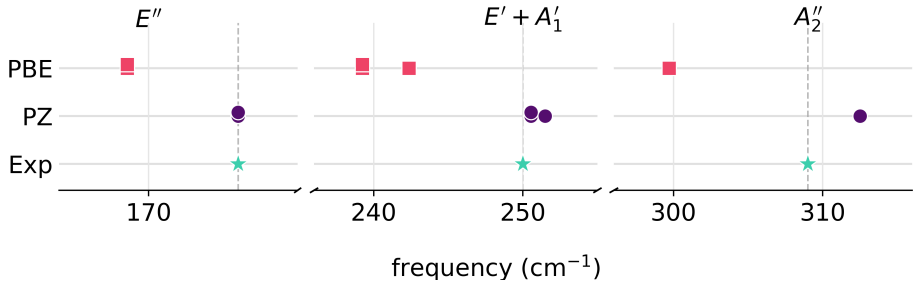


Figure 3.3: *Comparison of experimental [73] and DFT phonon frequencies at the Γ point for monolayer WSe_2, in agreement with [74]. The labels indicate the corresponding D_{3h} irreducible representations, and the frequency axis is broken to focus on the three optical-mode regions. The highest frequency A_2'' mode is only infrared active, but it can be observed in Raman spectra lowering the symmetry of the crystal. The mode A_1' is slightly more energetic than the E' mode, but the frequency difference is smaller than the experimental peak width. The mode E'' is Raman active, but its Raman tensor involves out-of-plane polarization components, which are suppressed in backscattering geometry. They are seen in multilayer samples, as in [75].*

called flexural mode, displays a quadratic dispersion near Γ. By imposing the correct rotational invariance and stress free Huang conditions [77], the phonon dispersion

$$\omega(\boldsymbol{q})^2 = aq^2 + bq^4 + \mathcal{O}(q^6) \tag{3.5}$$

enforces the quadratic term a to be zero. The condition must be imposed on the total force constants $\Phi^{SR} + \Phi^{LR}$, the long range contribution satisfying these requirements

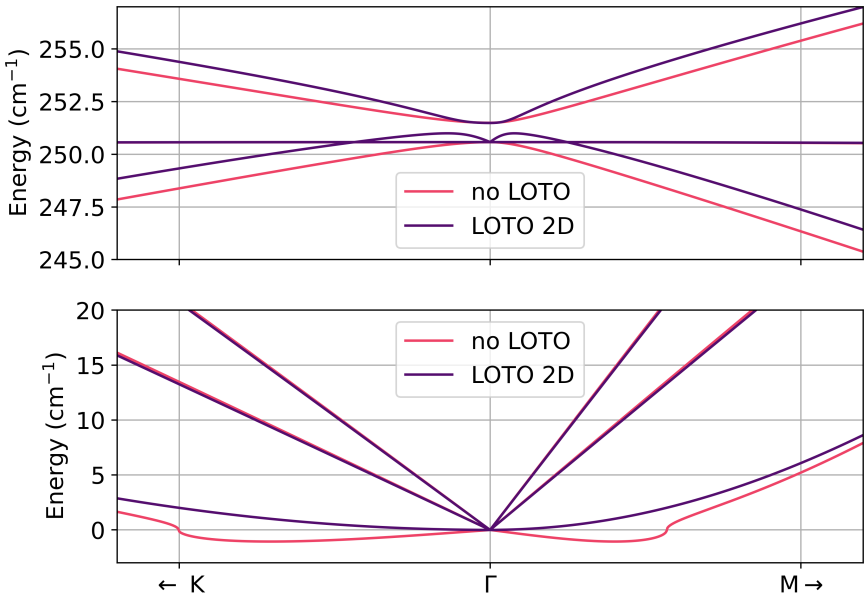


Figure 3.4: *Phonon dispersion of monolayer WSe_2 near the Γ point. LO modes display a finite slope at $\boldsymbol{q} = \mathbf{0}$ due to 2D long range interaction, ZA mode has the correct quadratic behaviour only after long range corrections.*

yields the correct quadratic behaviour of the ZA mode. The mismatch is shown in Fig. (3.4) for LO, TO and ZA modes. The Raman spectrum is obtained by multiplying the mode-resolved spectral function $A_j(\omega)$ at Γ with the corresponding Raman intensity, calculated in the orientational average [78]. Anharmonic effects are included only in the imaginary part of the self-energy, since the real part of the cubic anharmonic bubble self-energy is often partially compensated by the quartic loop contribution [79, 80]. The spectral function is reconstructed as:

$$A_j(\omega) = -\frac{1}{\pi}\operatorname{Im} G_{jj}(\omega) = -\frac{1}{\pi}(\omega^2 - \omega^2_{\mathbf{0}j}\delta_{jj'} - \Sigma_{jj'}(\omega))^{-1}_{jj} \tag{3.6}$$

$$\approx \frac{2\omega_{\mathbf{0}j}}{\pi}\frac{\Gamma_j(\omega)}{(\omega^2 - \omega^2_{\mathbf{0}j} - \Delta_j(\omega))^2 + \Gamma_j(\omega)^2}. \tag{3.7}$$

where $\Sigma_j(\omega) := \Delta_j(\omega) - i\Gamma_j(\omega)$ is the sum of the contributions

$$\Sigma_j(\omega) = \operatorname{Im}\Sigma^{anh}_{jj}(\omega) + \Sigma^{def}_{jj}(\omega). \tag{3.8}$$

The approximation holds if the off-diagonal elements of the self-energy are negligible, which is not always the case for strong phonon-defect scattering. The Raman intensities are $I(\mathrm{E}') = 103$ and $I(\mathrm{A}'_1) = 96$, while the other modes are suppressed. Transition metal dichalcogenides (TMDs) often display higher order Raman processes. The strong acoustic electron-phonon coupling together with the strong phonon DOS near zone-edge critical points can give rise to second order Raman peaks. As in Fig. (3.5), WSe_2 shows a

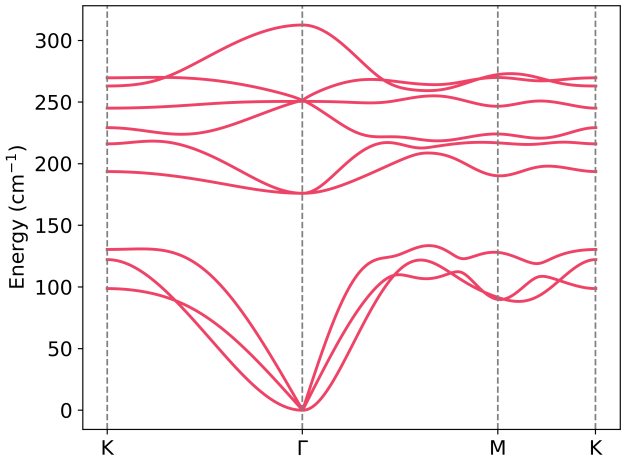


Figure 3.5: *Phonon dispersion of monolayer WSe_2 along a high symmetry path.*

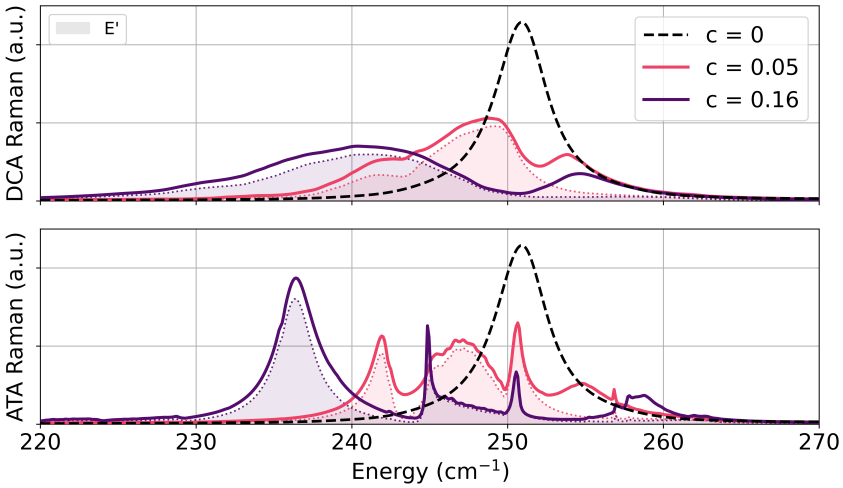


Figure 3.6: *Calculated Raman spectrum of Te-doped monolayer WSe_2 DCA (upper) and ATA (lower). The filled area gives the contribution of the E′ mode, the unfilled area is due to A_1' mode. The self-energy is given by the anharmonic and phonon-defect terms.*

nearly flat phonon dispersion around the M point for the LA mode. The 2D phonon DOS in Fig. (3.8) displays Van Hove singularities in the region 100–150 cm^{-1} as logarithmic divergences. In particular, a second order Raman peak, due to the combination of two LA modes at the M and −M point is seen experimentally at around 260 cm^{-1}, its intensity is highly dependent on the photon wavelength. The authors of [81] attributes the enhancement of the Raman intensity to the resonance with three excitonic states, origi-

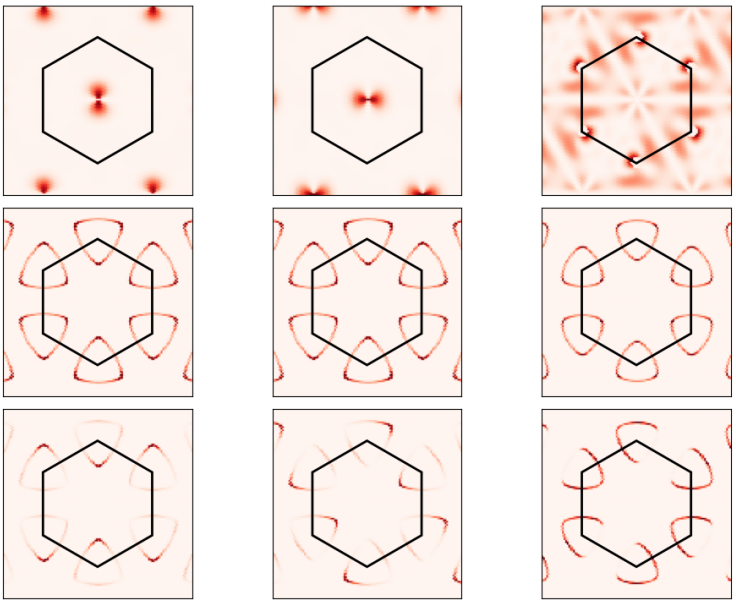


Figure 3.7: *Potential (upper line) and iso-energy surface with tetrahedron method (center line) for the scattering* $\Gamma E' \rightarrow \boldsymbol{q}j$*. For the first two columns* $j = E'$*, for the last column* $j = A_1'$*. The third row shows the scattering strength, given by multiplying the matrix element* V *with the iso-energy surface.*

nated by direct transitions, spin orbit coupling and nearly degenerate transitions in the proximity of Γ. The 2LA(M) peak is not seen in the calculated Raman spectrum, since we only considered first order processes, but the phonon frequency $\omega_{LA(M)} = 128$ cm^{-1} is in good agreement with the experimental value of $2\omega_{LA(M)} = 260$ cm^{-1}. The density of states shows a Te impurity state in the gap at 165 cm^{-1}, as in Fig. (3.8). The same peak is seen in ATA and DCA calculations. The DCA has a broader peak and generally a smoother density of states: while the single-defect localized state solves Eq. (1.118), the DCA self-consistent equation is

$$1 - \mathcal{G}(\omega)(V - \Sigma(\omega)) = 0, \tag{3.9}$$

which can be also solved for complex energies, yielding resonant states. The same behaviour is seen in the Raman spectrum in Fig. (3.6): the DCA has broader primary peaks, where in ATA the E′ splits into sharper peaks. For both approximations, the A_1' (E′) shifts to higher (lower) frequencies, with respect to the pristine crystal, where the distance between the two peaks is much smaller than the natural widths of the two modes.

3.2.2 P and B doping in Silicon

Silicon is an indirect band gap semiconductor with a diamond structure. It's space group is Fd$\bar{3}$m, with point group O_h. The valence band maximum is at the Γ point, while the

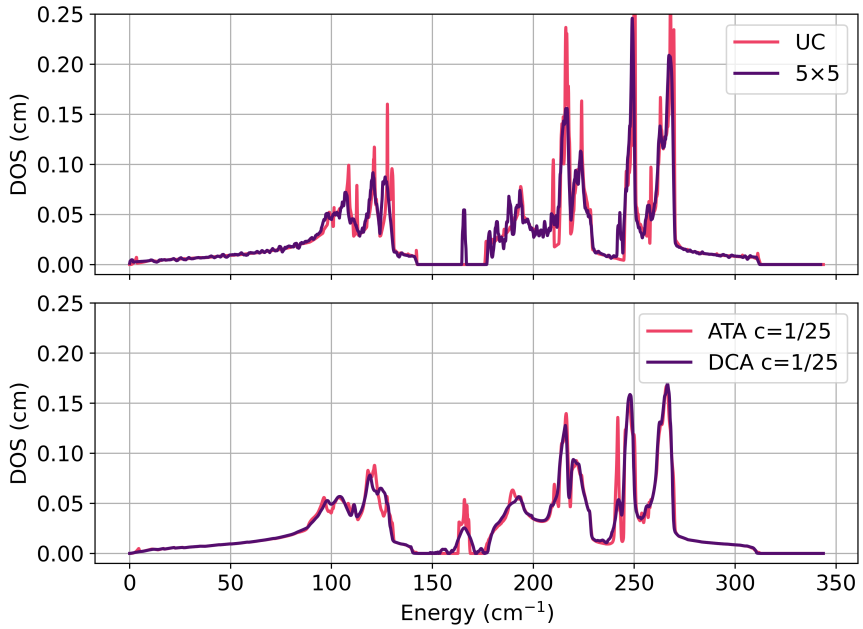


Figure 3.8: *Density of states of monolayer WSe_2. The spectrum features Van Hove singularities in various nearly flat regions of the phonon dispersion. A Te impurity state appears in the gap at around 165 cm^{-1}, both in the periodic 5×5 supercell (upper) and in the non-perturbative calculations ATA and DCA (lower) with the same concentration $c = 4\%$.*

conduction band minimum is near the X point. The phonon dispersion is shown in Fig. (3.9). The most common dopants are Boron for p-type and Phosphorus for n-type doping. The dopants are substitutional, replacing one of the two equivalent Si atoms in the unit cell. The dopants are shallow, meaning that they introduce a donor or acceptor state very close to the band edges. The corresponding bound state has then a large Bohr radius. Boron has a small mass and smaller covalent radius, hence it introduces a bigger mass perturbation and produces a stronger relaxation. All the useful details are summarized in the following Table:

Species	Role in Si	m (u)	r_{cov} (pm)	$\Delta r/r_{\mathrm{Si}}$	Electronic level	a^*/a_0
Si	host	28.085	111	0	–	–
B_{Si}	acceptor, p-type	10.81	84	-24.3%	$E_{\mathrm{A}} - E_{\mathrm{v}} \simeq 45\,\mathrm{meV}$	24.57
P_{Si}	donor, n-type	30.974	107	-3.6%	$E_{\mathrm{c}} - E_{\mathrm{D}} \simeq 45.6\,\mathrm{meV}$	34.02

References: atomic weights from CIAAW [82]; covalent radii from Cordero *et al.* [83]; shallow B and P levels from low-temperature dopant spectroscopy and donor calculations [84, 85]. Bohr radii are in [86, 87].

Shallow dopants in Si can be understood, to first approximation, within the effective-mass approximation. Far from the impurity core, the extra electron bound to a donor such as P, or the hole bound to an acceptor such as B, sees a Coulomb potential screened

by the static dielectric constant of the host,ϵ_{Si}, rather than the bare vacuum Coulomb potential. At the same time, the kinetic energy is governed not by the free-electron mass m_0, but by the curvature of the parabolic band near the edge, yielding an effective mass m^*. The envelope function ϕ therefore obeys a hydrogen-like Schrödinger equation,

$$\left[-\frac{\hbar^2}{2m^*}\nabla^2 - \frac{e^2}{4\pi\epsilon_0\epsilon_{\mathrm{Si}}r}\right]\phi(\boldsymbol{r}) = E\phi(\boldsymbol{r}), \tag{3.10}$$

giving an effective Bohr radius $a^* = \epsilon_{\mathrm{Si}}\frac{m_0}{m^*}a_0$, and an effective binding energy reduced by the factor $\frac{m^*}{m_0}\epsilon_{\mathrm{Si}}^{-2}$. This explains why shallow dopant states in Si extend over several lattice constants rather than being localized on one atom. In real Si the situation is not exactly hydrogenic: the conduction band has anisotropic, multivalley minima, while acceptor states involve the degenerate valence-band structure and spin-orbit coupling. Therefore the quoted a^* should be interpreted as an effective envelope length scale, or as a fitted parameter of the impurity wavefunction, rather than as a sharp radius of the bound state. Both the bound states extend over multiple cells, making impossible for the $4 \times 4 \times 4$ supercell we considered to capture the electron density of the localized state without spurious interactions with the periodic images. This interaction generates an artificial dispersion, seen as a broadening of the localized state in the DOS. Being the binding energies small compared to the band gap and since DFT is known to underestimate bandgaps [88, 89], these peaks are usually not seen in the supercell's DOS. The ionization energy $E(N-1) - E(N)$ is given by:

$$\int_0^1 df\, \frac{\partial E}{\partial f} = \int_0^1 df\, \epsilon(f) \simeq \epsilon(f = 1/2), \tag{3.11}$$

where $\epsilon(f)$ is the Kohn-Sham eigenvalue at occupation f and Janak's theorem is used in the first equality. Through half-filling atomic potentials, the authors of [90] were able to better reproduce the ionization energies, but this goes beyond the scope of this work. The additional electron or hole has a strong effect on the force constants. Luckily, the highly symmetric structure of Silicon doesn't allow the crystal to develop long range dipole-dipole interactions: the two equivalent sites must have the same Born effective charges Z^*, but acoustic sum rule requires that the sum of all Z^* in the unit cell is zero. The only contribution is the last term in Eq. (1.87), which is zero for isolated defects. Since the binding energy of both B and P impurity states is comparable with the room temperature thermal energy of 26 meV, we could expect to encounter ionized impurities, with the electron/hole in the conduction/valence band. These calculations are performed in a neutral supercell, where the charge difference is compensated by a uniform jellium background to avoid the Coulomb divergence. This approach is justified near the Γ point: the particle wavefunction is delocalized over the entire supercell, being the one of a nearly free electron, for parabolic bands. The Bloch condition imposes $\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = u_{n\boldsymbol{k}}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, with $u_{n\boldsymbol{k}}$ the nearly-free envelope function, periodic in the supercell. For phonons, the impurity state is instead well separated from the continuum: in Fig. (3.10) the B impurity shows up as a sharp peak at around 630 cm^{-1}, while the P impurity state is only able to create a resonance somewhere in

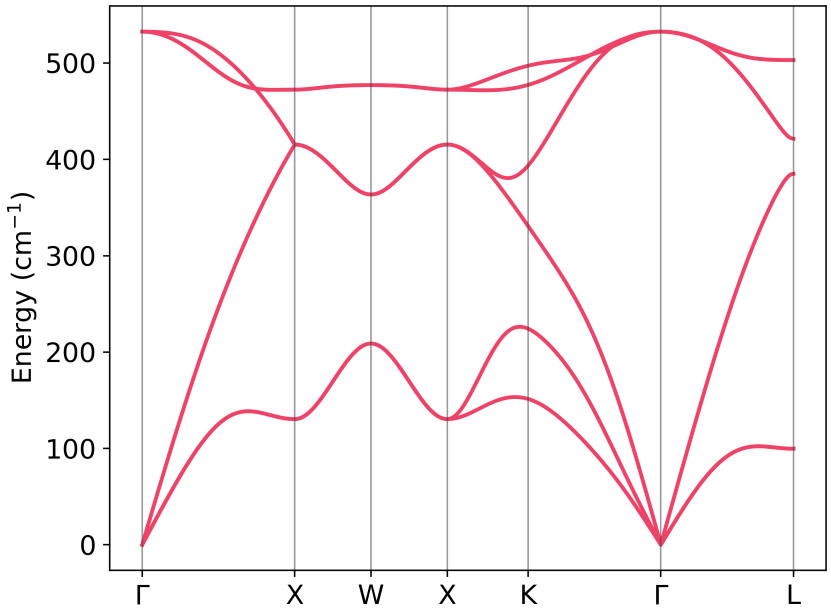


Figure 3.9: *Phonon dispersion of silicon along a high symmetry path*

the optical phonon band. IR absorption measurements of B–doped silicon show two impurity states at 619 and 642 cm^{-1}, which are attributed to the two isotopes of boron [91]. To explore doping above 10^{20} cm^{-3}, which corresponds to 0.2% of substitutions, we need to resort to the DCA, which is able to capture the defect-defect correlations. In Fig. (3.11) we show the imaginary part of the diagonal DCA self-energy, for a cluster grid of $6 \times 6 \times 6$ and an internal $\boldsymbol{q}$ grid of $30 \times 30 \times 30$. The self-energy is calculated for a concentration of $c = 2.5\%$ of P substitutions. The small positive imaginary part comes from the projection of the full matrix onto the diagonal elements: invariant quantities such as the trace show no causality violation. The optical branches are the main contributors for the linewidths, especially near high-symmetry points.

3.3 Lattice thermal conductivity and thermoelectricity

The full resolvent is given by the total self-energy, which should include all the contributions to the phonon scattering. The most important contributions come from anharmonicity (phonon-phonon), disorder (phonon-defect), electron-phonon and phonon-boundary scattering. These self-energies are often summed up in the Matthiessen's rule [92], neglecting the interference between different scattering mechanisms. This assumption leaves out coupled scattering channels, such as phonon-defect scattering in the intermediate states of the three-phonon vertex, which give non-negligible corrections [93]. The electron-phonon contribution is generally calculated in the lowest order, without vertex corrections. The phonon is able to perturb the Fermi sea, creating

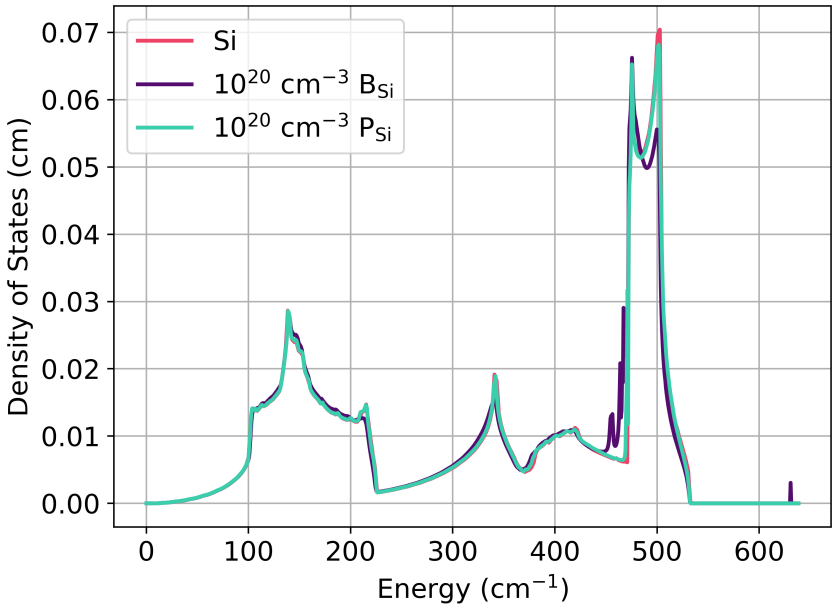


Figure 3.10: *ATA density of states of silicon with P and B substitutional defects. The B impurity state is seen as a sharp peak at around 630 cm^{-1}. At the same concentration, the Boron defect has a stronger effect on the phonon DOS, due to larger mass and covalent radius mismatch.*

an electron-hole pair, which then recombines emitting a phonon. The *bubble diagram* is then the lowest order contribution, and it's given by:

$$\Pi^{e-ph}_{jj'}(\boldsymbol{q},\omega) = \frac{2}{N_k}\sum_{\boldsymbol{k}nm} g^*_{mnj}(\boldsymbol{k},\boldsymbol{q}) g_{mnj'}(\boldsymbol{k},\boldsymbol{q}) \frac{f(\epsilon_{n\boldsymbol{k}}) - f(\epsilon_{m\boldsymbol{k}+\boldsymbol{q}})}{\epsilon_{n\boldsymbol{k}} - \epsilon_{m\boldsymbol{k}+\boldsymbol{q}} + \omega + i\eta}. \tag{3.12}$$

The matrix element $g_{mnj}(\boldsymbol{k},\boldsymbol{q}) = \langle m\boldsymbol{k}+\boldsymbol{q}|\partial_{\boldsymbol{q}j} V_{SCF}|n\boldsymbol{k}\rangle$ contains the electronic screening throught the self-consisten potential [94, 95], but it does not include higher-order vertex corrections. Each new order scales approximately as $\omega_{\boldsymbol{q}j}/E_F$, which is a small parameter in metals, but not in degenerate or room temperature semiconductors. The vertex correction accounts for phonon creations inside the bubble, or electron-electron interactions. The function f is the Fermi-Dirac distribution, it gives a strong scattering amplitude for states near the Fermi level. The denominator accounts for the energy conservation: for well separated branches at fixed $\boldsymbol{q}$, the mode mixing $j \to j'$ is strongly suppressed and the self-energy is diagonal. The DFPT force constants also include the static adiabatic response, hence actual correction in the dynamical matrix is $\Sigma(\omega) - \Sigma(0)$, where

$$\Sigma^{e-ph}_{jj'}(\boldsymbol{q},\omega) = 2\sqrt{\omega_{\boldsymbol{q}j}\omega_{\boldsymbol{q}j'}}\,\Pi^{e-ph}_{jj'}(\boldsymbol{q},\omega). \tag{3.13}$$

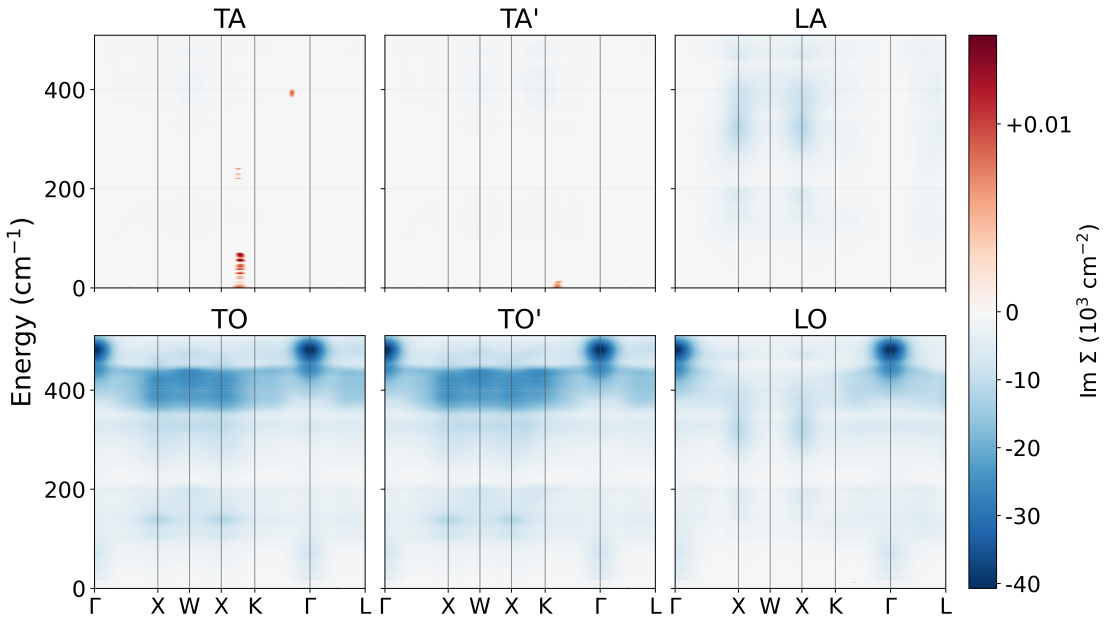


Figure 3.11: *Imaginary part of the DCA self-energy for P-doped silicon, with* $c = 2.5 \times 10^{21}$ *cm*$^{-3}$*. Optical branches contribute the most near high-symmetry points. The small positive imaginary part is not due causality violation: for non-diagonal self-energies, the diagonal elements do not correspond to the eigenvalues of the full matrix.*

Boundary scattering is important in nanostructures, where the phonon mean free path is comparable with the sample size. Phonons with higher mean free path reflections are fully diffused, the scattering rate is $\tau^{-1} = v/L$, where v is the phonon group velocity and L is the sample size. Corrections to fully account for the momentum diffusion are in [96]. Another source of scattering is the isotope disorder, which is a mass perturbation applied over the defect-free crystal. The isotope disorder posses a very small mass perturbation, it can be treated perturbatively as in the Tamura model [52]. Up to non-degenerate doping, the three phonon (3ph) and electron-phonon (e-ph) are the main contributions to the lattice thermal conductivity [97]. The ATA contribution is instead needed for higher concentrations [49, 98]. Defect scattering can be significative also in the electronic transport, where electron-defect can be found by first principles by taking the self-consisten potential difference ΔV_{SCF} or by analytical models [99]. Other mechanisms, such as the phonon drag effect, are important in the calculation of mobilities and Seebeck coefficients [100]. A non-equilibrium distribution of phonons contributes to the electron scattering, and it can be calculated by solving the coupled Boltzmann transport equations or through ad hoc terms [101]. The mobility itself needs major corrections to match the experimental value, such as spin orbit coupling for the hole mobility, experimental screening and electronic many-body effects [102]. Putting

together all the contributions, the thermoelectric figure of merit

$$ZT = \frac{S^2 \sigma T}{\kappa_e + \kappa_l}, \tag{3.14}$$

where S is the Seebeck coefficient, κ is the electronic and lattice and thermal conductivity and σ is the electrical conductivity, can be estimated by fully *ab initio* methods. Lattice thermal conductivity emerges from the steady state Boltzmann transport equation (BTE)

$$\boldsymbol{v}_\lambda \cdot \frac{\partial n_\lambda}{\partial \boldsymbol{r}} = \left(\frac{\partial n_\lambda}{\partial t}\right)_{\text{coll}}, \tag{3.15}$$

where $\lambda = (\boldsymbol{q}, j)$ and the collision term is built from the combined scattering rates of all the mechanisms. The scattering rate $W_{\lambda\lambda'}$ due to defects must be able to scatter phonons with different wavevectors, it is proportional to $|T_{\lambda\lambda'}|^2 \delta(\omega_\lambda - \omega_{\lambda'})$. The space dependence of the occupations n is given by the temperature gradient. By expanding the occupation at linear order $n_\lambda = n^0_\lambda(T(\boldsymbol{r})) + \delta n_\lambda$, the scattering term

$$\left(\frac{\partial n_\lambda}{\partial t}\right)_{\text{coll}} = \sum_{\lambda'} W_{\lambda\lambda'}(n_{\lambda'} - n_\lambda) \tag{3.16}$$

and the BTE is linear in δn_λ. The heat current is known after solving for the distribution functions:

$$J^\alpha = \frac{1}{V}\sum_\lambda \omega_\lambda v^\alpha_\lambda \delta n_\lambda = -\kappa_l^{\alpha\beta}\frac{\partial T}{\partial r^\beta}. \tag{3.17}$$

This reduces to single-mode relaxation time approximation (SMA) by enforcing the scattering rate to be diagonal in λ, for which the thermal conductivity is directly given by:

$$\kappa_l^{\alpha\beta} = \frac{1}{V}\sum_\lambda \omega_\lambda \frac{\partial n^0_\lambda}{\partial T} v^\alpha_\lambda v^\beta_\lambda \tau_\lambda \tag{3.18}$$

Another more useful approximation, which includes the off-diagonal scattering rates between different modes with the same $\boldsymbol{q}$ is to start with the Green–Kubo formula for the thermal conductivity, linking the thermal conductivity to the heat-current autocorrelation function $\left\langle J^\alpha J^\beta \right\rangle \propto v^\alpha v^\beta \left\langle AA \right\rangle$. Since our theory gives no information about two particle operators, we make the *bubble* approximation $\langle AA \rangle \simeq \langle A \rangle \langle A \rangle$, neglecting the vertex corrections. The frequency- and wavevector-resolved thermal conductivity is then proportional to $\mathrm{Tr}\left[v^\alpha A v^\beta A\right]$, where $A_{jj'}(\boldsymbol{q},\omega) = -\frac{1}{\pi}\operatorname{Im} G_{jj'}(\boldsymbol{q},\omega)$ is the spectral function and the velocity operator is:

$$v^\alpha_{jj'}(\boldsymbol{q}) = \frac{1}{2\sqrt{\omega_{\boldsymbol{q}j}\omega_{\boldsymbol{q}j'}}} \boldsymbol{e}^\dagger_{\boldsymbol{q}j} \cdot \frac{\partial D^0(\boldsymbol{q})}{\partial q_\alpha} \cdot \boldsymbol{e}_{\boldsymbol{q}j'}(\eta'). \tag{3.19}$$

This approximation captures the Wigner correction to thermal conductivity for the quasiparticle transport [103], in which $A_{jj'}$ is diagonal.

3.4 Alloying and thermal conductivity

While the concentration of defects is low in doped materials, in random alloys the concentration of one species spans the entire range from 0 to 100%. For concentrations around 10% or higher, multisite scattering makes a non-negligible contribution, and the ATA and DCA predictions begin to differ. The latter can be applied instead to a wide variety of alloys over a broad concentration range: the sampling can take into account a non-uniform distribution in the case of clustering, ordering, or segregation. For ternary alloys, the sampling could include two "defective" species, or the scattering can be separated into two distinct contributions and summed through Matthiessen's rule. The DCA becomes less reliable when the species have similar concentrations, where it is difficult to tell which species is "defective" and which is "host". One can compare the limits obtained by treating either A or B as the host to assess the DCA in the central region of the alloy phase diagram. Otherwise, we can also start from the VCA, by constructing a virtual crystal whose pseudopotentials are given by the weighted average of the two species. This approach can be used to compute other scattering rates, such as phonon-phonon and phonon-electron scattering, and as a starting point for the DCA.

3.4.1 Silicon-germanium alloys

Among semiconductor alloys, $\mathrm{Si}_{1-x}\mathrm{Ge}_x$ is a particularly useful test case for theories of vibrational disorder. Its constituents have the same crystal structure and similar chemical bonding, yet differ substantially in mass. Substitution of Si by Ge therefore produces a strong mass perturbation for phonons. This alloy is commonly treated as a random mixture, although the authors of [104] have shown that strain-free thermodynamics can favor Si-rich and Ge-rich phase separation. A common strain due to lattice mismatch instead strongly stabilizes homogeneous mixing.

The resulting reduction of lattice thermal conductivity is also technologically relevant: SiGe has long been employed as a high-temperature thermoelectric material [105], where phonon scattering by alloy disorder helps reduce the thermal conductivity, while the material remains a dopable semiconductor with reasonably good electronic transport. Fig. (3.12) shows the scattering rates for 10% Ge in Si, computed with the DCA. Mass disorder is the dominant contribution, isotopic scattering can instead be neglected. Being a weak perturbation, it can be treated within the Tamura model, equivalent to second-order perturbation theory of the mass disorder due to the presence of isotopes. The mass perturbation is point-like in real space; it therefore has no dispersion in reciprocal space. In crystals like silicon, in which each atom in the unit cell can host a defect, the single-site symmetrized self energy in the cartesian basis is

$$\Sigma_{ATA} \propto S I_3 S^{-1} \propto I_6, \tag{3.20}$$

where I_3 is a half identity matrix identifying which site hosts the mass defect in the three cartesian directions. I_6 is the identity matrix obtained after the symmetrization. The self-energy is proportional to the identity also in the phonon basis, independently

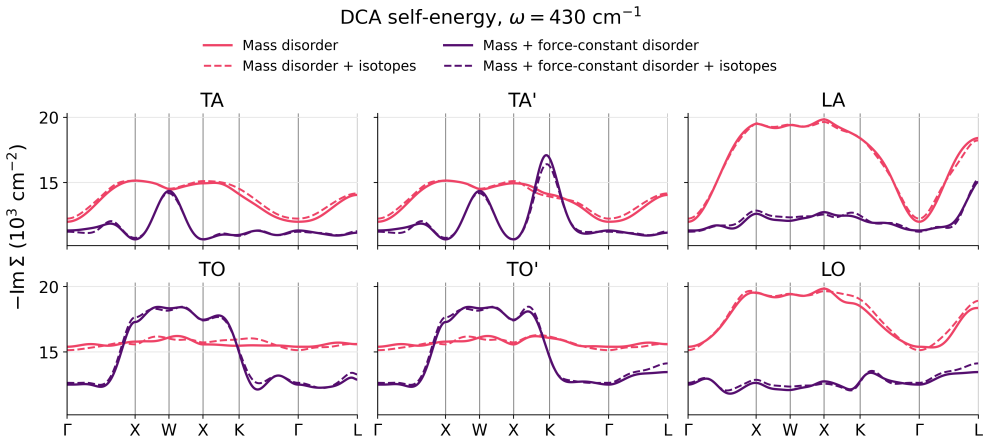


Figure 3.12: *Comparison between DCA calculations of $Si_{0.9}Ge_{0.1}$ for separate perturbations. The scattering rates are plotted for the six phonon branches. The isotopic scattering is the weaker contribution, mass disorder contributes the most.*

of the phonon rotation matrix $U(\boldsymbol{q})$, hence the scattering rates show no dispersion in reciprocal space. DCA instead well treats defect correlations: the self-energy gains a dependence on the defect-defect distance and hence acquires a $\boldsymbol{q}$-dependence even for the mass disorder case. It is also interesting to note that the isotopic scattering does not simply translate rigidly the profiles: the Matthiessen rule does not hold perfectly in this case, and the total scattering rate is not exactly the sum of the two contributions. The effect is evident in the TA′ branch, where the isotopic scattering tends to reduce the scattering rate in the $V_M + V_K$ case.

3.4.2 Scattering from virtual crystal

The ATA self-energy is nearly proportional to the concentration in the dilute limit: the scattering rates tend to increase linearly, making the thermal conductivity κ decrease monotonically with the concentration. If κ becomes lower than the limiting values at $c = 0$ and $c = 1$, it will never be possible to connect continuously these two limits. The overcounting correction in Eq. (1.175) reproduces correctly the limiting behaviour, because the $(1 - c)$ factor in the denominator makes the perturbation vanish in the $c \to 1$ limit. However, the profiles $0 \to 1$ and $1 \to 0$ are expected to be different. An alternative approach is to start with a virtual crystal state, in which the force constants, lattice parameters and masses are linearly interpolated between the two species:

$$\bar{m} = (1 - c)m_A + cm_B, \tag{3.21}$$

$$\bar{\Phi} = (1 - c)\Phi_A + c\Phi_B, \tag{3.22}$$

$$\bar{a} = (1 - c)a_A + ca_B. \tag{3.23}$$

One could be tempted to define the perturbation as the difference between the defective force constants $\Phi_{A|B}$ — where A is the defect and B is the host — and $\bar{\Phi}$, but this would lead to a perturbation which does not decay with the distance from the defect, since the two backgrounds are not the same. In the virtual crystal spirit, we need to define two perturbations $\Delta\Phi_A$ and $\Delta\Phi_B$ with zero average:

$$(1-c)\Delta\Phi_A + c\Delta\Phi_B = 0. \tag{3.24}$$

This equation can be satisfied by defining $\Delta\Phi_A = -c\Delta\Phi$ and $\Delta\Phi_B = (1-c)\Delta\Phi$, where $\Delta\Phi$ is an unknown perturbation. For the mass disorder, the perturbation $V_i = (1 - m_i/\bar{m})\omega^2$ already satisfies the zero-average condition:

$$(1-c)(1-\frac{m_A}{\bar{m}}) + c(1-\frac{m_B}{\bar{m}}) = 0. \tag{3.25}$$

The perturbation $\Delta\Phi$ must be a function of the decaying force-constant perturbations $\delta\Phi_d = \Phi_{d|h} - \Phi_h$, where d is the defect and h the host. It must also reproduce the correct limiting behaviour: $\Delta\Phi_B \to \delta\Phi_B$ for $c \to 0$ and $\Delta\Phi_A \to \delta\Phi_A$ for $c \to 1$. The choice $\Delta\Phi = (1-c)\delta\Phi_B - c\delta\Phi_A = (1-c)\Phi_{B|A} - c\Phi_{A|B} - \bar{\Phi}$ satisfies all these conditions:

$$\Delta\Phi_A = -c\Delta\Phi = c^2\delta\Phi_A - c(1-c)\delta\Phi_B, \tag{3.26}$$

$$\Delta\Phi_B = (1-c)\Delta\Phi = (1-c)^2\delta\Phi_B - c(1-c)\delta\Phi_A. \tag{3.27}$$

The (single-site) average T-matrix is weighted by the concentration of the two species, as:

$$\langle t \rangle = (1-c)t[V_A] + ct[V_B], \tag{3.28}$$

where $V_i = (1 - m_i/\bar{m})\omega^2 + \bar{m}^{-1/2}\Delta\Phi_i\bar{m}^{-1/2}$ is the perturbation for the i-th species. It is useful to check whether the limiting behaviour agrees with the ATA. The two full potentials can be written as

$$V_A = -c\bar{m}^{-1/2}[(m_A - m_B)\omega^2 + \Delta\Phi]\bar{m}^{-1/2} = -cK(c) \tag{3.29}$$

$$V_B = (1-c)\bar{m}^{-1/2}[(m_A - m_B)\omega^2 + \Delta\Phi]\bar{m}^{-1/2} = (1-c)K(c), \tag{3.30}$$

where $K(c)$ is a perturbation which depends on the concentration through $\Delta\Phi$. We start by expanding Eq. (3.28) in the $c \to 0$ limit:

$$\langle t \rangle = (1-c)t[-cK(c)] + ct[(1-c)K(c)] = -cK(c) + ct[K(c)] + \mathcal{O}(c^2), \tag{3.31}$$

$$K(c) = (1 - \frac{m_B}{\bar{m}_A})\omega^2 + m_A^{-1/2}\delta\Phi_B m_A^{-1/2} + \mathcal{O}(c). \tag{3.32}$$

The operator K tends to the ATA perturbation for a single B defect embedded in the A host. The average T-matrix tends to the ATA result, apart from a real shift $-cK(c)$ given by the majority species. In the limit $c \to 1$, we define $x = 1 - c$ to obtain:

$$\langle t \rangle = xt[-(1-x)K(x)] + (1-x)t[xK(x)] = xt[-K(x)] + xK(x) + \mathcal{O}(x^2), \tag{3.33}$$

$$-K(x) = (1 - \frac{m_A}{\bar{m}_B})\omega^2 + m_B^{-1/2}\delta\Phi_A m_B^{-1/2} + \mathcal{O}(x). \tag{3.34}$$

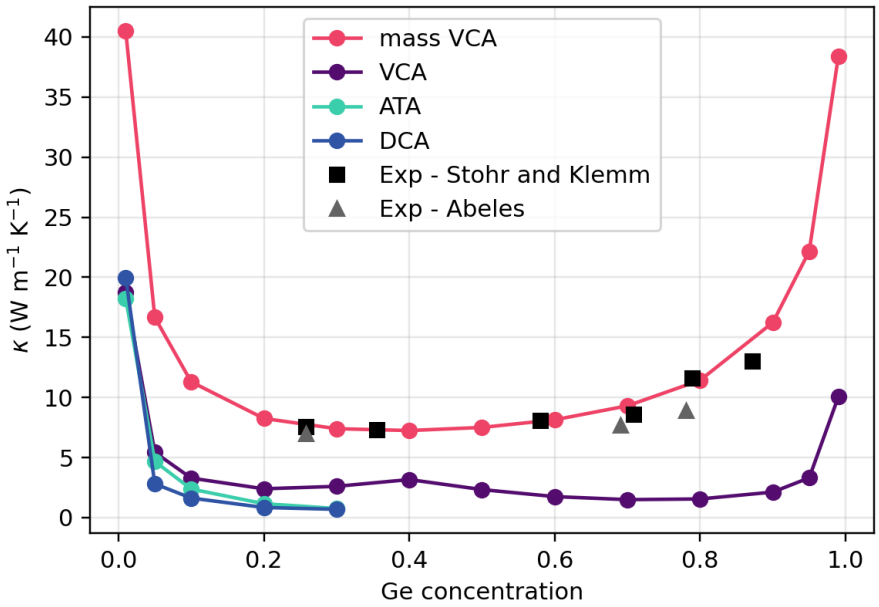


Figure 3.13: *Comparison of the thermal conductivity in the range* $[0.01, 0.99]$ *of Ge concentration, computed with ATA, DCA and the new approach based on the VCA. The presence of force constants disorder produces a strong reduction of the thermal conductivity for each approximation. The calculated pristine values for* κ *are 130 W/mK for Si and 60 W/mK for Ge. Experiments are from [106], while Stohr and Klemm was taken from [107].*

This gives the previous result with swapped indices. This approach has the advantage of being symmetric with respect to the two species, providing a good interpolation between the two limits. The results for $Si_{1-x}Ge_x$ are shown in Fig. (3.13) for the lattice thermal conductivity of the alloy, computed in the SMRTA through Eq. (3.18). The anharmonic linewidths are computed for pure Si and Ge, and then interpolated linearly to the virtual crystal. The volume is obtained by linear interpolation of the lattice parameters, according to Vegard's law in Eq. (3.23). Velocities and frequencies are instead obtained from the virtual crystal force constants. The defective linewidths are computed with ATA and DCA on both sides at low-to-moderate concentrations, and with this new approach for mass and full disorder. The self-energy is computed explicitly through Eq. (1.168), since there is no closed formula as in the ATA in Eq. (1.175). For relaxation and phonons we used the same parameters as in Section (3.1.1), the d-orbital is frozen in germanium PZ [72] pseudopotential, to be able to calculate the force constants in the usual $4 \times 4 \times 4$ supercell in a reasonable time. For scattering rates, the inner and outer grid are $30 \times 30 \times 30$, the DCA cluster is $6 \times 6 \times 6$. These results are comparable with experimental measurements only for the VCA mass disorder, as highlighted in [107]. The authors used the same VCA approach for mass perturbation only, up to second order. The most obvious explanation for this behaviour is that the force constant dis-

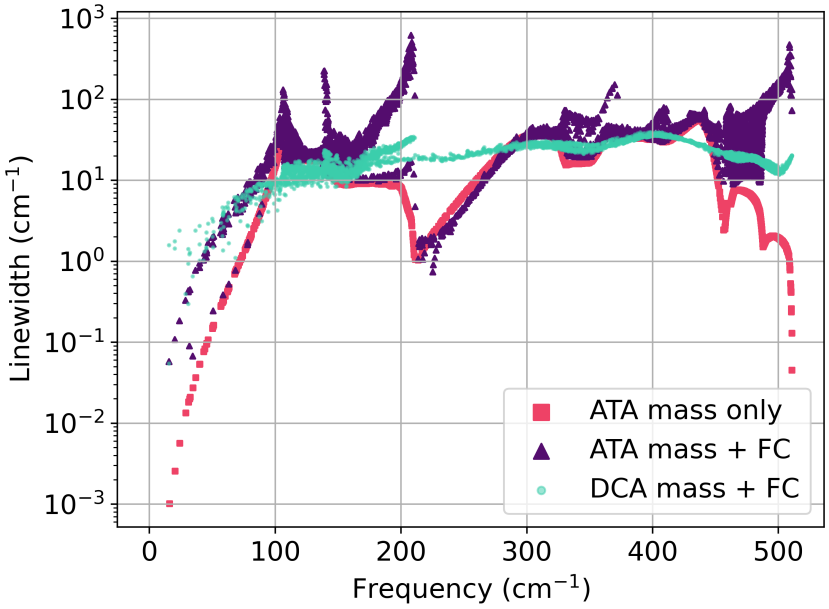


Figure 3.14: *Linewidths for $Si_{0.9}Ge_{0.1}$, computed with ATA and DCA. For ATA, force constants disorder produces a strong scattering near resonances: at around 200 cm^{-1} the linewidth is larger than the reference frequency, and hence the phonon is no longer a quasiparticle with a well-defined energy. DCA instead smears the resonances, but still does not give the experimental thermal conductivity.*

order introduces a strong scattering near resonances, yielding a linewidth larger than the reference frequency, as shown in Fig. (3.14). The DCA also fails to reproduce the experimental thermal conductivity, even though the resonance peaks are smeared out by self-consistency. The reason is that the presence of force constants introduces strong scattering in the low-frequency region: Fig. (3.15) shows that the thermal conductivity is dominated by the low-frequency modes in the mass disorder case, while for force constants disorder this peak is absent.

The first guess for this discrepancy would be to check for failures of the low-frequency elastic scattering limit $\Sigma \propto \omega^5$. This limit is correctly reproduced by the mass term, since the first non-zero perturbation term is the second order and then $\Sigma \propto VgV \propto \omega^4 g \propto \omega^3 \rho(\omega) \propto \omega^5$. The force constants term should also display the same behaviour $V \propto \omega^2$;

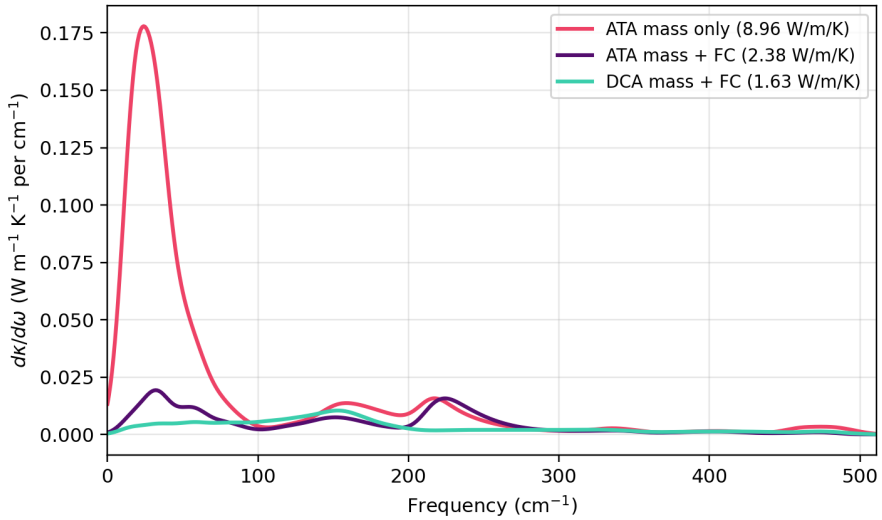


Figure 3.15: *Spectral decomposition of the thermal conductivity for $Si_{0.9}Ge_{0.1}$, computed with ATA and DCA. The presence of force constants disorder strongly reduces the contribution of low-frequency modes, which dominate the thermal conductivity in the mass disorder case.*

at low frequencies, the acoustic phonons have a linear dispersion $\omega \propto q$ and then

$$V(\boldsymbol{q}, \boldsymbol{q'}) = \sum_{\boldsymbol{RR'}} (1 - i\boldsymbol{q} \cdot \boldsymbol{R} + \mathcal{O}(\omega^2)) V(\boldsymbol{R}, \boldsymbol{R'}) (i\boldsymbol{q'} \cdot \boldsymbol{R'} + 1 + \mathcal{O}(\omega^2)) \tag{3.35}$$

$$= \boldsymbol{q} \cdot \left[\sum_{\boldsymbol{R},\boldsymbol{R'}} \boldsymbol{R} V(\boldsymbol{R}, \boldsymbol{R'}) \boldsymbol{R'} \right] \cdot \boldsymbol{q'} + \sum_{\boldsymbol{RR'}} V(\boldsymbol{R}, \boldsymbol{R'}) + \mathcal{O}(\omega^3) \tag{3.36}$$

$$= V(\boldsymbol{0}, \boldsymbol{0}) + \mathcal{O}(\omega^2). \tag{3.37}$$

The zero-momentum term $V(\boldsymbol{0}, \boldsymbol{0})$ is set to zero by acoustic sum rules. Centering slightly violates the ASR, but even by restoring it afterwards we cannot recover the low-frequency peak. The reason is that even if the ω^5 limit is satisfied, the prefactor is much larger than the mass disorder case.
Another possible explanation is that the alloy clusters into Si-rich and Ge-rich regions, for which the force constants are much closer to the pure Si and Ge values. The force constants disorder is then present only at the interfaces between the two regions, effectively reducing the total scattering.

Future perspectives

The aim of this thesis was to investigate the known non-perturbative approaches to treat the phonon-defect scattering problem, such as the average T-matrix approximation (ATA), and to develop a new approach called dynamical cluster approximation (DCA). These two approximations are used with great results in the study of systems in the presence of strong scattering centers in disordered systems. We have described in detail the theoretical framework and the workflow that yields the potential from first principles calculations of the force constants of the pristine and defective systems. We introduced new methods that impose the acoustic sum rule and generalized the centering method to the case of defective force constants, to obtain a smooth potential in reciprocal space. We then investigated the various methods linked with tetrahedron integration to obtain non-divergent Green's Functions. In the systems we studied, we noticed that ATA and DCA tend to overestimate the scattering strength and gave possible explanations for this behavior. Better approximations should proceed in various directions:

- The defect distribution in the system is assumed to be random, but in real systems clustering of defects can occur. This should be taken into account in the future, by sampling according to an a priori distribution in the DCA clusters. The distribution can be guessed with accurate studies of thermodynamic stability of the defects.
- The force constants difference is generally really informative: it scales not only the interaction strength between host and defective atoms with distance, but also between neighbouring host atoms. It is sufficient to solve the single-site problem, but does not take into account the presence of any other defects in the system. More accurate methods should fit the interaction strength with distance between atoms and defect and plug the correct interactions in the DCA cluster.
- The real part of the self-energy plays an important role, especially near resonance energies. The quasi-particle energies are then shifted and the scattering strength is modified by the renormalization factor. Off-diagonal branch elements can yield unphysical scattering rates, even if the self-energy is causal. This is usually a signature of strong mixing between branches, that must be taken into account in future calculations.
- Great attention should be paid in the low-energy region, which contributes mostly to the lattice thermal conductivity. Interaction between centering methods and acoustic sum rules should be deeply investigated. The acoustic sum rule should retain the locality of the force constants, without breaking the derivative-exchange symmetry.

- While it was possible to adapt the ATA with in a virtual crystal, both for mass and force constant disorder, the DCA still lacks the introduction of multiple type scatterers in the cluster, taking the information of many species in alloy systems.

Alternative approaches would open the door to the study of high entropy alloys and the ad hoc synthesis of new classes of thermoelectric materials with negligible thermal conductivity driven by defect scattering. Further analysis should isolate the role of mass, chemical bonding and strain changes in the final scattering strength, and especially the role of defect correlations in more complex systems.

Bibliography

[1] James C. Phillips and Leonard Kleinman. “New Method for Calculating Wave Functions in Crystals and Molecules”. In: *Phys. Rev.* 116 (2 Oct. 1959), pp. 287–294. DOI: 10.1103/PhysRev.116.287. URL: https://link.aps.org/doi/10.1103/PhysRev.116.287.

[2] P. Hohenberg and W. Kohn. “Inhomogeneous Electron Gas”. In: *Phys. Rev.* 136 (3B Nov. 1964), B864–B871. DOI: 10.1103/PhysRev.136.B864. URL: https://link.aps.org/doi/10.1103/PhysRev.136.B864.

[3] Mel Levy. “Universal variational functionals of electron densities, first-order density matrices, and natural spin-orbitals and solution of the V-representability problem”. In: *Proceedings of the National Academy of Sciences* 76.12 (1979), pp. 6062–6065. DOI: 10.1073/pnas.76.12.6062.

[4] Elliott H. Lieb. “Density functionals for coulomb systems”. In: *International Journal of Quantum Chemistry* 24.3 (1983), pp. 243–277. DOI: https://doi.org/10.1002/qua.560240302.

[5] W. Kohn and L. J. Sham. “Self-Consistent Equations Including Exchange and Correlation Effects”. In: *Phys. Rev.* 140 (4A Nov. 1965), A1133–A1138. DOI: 10.1103/PhysRev.140.A1133. URL: https://link.aps.org/doi/10.1103/PhysRev.140.A1133.

[6] Paolo Giannozzi and Stefano Baroni. “Density-Functional Perturbation Theory”. In: *Handbook of Materials Modeling: Methods*. Ed. by Sidney Yip. Dordrecht: Springer Netherlands, 2005, pp. 195–214. ISBN: 978-1-4020-3286-8. DOI: 10.1007/978-1-4020-3286-8_11.

[7] Stefano Baroni et al. “Phonons and related crystal properties from density-functional perturbation theory”. In: *Reviews of Modern Physics* 73.2 (July 2001), pp. 515–562. DOI: 10.1103/revmodphys.73.515. URL: https://doi.org/10.1103%5C%2Frevmodphys.73.515.

[8] Paolo Giannozzi et al. “QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials”. In: *Journal of Physics: Condensed Matter* 21.39 (2009), p. 395502. ISSN: 1361-648X. DOI: 10.1088/0953-8984/21/39/395502. URL: http://dx.doi.org/10.1088/0953-8984/21/39/395502.

[9] P Giannozzi et al. “Advanced capabilities for materials modelling with Quantum ESPRESSO”. In: *Journal of Physics: Condensed Matter* 29.46 (Oct. 2017), p. 465901. DOI: 10.1088/1361-648X/aa8f79. URL: https://doi.org/10.1088/1361-648X/aa8f79.

[10] R. D. King-Smith and David Vanderbilt. “Theory of polarization of crystalline solids”. In: *Phys. Rev. B* 47 (3 Jan. 1993), 1651(R)–1654(R). DOI: 10.1103/PhysRevB.47.1651. URL: https://link.aps.org/doi/10.1103/PhysRevB.47.1651.

[11] Harald Böttger. “Principles of the Theory of Lattice Dynamics”. In: 1983.

[12] X. Gonze et al. “Interatomic force constants from first principles: The case of α-quartz”. In: *Phys. Rev. B* 50 (17 Nov. 1994), 13035(R)–13038(R). DOI: 10.1103/PhysRevB.50.13035. URL: https://link.aps.org/doi/10.1103/PhysRevB.50.13035.

[13] Xavier Gonze and Changyol Lee. “Dynamical matrices, Born effective charges, dielectric permittivity tensors, and interatomic force constants from density-functional perturbation theory”. In: *Phys. Rev. B* 55 (16 Apr. 1997), pp. 10355–10368. DOI: 10.1103/PhysRevB.55.10355. URL: https://link.aps.org/doi/10.1103/PhysRevB.55.10355.

[14] Paolo Giannozzi et al. “Ab initio calculation of phonon dispersions in semiconductors”. In: *Phys. Rev. B* 43 (9 Mar. 1991), pp. 7231–7242. DOI: 10.1103/PhysRevB.43.7231. URL: https://link.aps.org/doi/10.1103/PhysRevB.43.7231.

[15] W. Hayes and R. Loudon. *Scattering of Light by Crystals*. New York: Wiley, 1978.

[16] Peter Brüesch. “Raman Spectroscopy”. In: *Phonons: Theory and Experiments II: Experiments and Interpretation of Experimental Results*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1986, pp. 65–105. ISBN: 978-3-642-52263-5. DOI: 10.1007/978-3-642-52263-5_3. URL: https://doi.org/10.1007/978-3-642-52263-5_3.

[17] Michele Lazzeri and Francesco Mauri. “First-Principles Calculation of Vibrational Raman Spectra in Large Systems: Signature of Small Rings in Crystalline SiO_2”. In: *Phys. Rev. Lett.* 90 (3 Jan. 2003), p. 036401. DOI: 10.1103/PhysRevLett.90.036401. URL: https://link.aps.org/doi/10.1103/PhysRevLett.90.036401.

[18] Michele Lazzeri and Francesco Mauri. “High-order density-matrix perturbation theory”. In: *Phys. Rev. B* 68 (16 Oct. 2003), p. 161101. DOI: 10.1103/PhysRevB.68.161101. URL: https://link.aps.org/doi/10.1103/PhysRevB.68.161101.

[19] Lorenzo Paulatto, Francesco Mauri, and Michele Lazzeri. “Anharmonic properties from a generalized third-order ab initio approach: Theory and applications to graphite and graphene”. In: *Phys. Rev. B* 87 (21 June 2013), p. 214303. DOI: 10.1103/PhysRevB.87.214303. URL: https://link.aps.org/doi/10.1103/PhysRevB.87.214303.

[20] I-Te Lu et al. “Ab initio electron-defect interactions using Wannier functions”. In: *npj Computational Materials* 6.1 (2020), p. 17. ISSN: 2057-3960. DOI: 10.1038/s41524-020-0284-y. URL: https://doi.org/10.1038/s41524-020-0284-y.

[21] Rongjing Guo et al. “Accurate Ab Initio Method for Charged Defect Scattering”. In: *Phys. Rev. Lett.* 135 (12 Sept. 2025), p. 126302. DOI: 10.1103/42pc-t2bx. URL: https://link.aps.org/doi/10.1103/42pc-t2bx.

[22] Miles V. Klein. “Phonon Scattering by Lattice Defects”. In: *Phys. Rev.* 131 (4 Aug. 1963), pp. 1500–1510. DOI: 10.1103/PhysRev.131.1500. URL: https://link.aps.org/doi/10.1103/PhysRev.131.1500.

[23] David Borwein, Jonathan M. Borwein, and Keith F. Taylor. “Convergence of lattice sums and Madelung’s constant”. In: *Journal of Mathematical Physics* 26.11 (Nov. 1985), pp. 2999–3009. ISSN: 0022-2488. DOI: 10.1063/1.526675. eprint: https://pubs.aip.org/aip/jmp/article-pdf/26/11/2999/19092169/2999_1_online.pdf. URL: https://doi.org/10.1063/1.526675.

[24] Christoph Freysoldt, Jörg Neugebauer, and Chris G. Van de Walle. “Fully Ab Initio Finite-Size Corrections for Charged-Defect Supercell Calculations”. In: *Phys. Rev. Lett.* 102 (1 Jan. 2009), p. 016402. DOI: 10.1103/PhysRevLett.102.016402. URL: https://link.aps.org/doi/10.1103/PhysRevLett.102.016402.

[25] Yu Kumagai and Fumiyasu Oba. “Electrostatics-based finite-size corrections for first-principles point defect calculations”. In: *Phys. Rev. B* 89 (19 May 2014), p. 195205. DOI: 10.1103/PhysRevB.89.195205. URL: https://link.aps.org/doi/10.1103/PhysRevB.89.195205.

[26] G. Makov and M. C. Payne. “Periodic boundary conditions in ab initio calculations”. In: *Phys. Rev. B* 51 (7 Feb. 1995), pp. 4014–4022. DOI: 10.1103/PhysRevB.51.4014. URL: https://link.aps.org/doi/10.1103/PhysRevB.51.4014.

[27] W. Kohn and J. M. Luttinger. “Theory of Donor States in Silicon”. In: *Phys. Rev.* 98 (4 May 1955), pp. 915–922. DOI: 10.1103/PhysRev.98.915. URL: https://link.aps.org/doi/10.1103/PhysRev.98.915.

[28] Fusayoshi J. Ohkawa. “Theory of Shallow Donor States in Silicon Substitutional Donors”. In: *Journal of the Physical Society of Japan* 46.5 (1979), pp. 1529–1537. DOI: 10.1143/JPSJ.46.1529.

[29] I-Te Lu et al. “First-principles ionized-impurity scattering and charge transport in doped materials”. In: *Phys. Rev. Mater.* 6 (1 Jan. 2022), p. L010801. DOI: 10.1103/PhysRevMaterials.6.L010801. URL: https://link.aps.org/doi/10.1103/PhysRevMaterials.6.L010801.

[30] W. Kohn. “Image of the Fermi Surface in the Vibration Spectrum of a Metal”. In: *Phys. Rev. Lett.* 2 (9 May 1959), pp. 393–394. DOI: 10.1103/PhysRevLett.2.393. URL: https://link.aps.org/doi/10.1103/PhysRevLett.2.393.

[31] J. Friedel. “XIV. The distribution of electrons round impurities in monovalent metals”. In: *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 43.337 (1952), pp. 153–189. DOI: 10.1080/14786440208561086. eprint: https://doi.org/10.1080/14786440208561086. URL: https://doi.org/10.1080/14786440208561086.

[32] Peter J. Feibelman. “Friedel oscillations in the force constants of metals”. In: *Phys. Rev. B* 55 (14 Apr. 1997), pp. 8821–8828. DOI: 10.1103/PhysRevB.55.8821. URL: https://link.aps.org/doi/10.1103/PhysRevB.55.8821.

[33] J Lindhard. “On the properties of a gas of charged particles”. In: *Kgl. Danske Videnskab. Selskab Mat.-fys. Medd.* Vol: 28, No. 8 (Jan. 1954). URL: https://www.osti.gov/biblio/4405425.

[34] Matteo Calandra, Gianni Profeta, and Francesco Mauri. “Adiabatic and nonadiabatic phonon dispersion in a Wannier function approach”. In: *Physical Review B* 82.16 (Oct. 2010). ISSN: 1550-235X. DOI: 10.1103/physrevb.82.165111. URL: http://dx.doi.org/10.1103/PhysRevB.82.165111.

[35] L. Paulatto A. Ciavatta. *D3Q-def: A code for the calculation of phonon scattering by defects*. 2026. URL: https://github.com/lord-cepo/D3Q.

[36] Paul L. Leath and Bernard Goodman. “Displacement Correlations and Frequency Spectra for Mass-Disordered Lattices. II”. In: *Phys. Rev.* 175 (3 Nov. 1968), pp. 963–971. DOI: 10.1103/PhysRev.175.963. URL: https://link.aps.org/doi/10.1103/PhysRev.175.963.

[37] P. L. Leath. “Self-Consistent-Field Approximations in Disordered Alloys”. In: *Phys. Rev.* 171 (3 July 1968), pp. 725–727. DOI: 10.1103/PhysRev.171.725. URL: https://link.aps.org/doi/10.1103/PhysRev.171.725.

[38] Peter Staar, Thomas Maier, and Thomas C. Schulthess. “Dynamical cluster approximation with continuous lattice self-energy”. In: *Physical Review B* 88.11 (Sept. 2013). ISSN: 1550-235X. DOI: 10.1103/physrevb.88.115101. URL: http://dx.doi.org/10.1103/PhysRevB.88.115101.

[39] Mitsuaki Kawamura, Yoshihiro Gohda, and Shinji Tsuneyuki. “Improved tetrahedron method for the Brillouin-zone integration applicable to response functions”. In: *Physical Review B* 89.9 (Mar. 2014). ISSN: 1550-235X. DOI: 10.1103/physrevb.89.094515. URL: http://dx.doi.org/10.1103/PhysRevB.89.094515.

[40] N. E. Brener, J. L. Fry, and R. A. Johnson. “Matrix elements in the analytic tetrahedron method”. In: *International Journal of Quantum Chemistry* 20.S15 (1981), pp. 459–468. DOI: https://doi.org/10.1002/qua.560200850. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/qua.560200850. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/qua.560200850.

[41] A H MacDonald, S H Vosko, and P T Coleridge. “Extensions of the tetrahedron method for evaluating spectral properties of solids”. In: *Journal of Physics C: Solid State Physics* 12.15 (Aug. 1979), p. 2991. DOI: 10.1088/0022-3719/12/15/008. URL: https://doi.org/10.1088/0022-3719/12/15/008.

[42] Minsu Ghim and Cheol-Hwan Park. “Converging tetrahedron method calculations for the nondissipative parts of spectral functions”. In: *Phys. Rev. B* 106 (7 Aug. 2022), p. 075126. DOI: 10.1103/PhysRevB.106.075126. URL: https://link.aps.org/doi/10.1103/PhysRevB.106.075126.

[43] Ph. Lambin and J. P. Vigneron. “Computation of crystal Green’s functions in the complex-energy plane with the use of the analytical tetrahedron method”. In: *Phys. Rev. B* 29 (6 Mar. 1984), pp. 3430–3437. DOI: 10.1103/PhysRevB.29.3430. URL: https://link.aps.org/doi/10.1103/PhysRevB.29.3430.

[44] S Kaprzyk and P E Mijnarends. “A simple linear analytic method for Brillouin zone integration of spectral functions in the complex energy plane”. In: *Journal of Physics C: Solid State Physics* 19.9 (Mar. 1986), p. 1283. DOI: 10.1088/0022-3719/19/9/002. URL: https://doi.org/10.1088/0022-3719/19/9/002.

[45] S. Kaprzyk. “An algebraic formulation and implementation of the tetrahedron linear method for the Brillouin zone integration of spectral functions”. In: *Computer Physics Communications* 183.2 (2012), pp. 347–362. ISSN: 0010-4655. DOI: https://doi.org/10.1016/j.cpc.2011.08.022. URL: https://www.sciencedirect.com/science/article/pii/S0010465511003109.

[46] Roger James - Na6211 Elliott and D. W. Taylor. “Vibrations of random dilute alloys”. In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 296.1445 (Jan. 1967), pp. 161–188. ISSN: 0080-4630. DOI: 10.1098/rspa.1967.0012. eprint: https://royalsocietypublishing.org/rspa/article-pdf/296/1445/161/56657/rspa.1967.0012.pdf. URL: https://doi.org/10.1098/rspa.1967.0012.

[47] P. L. Taylor and Shi-Yu Wu. “Dynamics of Disordered Alloys and Glasses”. In: *Phys. Rev. B* 2 (6 Sept. 1970), pp. 1752–1757. DOI: 10.1103/PhysRevB.2.1752. URL: https://link.aps.org/doi/10.1103/PhysRevB.2.1752.

[48] Yu. Kagan and Ya. A. Iosilevskii. “Scattering of Neutrons by Crystals with Impurity Nuclei and the Problem of Reconstructing the Vibration Spectrum”. In: *ZhETF* 44 (1963). English translation: Sov. Phys. JETP 17, 926 (1963), p. 1375.

[49] Bonny Dongre et al. “Combined treatment of phonon scattering by electrons and point defects explains the thermal conductivity reduction in highly-doped Si”. In: *J. Mater. Chem. A* 8 (3 2020), pp. 1273–1278. DOI: 10.1039/C9TA11424F. URL: http://dx.doi.org/10.1039/C9TA11424F.

[50] R. Guo and S. Lee. “Mie scattering of phonons by point defects in IV-VI semiconductors PbTe and GeTe”. In: *Materials Today Physics* 12 (Mar. 2020), p. 100177. ISSN: 2542-5293. DOI: 10.1016/j.mtphys.2020.100177. URL: http://dx.doi.org/10.1016/j.mtphys.2020.100177.

[51] Marco Arrigoni et al. “First-principles quantitative prediction of the lattice thermal conductivity in random semiconductor alloys: The role of force-constant disorder”. In: *Phys. Rev. B* 98 (11 Sept. 2018), p. 115205. DOI: 10.1103/PhysRevB.98.115205. URL: https://link.aps.org/doi/10.1103/PhysRevB.98.115205.

[52] Shin-ichiro Tamura. “Isotope scattering of dispersive phonons in Ge”. In: *Phys. Rev. B* 27 (2 Jan. 1983), pp. 858–866. DOI: 10.1103/PhysRevB.27.858. URL: https://link.aps.org/doi/10.1103/PhysRevB.27.858.

[53] Nakib H. Protik and Claudia Draxl. “Beyond the Tamura model of phonon-isotope scattering”. In: *Phys. Rev. B* 109 (16 Apr. 2024), p. 165201. DOI: 10.1103/PhysRevB.109.165201. URL: https://link.aps.org/doi/10.1103/PhysRevB.109.165201.

[54] D. D. Johnson. “Modified Broyden’s method for accelerating convergence in self-consistent calculations”. In: *Phys. Rev. B* 38 (18 Dec. 1988), pp. 12807–12813. DOI: 10.1103/PhysRevB.38.12807. URL: https://link.aps.org/doi/10.1103/PhysRevB.38.12807.

[55] David Vanderbilt and Steven G. Louie. “Total energies of diamond (111) surface reconstructions by a linear combination of atomic orbitals method”. In: *Phys. Rev. B* 30 (10 Nov. 1984), pp. 6118–6130. DOI: 10.1103/PhysRevB.30.6118. URL: https://link.aps.org/doi/10.1103/PhysRevB.30.6118.

[56] P. G. Klemens. “The Scattering of Low-Frequency Lattice Waves by Static Imperfections”. In: *Proceedings of the Physical Society. Section A* 68 (1955), pp. 1113–1128. DOI: 10.1088/0370-1298/68/12/303.

[57] C. A. Ratsifaritana and P. G. Klemens. “Scattering of phonons by vacancies”. In: *International Journal of Thermophysics* 8.6 (Nov. 1987), pp. 737–750. DOI: 10.1007/BF00500791.

[58] Carlos A. Polanco and Lucas Lindsay. “Ab initio phonon point defect scattering and thermal transport in graphene”. In: *Phys. Rev. B* 97 (1 Jan. 2018), p. 014303. DOI: 10.1103/PhysRevB.97.014303. URL: https://link.aps.org/doi/10.1103/PhysRevB.97.014303.

[59] N. A. Katcho et al. “Effect of nitrogen and vacancy defects on the thermal conductivity of diamond: An ab initio Green’s function approach”. In: *Phys. Rev. B* 90 (9 Sept. 2014), p. 094117. DOI: 10.1103/PhysRevB.90.094117. URL: https://link.aps.org/doi/10.1103/PhysRevB.90.094117.

[60] G Bouzerar et al. “Drastic effects of vacancies on phonon lifetime and thermal conductivity in graphene”. In: *Journal of Physics: Condensed Matter* 32.29 (Apr. 2020), p. 295702. DOI: 10.1088/1361-648X/ab7f70. URL: https://doi.org/10.1088/1361-648X/ab7f70.

[61] Joseph Callaway. “Model for Lattice Thermal Conductivity at Low Temperatures”. In: *Phys. Rev.* 113 (4 Feb. 1959), pp. 1046–1051. DOI: 10.1103/PhysRev.113.1046. URL: https://link.aps.org/doi/10.1103/PhysRev.113.1046.

[62] G D Watkins. “AN EPR STUDY OF THE LATTICE VACANCY IN SILICON”. In: *J. Phys. Soc. Japan* Vol: 18: Suppl. II (Mar. 1963). URL: https://www.osti.gov/biblio/4638373.

[63] Fedwa El-Mellouhi, Normand Mousseau, and Pablo Ordejón. “Sampling the diffusion paths of a neutral vacancy in silicon with quantum mechanical calculations”. In: *Phys. Rev. B* 70 (20 Nov. 2004), p. 205202. DOI: 10.1103/PhysRevB.70.205202. URL: https://link.aps.org/doi/10.1103/PhysRevB.70.205202.

[64] Yongjin Lee, Sangheon Lee, and Gyeong S. Hwang. “Effects of vacancy defects on thermal conductivity in crystalline silicon: A nonequilibrium molecular dynamics study”. In: *Phys. Rev. B* 83 (12 Mar. 2011), p. 125202. DOI: 10.1103/PhysRevB.83.125202. URL: https://link.aps.org/doi/10.1103/PhysRevB.83.125202.

[65] Fabiano Corsetti. “System-size convergence of point defect properties: The case of the silicon vacancy”. In: *Phys. Rev. B* 84 (Oct. 2010). DOI: 10.1103/PhysRevB.84.035209.

[66] A. Wright. “Density-functional-theory calculations for the silicon vacancy”. In: *Phys. Rev. B* 74 (Oct. 2006). DOI: 10.1103/PhysRevB.74.165116.

[67] Sholihun Sholihun et al. “Density-functional-theory-based calculations of formation energy and concentration of the silicon monovacancy”. In: *Japanese Journal of Applied Physics* 54 (Feb. 2015), p. 041301. DOI: 10.7567/JJAP.54.041301.

[68] D. R. Hamann. “Optimized norm-conserving Vanderbilt pseudopotentials”. In: *Phys. Rev. B* 88 (8 Aug. 2013), p. 085117. DOI: 10.1103/PhysRevB.88.085117. URL: https://link.aps.org/doi/10.1103/PhysRevB.88.085117.

[69] John P. Perdew, Kieron Burke, and Matthias Ernzerhof. “Generalized Gradient Approximation Made Simple”. In: *Physical Review Letters* 77.18 (1996), pp. 3865–3868. DOI: 10.1103/PhysRevLett.77.3865.

[70] Martin Schlipf and François Gygi. “Optimization algorithm for the generation of ONCV pseudopotentials”. In: *Computer Physics Communications* 196 (2015), pp. 36–44. ISSN: 0010-4655. DOI: 10.1016/j.cpc.2015.05.011. URL: http://www.sciencedirect.com/science/article/pii/S0010465515001897.

[71] Thibault Sohier, Matteo Calandra, and Francesco Mauri. “Density functional perturbation theory for gated two-dimensional heterostructures: Theoretical developments and application to flexural phonons in graphene”. In: *Phys. Rev. B* 96 (7 Aug. 2017), p. 075448. DOI: 10.1103/PhysRevB.96.075448. URL: https://link.aps.org/doi/10.1103/PhysRevB.96.075448.

[72] J. P. Perdew and A. Zunger. “Self-interaction correction to density-functional approximations for many-electron systems”. In: *Physical Review B* 23.10 (1981), pp. 5048–5079. DOI: 10.1103/PhysRevB.23.5048.

[73] Hualing Zeng et al. “Optical signature of symmetry variations and spin-valley coupling in atomically thin tungsten dichalcogenides”. In: *Scientific Reports* 3.1 (2013), p. 1608. ISSN: 2045-2322. DOI: 10.1038/srep01608. URL: https://doi.org/10.1038/srep01608.

[74] H. Terrones et al. “New First Order Raman-active Modes in Few Layered Transition Metal Dichalcogenides”. In: *Scientific Reports* 4.1 (2014), p. 4215. ISSN: 2045-2322. DOI: 10.1038/srep04215. URL: https://doi.org/10.1038/srep04215.

[75] Claire Blaga et al. “Unveiling the complex phonon nature and phonon cascades in 1L to 5L WSe2 using multiwavelength excitation Raman scattering”. In: *Nanoscale Adv.* 6 (18 2024), pp. 4591–4603. DOI: 10.1039/D4NA00399C. URL: http://dx.doi.org/10.1039/D4NA00399C.

[76] Miquel Royo and Massimiliano Stengel. “Exact Long-Range Dielectric Screening and Interatomic Force Constants in Quasi-Two-Dimensional Crystals”. In: *Physical Review X* 11 (2021), p. 041027. DOI: 10.1103/PhysRevX.11.041027.

[77] Changpeng Lin, Samuel Poncé, and Nicola Marzari. “General invariance and equilibrium conditions for lattice dynamics in 1D, 2D, and 3D materials”. In: *npj Computational Materials* 8.1 (2022), p. 236. ISSN: 2057-3960. DOI: 10.1038/s41524-022-00920-6. URL: https://doi.org/10.1038/s41524-022-00920-6.

[78] G. Placzek. “Rayleigh-Streuung und Raman-Effekt”. German. In: *Handbuch der Radiologie*. Ed. by E. Marx. Vol. 6. 2. Leipzig: Akademische Verlagsgesellschaft, 1934, pp. 205–374.

[79] Michele Lazzeri, Matteo Calandra, and Francesco Mauri. “Anharmonic phonon frequency shift in MgB2”. In: *Physical Review B* 68.22 (2003), p. 220509. DOI: 10.1103/PhysRevB.68.220509. arXiv: cond-mat/0306650.

[80] Enda Xiao and Chris A. Marianetti. “Anharmonic phonon behavior via irreducible derivatives: Self-consistent perturbation theory and molecular dynamics”. In: *Physical Review B* 107.9 (2023), p. 094303. DOI: 10.1103/PhysRevB.107.094303.

[81] E. del Corro et al. “Atypical Exciton–Phonon Interactions in WS2 and WSe2 Monolayers Revealed by Resonance Raman Spectroscopy”. In: *Nano Letters* 16.4 (2016). PMID: 26998817, pp. 2363–2368. DOI: 10.1021/acs.nanolett.5b05096. eprint: https://doi.org/10.1021/acs.nanolett.5b05096. URL: https://doi.org/10.1021/acs.nanolett.5b05096.

[82] Commission on Isotopic Abundances and Atomic Weights. *Standard Atomic Weights 2024*. IUPAC/CIAAW table of standard atomic weights. 2024.

[83] Beatriz Cordero et al. “Covalent Radii Revisited”. In: *Dalton Transactions* 21 (2008), pp. 2832–2838. DOI: 10.1039/B801115J.

[84] M. Gasseller et al. “Single-Electron Capacitance Spectroscopy of Individual Dopants in Silicon”. In: *Nano Letters* 11.12 (2011), pp. 5208–5212. DOI: 10.1021/nl2025163.

[85] H. C. Alt. “Determination of Shallow Dopants in Silicon by Low-Temperature Infrared Spectroscopy”. In: *AIP Conference Proceedings*. Vol. 449. 1998, pp. 201–206. DOI: 10.1063/1.56861.

[86] S. M. Sze and Kwok K. Ng. *Physics of Semiconductor Devices (Appendix G)*. 3rd ed. Hoboken, New Jersey: John Wiley & Sons, 2007.

[87] J. S. Smith et al. “Ab initio calculation of energy levels for phosphorus donors in silicon”. In: *Scientific Reports* 7.1 (2017), p. 6010. ISSN: 2045-2322. DOI: 10.1038/s41598-017-06296-8. URL: https://doi.org/10.1038/s41598-017-06296-8.

[88] John P. Perdew et al. “Understanding band gaps of solids in generalized Kohn–Sham theory”. In: *Proceedings of the National Academy of Sciences* 114.11 (Mar. 2017), pp. 2801–2806. ISSN: 1091-6490. DOI: 10.1073/pnas.1621352114. URL: http://dx.doi.org/10.1073/pnas.1621352114.

[89] Paula Mori-Sánchez, Aron J. Cohen, and Weitao Yang. “Localization and Delocalization Errors in Density Functional Theory and Implications for Band-Gap Prediction”. In: *Phys. Rev. Lett.* 100 (14 Apr. 2008), p. 146401. DOI: 10.1103/PhysRevLett.100.146401. URL: https://link.aps.org/doi/10.1103/PhysRevLett.100.146401.

[90] Joshua Claes et al. *An accurate DFT-1/2 approach for shallow defect states: Efficient calculation of donor binding energies in silicon*. 2025. arXiv: 2508.14738 [cond-mat.mtrl-sci]. URL: https://arxiv.org/abs/2508.14738.

[91] J. F. Angress, A. R. Goodwin, and S. D. Smith. “A study of vibrations of boron and phosphorus in silicon by infra-red absorption”. In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 287.1408 (Aug. 1965), pp. 64–88. ISSN: 0080-4630. DOI: 10.1098/rspa.1965.0169. eprint: https://royalsocietypublishing.org/rspa/article-pdf/287/1408/64/55647/rspa.1965.0169.pdf. URL: https://doi.org/10.1098/rspa.1965.0169.

[92] A. Matthiessen and C. Vogt. “On the Influence of Temperature on the Electric Conducting-Power of Alloys”. In: *Philosophical Transactions of the Royal Society of London* 154 (1864), pp. 167–200.

[93] Tianli Feng, Bo Qiu, and Xiulin Ruan. “Coupling between phonon-phonon and phonon-impurity scattering: A critical revisit of the spectral Matthiessen’s rule”. In: *Physical Review B* 92 (2015), p. 235206. DOI: 10.1103/PhysRevB.92.235206.

[94] Feliciano Giustino. “Electron-phonon interactions from first principles”. In: *Rev. Mod. Phys.* 89 (1 Feb. 2017), p. 015003. DOI: 10.1103/RevModPhys.89.015003. URL: https://link.aps.org/doi/10.1103/RevModPhys.89.015003.

[95] Hyungjun Lee et al. “Electron–phonon physics from first principles using the EPW code”. In: *npj Computational Materials* 9.1 (2023), p. 156. ISSN: 2057-3960. DOI: 10.1038/s41524-023-01107-3. URL: https://doi.org/10.1038/s41524-023-01107-3.

[96] J.M. Ziman. *Electrons and Phonons: The Theory of Transport Phenomena in Solids*. Oxford University Press, Feb. 2001. ISBN: 9780198507796. DOI: 10.1093/acprof:oso/9780198507796.001.0001. URL: https://doi.org/10.1093/acprof:oso/9780198507796.001.0001.

[97] Hiroshi Uchiyama et al. “Quantifying doping-dependent electron-phonon scattering rates in silicon by inelastic x-ray scattering and first-principles lattice dynamics”. In: *Phys. Rev. Mater.* 7 (10 Oct. 2023), p. 104601. DOI: 10.1103/PhysRevMaterials.7.104601. URL: https://link.aps.org/doi/10.1103/PhysRevMaterials.7.104601.

[98] Raja Sen et al. “Competing effects of charge-carrier and impurity scattering limiting phonon heat conduction in heavily-doped silicon”. working paper or preprint. Jan. 2026. URL: https://cnrs.hal.science/hal-05451267.

[99] O. D. Restrepo, K. Varga, and S. T. Pantelides. “First-principles calculations of electron mobilities in silicon: Phonon and Coulomb scattering”. In: *Applied Physics Letters* 94.21 (May 2009), p. 212103. ISSN: 0003-6951. DOI: 10.1063/1.3147189. eprint: https://pubs.aip.org/aip/apl/article-pdf/doi/10.1063/1.3147189/14412442/212103_1_online.pdf. URL: https://doi.org/10.1063/1.3147189.

[100] Mattia Fiorentini and Nicola Bonini. “Thermoelectric coefficients of n-doped silicon from first principles via the solution of the Boltzmann transport equation”. In: *Phys. Rev. B* 94 (8 Aug. 2016), p. 085204. DOI: 10.1103/PhysRevB.94.085204. URL: https://link.aps.org/doi/10.1103/PhysRevB.94.085204.

[101] D G Cantrell and P N Butcher. “A calculation of the phonon-drag contribution to the thermopower of quasi-2D electrons coupled to 3D phonons. I. General theory”. In: *Journal of Physics C: Solid State Physics* 20.13 (May 1987), p. 1985. DOI: 10.1088/0022-3719/20/13/014. URL: https://doi.org/10.1088/0022-3719/20/13/014.

[102] Samuel Poncé, Elena R. Margine, and Feliciano Giustino. “Towards predictive many-body calculations of phonon-limited carrier mobilities in semiconductors”. In: *Phys. Rev. B* 97 (12 Mar. 2018), 121201(R). DOI: 10.1103/PhysRevB.97.121201. URL: https://link.aps.org/doi/10.1103/PhysRevB.97.121201.

[103] Michele Simoncelli, Nicola Marzari, and Francesco Mauri. “Wigner Formulation of Thermal Transport in Solids”. In: *Phys. Rev. X* 12 (4 Oct. 2022), p. 041011. DOI: 10.1103/PhysRevX.12.041011. URL: https://link.aps.org/doi/10.1103/PhysRevX.12.041011.

[104] Sesha Sai Behara et al. “Chemo-mechanics in alloy phase stability”. In: (2023). arXiv: 2310.20085 [cond-mat.mtrl-sci]. URL: https://arxiv.org/abs/2310.20085.

[105] R. Basu and A. Singh. “High temperature Si–Ge alloy towards thermoelectric applications: A comprehensive review”. In: *Materials Today Physics* 21 (2021), p. 100468. ISSN: 2542-5293. DOI: https://doi.org/10.1016/j.mtphys.2021.100468. URL: https://www.sciencedirect.com/science/article/pii/S2542529321001292.

[106] B. Abeles. “Lattice Thermal Conductivity of Disordered Semiconductor Alloys at High Temperatures”. In: *Phys. Rev.* 131 (5 Sept. 1963), pp. 1906–1911. DOI: 10.1103/PhysRev.131.1906. URL: https://link.aps.org/doi/10.1103/PhysRev.131.1906.

[107] Jivtesh Garg et al. “Role of Disorder and Anharmonicity in the Thermal Conductivity of Silicon-Germanium Alloys: A First-Principles Study”. In: *Phys. Rev. Lett.* 106 (4 Jan. 2011), p. 045901. DOI: 10.1103/PhysRevLett.106.045901. URL: https://link.aps.org/doi/10.1103/PhysRevLett.106.045901.